

GTOS Detailed Volume - Volume 03

End To End Global Trade Lifecycle

GTOS Enterprise Engineering Handbook - Final Edition 1.0

Source Chapters

- Publication Chapter 10 — Proforma Invoice Lifecycle
- Publication Chapter 11 — Payment and Risk Approval
- Publication Chapter 12 — Purchase Order Creation and Acknowledgement
- Publication Chapter 13 — Production Planning and Capacity Commitment
- Publication Chapter 14 — Production Execution and Milestone Observation
- Publication Chapter 15 — Quality Inspection, Nonconformance, and Release
- Publication Chapter 16 — Packing List and Commercial Invoice
- Publication Chapter 17 — Export Customs and Legal Release
- Publication Chapter 18 — Transport Planning, Booking, and Capacity
- Publication Chapter 19 — Pickup, Custody Transfer, and Handover
- Publication Chapter 20 — International Transport, Tracking, and Exceptions
- Publication Chapter 21 — Import Customs, Duties, and Legal Release
- Publication Chapter 22 — Delivery and Warehouse Receiving
- Publication Chapter 23 — Inventory Synchronization, Reservation, and Availability
- Publication Chapter 24 — Product Publication, Marketplace, and Channel Eligibility
- Publication Chapter 25 — Checkout, Customer Order, and Commitment
- Publication Chapter 26 — Allocation, Fulfilment, and Last-Mile Delivery
- Publication Chapter 27 — Returns, Claims, Refunds, and Disputes
- Publication Chapter 28 — Settlement, Reporting, ESG, and Learning
- Batch 01 — Canonical Stage Map and Publication Numbering

Publication Chapter 10 — Proforma Invoice Lifecycle

Metadata	Value
Publication Chapter	10
Canonical Lifecycle Stage	09
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE with explicit repository notes
Accountable capability	Trade Documents / Contract / Finance Control
Canon dependencies	CANON-001 through CANON-011
Constitutional dependencies	ART-001 through ART-016
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This stage exists to **turn accepted commercial terms into a governed, versioned supplier-issued commercial document without confusing it with payment, a commercial invoice, or a purchase order.**

Its output is not merely a screen state or document download. It is a governed set of identified subjects, Relationships, Decisions, Intent, Commands, completed facts, external observations, evidence, and correction history.

Reality or external outcome
≠ platform Observation
≠ derived Perspective
≠ AI Prediction
≠ Recommendation
≠ authorized Decision
≠ execution Intent

1.1 Accountability

The accountable capability is **Trade Documents / Contract / Finance Control**. Supporting Portals, workflow engines, reports, search indexes, integration adapters, and AI Agents may coordinate or present the work, but they do not silently become owners of the underlying business truth.

1.2 Primary business outcomes

- a stable subject identity and exact source-version lineage;
- explicit authority and effective mandate;
- a testable lifecycle outcome;
- evidence sufficient for downstream use and later reconstruction;
- visible failure, partial completion, uncertainty, and correction;
- a completed-fact Event or qualified external observation;
- downstream activation without shared-database ownership transfer.

1.3 Non-goals

- treating a registered module as proof of complete implementation;

- treating a frontend route as proof of backend behavior;
- allowing an AI recommendation to become a binding Decision;
- inventing external-provider or government-authority success;
- overwriting history during amendment or correction;
- combining unrelated states into one generic status field.

2. Entry and Exit Criteria

2.1 Entry criteria

1. Every upstream subject reference resolves to a stable identifier.
2. The exact approved, accepted, issued, or observed version is known.
3. The initiating human, service, organization, or AI Agent is authenticated.
4. The represented principal and mandate are valid at command time.
5. Required Evidence exists, or an explicit authorized exception is linked.
6. Applicable Policy and jurisdiction versions are fixed.
7. Duplicate, stale-version, and replay checks have run.
8. Any conflicting active process is resolved, suspended, or explicitly related.

2.2 Exit criteria

1. The authoritative Domain or external authority records the outcome.
2. State and version changes are durable.
3. A completed-fact Event or qualified observation is available.
4. Downstream consumers receive stable subject, correlation, and causation references.
5. Unknown, partial, disputed, or failed outcomes remain visible.
6. Corrections preserve the original record and reason.
7. Metrics, audit evidence, and operational signals are emitted.
8. A qualified reviewer can reconstruct who knew what, when, and why.

3. Ubiquitous Language and Subject Model

Primary subject families:

Subject	Required interpretation
ProformaInvoice	independent identified subject or governed subordinate entity; ownership must be declared
ProformaInvoiceVersion	independent identified subject or governed subordinate entity; ownership must be declared
ProformaInvoiceLine	independent identified subject or governed subordinate entity; ownership must be declared
PaymentTermSchedule	independent identified subject or governed subordinate entity; ownership must be declared
BankingInstruction	independent identified subject or governed subordinate entity; ownership must be declared
DocumentEvidenceSet	independent identified subject or governed subordinate entity; ownership must be declared

Reference aggregate:

```
ProformaInvoice {
  aggregateId
  tenantId
  organizationContext
  canonicalSubjectRefs[]
  sourceVersionRefs[]
  lifecycleState
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
```

```

effectiveAt
updatedAt
version
correctionOf?
correlationId
}

```

3.1 Identity and relationship rules

- a display name, document number, provider reference, barcode, or URL is not automatically canonical identity;
- relationships such as custody, mandate, allocation, publication, reservation, or obligation are independent facts;
- cross-Domain references use stable IDs and qualified snapshots, not shared-table mutation;
- external identifiers retain issuer and namespace;
- merge, split, correction, and supersession preserve lineage.

3.2 Invariants

1. Submitted, approved, accepted, issued, or signed versions are immutable.
2. State changes occur through authorized Commands.
3. Completed-fact Events use past-tense semantics.
4. A Recommendation cannot self-promote to a Decision.
5. An external outcome retains its source authority.
6. Derived projections cannot become authoritative by repeated copying.
7. All writes are tenant-, organization-, facility-, and authority-scoped as applicable.
8. Retried writes are idempotent.
9. Outcome Unknown is not converted to success or failure by assumption.
10. Corrections reference the corrected fact and downstream impact.

4. Lifecycle State Model

4.1 State vocabulary

State	Meaning
DRAFT	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
VALIDATION_PENDING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
APPROVAL_PENDING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
APPROVED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
ISSUED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
BUYER_ACCEPTED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
PAYMENT_PENDING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
PARTIALLY_PAID	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
PAID	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
EXPIRED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
CANCELLED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
SUPERSEDED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
DISPUTED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification

4.2 Reference state machine

```

stateDiagram-v2
    [*] --> DRAFT
    DRAFT --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING

```

```

APPROVAL_PENDING --> APPROVED
APPROVED --> ISSUED
ISSUED --> BUYER_ACCEPTED
BUYER_ACCEPTED --> PAYMENT_PENDING
PAYMENT_PENDING --> PARTIALLY_PAID
PARTIALLY_PAID --> PAID
PAID --> EXPIRED
EXPIRED --> CANCELLED
CANCELLED --> SUPERSEDED
SUPERSEDED --> DISPUTED

```

The diagram shows a readable primary path. A conformant implementation may include parallel approvals, amendment, suspension, timeout, appeal, rejection, partial completion, reconciliation, and correction paths, but it must preserve the same semantic distinctions.

4.3 Transition matrix

From	To	Minimum guard
DRAFT	VALIDATION_PENDING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
VALIDATION_PENDING	APPROVAL_PENDING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
APPROVAL_PENDING	APPROVED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
APPROVED	ISSUED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
ISSUED	BUYER_ACCEPTED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
BUYER_ACCEPTED	PAYMENT_PENDING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
PAYMENT_PENDING	PARTIALLY_PAID	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
PARTIALLY_PAID	PAID	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
PAID	EXPIRED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
EXPIRED	CANCELLED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
CANCELLED	SUPERSEDED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
SUPERSEDED	DISPUTED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard

4.4 Forbidden state behavior

- direct arbitrary assignment from a Portal request;
- using one status to represent lifecycle, approval, payment, document, provider, and dispute state simultaneously;
- deleting terminal history to reopen a process;
- inferring completion from elapsed time or an AI prediction;
- moving past a mandatory gate because a downstream system already acted.

5. Decision, Policy, and Authority

Material Decisions in this stage include readiness, eligibility, approval, exception, amendment, correction, and downstream activation decisions.

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Decision rules

- the Decision references exact source versions;
 - authority and delegation are evaluated at Decision time;
 - a Policy version and jurisdiction context are recorded;
 - mandatory-gate failure blocks progression unless a lawful exception exists;
 - dissent, uncertainty, and missing Evidence remain visible;
 - Decision correction does not erase the original Decision;
 - AI may prepare a briefing, but cannot manufacture authority.
-

6. Evidence, Documents, and Knowledge Cutoff

Every binding or high-impact action requires an Evidence Set containing:

- evidence identity and integrity hash where required;
- issuing, observing, or signing party;
- observed time, received time, and effective-time interval;
- subject and source-version relationships;
- qualification, confidence, and limitations;
- contradiction, retraction, supersession, and correction links;
- jurisdiction, classification, retention, and residency;
- chain of custody for physical or high-integrity evidence;
- the knowledge cutoff used by the Decision.

A document is not automatically truth. It is an Artifact containing Claims. Its legal or operational effect depends on issuer, mandate, signature, version, applicable rules, and external authority.

7. Commands, Events, and Queries

7.1 Commands

- CreateProformaInvoice
- ValidateProformaInvoice
- ApproveProformaInvoice
- IssueProformaInvoice
- AcceptProformaInvoice
- AmendProformaInvoice
- CancelProformaInvoice
- LinkPaymentIntent

Each Command shall contain an idempotency key, actor identity, represented principal, tenant and organization context, correlation ID, expected version where relevant, reason for privileged action, and exact target subject.

7.2 Events

- ProformaInvoiceCreated
- ProformaInvoiceValidated

- ProformaInvoiceApproved
- ProformaInvoiceIssued
- ProformaInvoiceAccepted
- ProformaInvoiceAmended
- ProformaInvoiceCancelled
- PaymentIntentLinked

Events describe completed facts or explicitly qualified observations. A request, proposed action, timer, or prediction is not a completed-fact Event.

7.3 Queries and perspectives

- GetProformaInvoice
- ListProformaInvoiceRecords
- GetProformaInvoiceTimeline
- GetProformaInvoiceEvidence
- GetProformaInvoiceDecisionPackage
- GetProformaInvoiceExceptions
- GetProformaInvoiceAuditTrail
- GetProformaInvoiceCurrentPerspective

Every query response states data freshness, source cutoff, version, and whether values are authoritative, observed, derived, predicted, or disputed.

8. Workflow, Timeout, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Evidence and policy gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Create Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and available compensation]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Trigger downstream stage]

```

Required workflow behavior:

- durable process and correlation identity;
- persisted approvals and timers;
- bounded retries and retry budgets;
- idempotent command submission;
- safe restart after deployment or outage;
- compensation that records new Intent instead of deleting history;
- visible stalled, orphaned, and manually intervened processes;
- operator access to source responses and downstream impact;
- reconciliation before repeating an external financial, legal, or physical action.

9. Target API Contract

Reference surface:

```

POST /api/v1/proforma_invoice_lifecycle/commands
GET  /api/v1/proforma_invoice_lifecycle/{id}
GET  /api/v1/proforma_invoice_lifecycle/{id}/timeline
GET  /api/v1/proforma_invoice_lifecycle/{id}/evidence
GET  /api/v1/proforma_invoice_lifecycle/{id}/decisions
POST /api/v1/proforma_invoice_lifecycle/{id}/exceptions
POST /api/v1/proforma_invoice_lifecycle/{id}/corrections

```

These are reference endpoints, not claims that the exact routes exist.

Contract requirements:

- typed request and response schemas;
- action-level authorization;

- trusted authentication context for tenant identity;
- idempotency for writes;
- optimistic concurrency or equivalent aggregate guard;
- RFC problem-details or governed equivalent;
- stable pagination and sorting;
- versioning and deprecation policy;
- trace, correlation, and causation propagation;
- explicit Outcome Unknown response when external completion is unresolved.

10. Target Persistence and Data Ownership

Reference table families:

Table family	Purpose
stage_aggregate	authoritative lifecycle subject
stage_version	immutable approved/submitted/issued versions
stage_decision	binding Decisions and rationale
stage_evidence_link	evidence and provenance relationships
stage_event_outbox	transactional Event publication
stage_idempotency	duplicate-write protection
stage_exception	exception, claim, hold, or dispute
stage_correction	correction and supersession lineage
stage_audit	privileged action evidence

Data rules:

- the owning service controls its schema and migrations;
- foreign keys stay inside an ownership boundary;
- cross-Domain relationships use stable IDs or contracts;
- effective-time and recorded-time are separate when needed;
- binding versions are immutable;
- indexes support tenant, owner, state, correlation, external reference, and effective time;
- retention, legal hold, privacy, residency, backup, and restore are documented;
- analytics, search, reports, and vector stores are projections.

11. Portal and Experience Architecture

Required Portal behavior:

- display exact version, owner, state, and last-updated time;
- distinguish authoritative, observed, derived, predicted, and disputed values;
- expose required action and authority;
- show loading, empty, partial, stale, error, timeout, Outcome Unknown, and permission-denied states;
- provide accessible keyboard and screen-reader behavior;
- handle locale, currency, units, dates, and RTL/LTR correctly;
- preserve evidence and download audit;
- never trust a browser-provided tenant header as authority;
- prevent double submission while still supporting safe retry;
- show correction and supersession history.

Operator surfaces additionally expose correlation, source responses, retry budget, replay status, compensation options, and affected downstream subjects.

12. AI Participation and Tool Governance

Permitted AI tasks:

- extract PI fields and compare them with the accepted quotation
- detect beneficiary or bank-detail changes
- detect arithmetic, currency, tax, and charge anomalies
- draft discrepancy explanations

AI may also summarize evidence, detect gaps, recommend next actions, and draft human-readable explanations.

AI shall not:

- own or directly mutate Domain databases;
- fabricate external-provider, carrier, bank, customs, inspection, or delivery outcomes;
- grant itself permission or mandate;
- silently change an approved or issued version;
- suppress uncertainty, contradictions, policy failures, or human dissent;
- execute high-impact actions outside typed tools and approval policy;
- create shadow records for canonical subjects.

Reference automation:

Task	Maximum default
extraction and classification	A2, automatic with reviewable evidence
anomaly detection and recommendation	A2/A3 according to risk
binding commercial, legal, financial, quality, or custody Decision	A0/A1, human authority required
external execution	typed tool, policy, approval, idempotency, and outcome observation
correction of authoritative truth	governed human-authorized workflow

Every Agent Run records prompt/model/tool versions, context sources, policy evaluation, approvals, tool calls, structured outputs, uncertainty, final observed outcome, and recovery actions.

13. Security, Privacy, and Trust Boundaries

Minimum controls:

- authenticated human, service, organization, device, and AI identities;
- represented principal and effective mandate validation;
- least privilege, ABAC/ReBAC where appropriate, and separation of duties;
- tenant, organization, facility, channel, and jurisdiction isolation;
- integrity protection for binding documents and Evidence;
- external callback signature and replay verification;
- confidential handling of commercial, personal, financial, transport, and regulatory data;
- secret isolation and no sensitive values in logs or prompts;
- privileged-action audit and emergency-access review;
- threat models for impersonation, tampering, replay, confused deputy, fraud, exfiltration, and prompt injection.

14. Reliability, SLOs, and Operations

Reference objectives:

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by the business deadline of the stage

Operational requirements include dashboards, alerts, runbooks, ownership, dependency health, retry budgets, DLQ/replay controls, Event outbox/inbox, workflow restart, reconciliation, capacity, backup/restore, incident command, and post-incident correction.

15. Failure Modes and Recovery

Representative failures:

- stale or wrong source version;
- duplicate submission or replay;
- external timeout after possible success;

- contradictory source observations;
- lost Event after committed state;
- approval or mandate revoked during execution;
- downstream action performed before a mandatory gate;
- partial completion across multiple subjects;
- evidence later retracted or corrected;
- tenant, facility, or jurisdiction mismatch.

Recovery principles:

1. Query the authoritative source before repeating an external action.
2. Preserve partial success.
3. Use compensating Intent instead of destructive history rewrite.
4. Publish correction Events when distributed facts change.
5. Quarantine high-impact ambiguity.
6. Escalate when authority or legal effect is unclear.
7. Record business impact, affected subjects, and final reconciliation.

16. Testing and Conformance

Required tests:

- aggregate invariant and transition tests;
- authorization, mandate, separation-of-duties, and tenant-isolation tests;
- idempotency and optimistic-concurrency tests;
- migration, constraint, index, and rollback tests;
- outbox/inbox, ordering, duplicate, replay, and DLQ tests;
- workflow restart, timeout, compensation, and manual-intervention tests;
- provider callback verification and Outcome Unknown tests;
- Portal E2E for success, partial, stale, denied, and error journeys;
- accessibility, locale, currency, unit, and RTL tests;
- security adversarial and data-exfiltration tests;
- AI evaluation, prompt-injection, policy-bypass, and unauthorized-tool tests;
- historical reconstruction and correction tests;
- performance, capacity, disaster-recovery, and operational exercise evidence.

17. Repository Review Boundary and Previously Inspected Evidence

- Registered supporting capabilities include contract, payment, billing, trade, and document-related modules.
- A complete canonical PI aggregate, immutable history, and verified quotation-to-PI backend path have not yet been established.
- This chapter therefore defaults to `REFERENCE_ARCHITECTURE` and `REQUIRES_REPOSITORY_REVIEW`.

Area	Classification for this chapter
Target aggregate, API, tables, state model, and workflow	<code>REFERENCE_ARCHITECTURE</code>
Registered supporting modules	<code>DOCUMENTATION_ONLY_CLAIM</code> until executable files are inspected
Concrete behavior already established in earlier repository-grounded chapters	retain the original classification and evidence
Exact integration paths not inspected for this stage	<code>REQUIRES_REPOSITORY_REVIEW</code>
External authority or provider outcome	qualified external observation, never platform-owned truth

No statement in this chapter upgrades an implementation status merely because a module name, frontend route, README, plan, or database catalog entry exists.

18. Worked Scenario

An upstream governed commitment enters the stage with exact subject and version references. The initiating principal is authenticated, and its mandate is checked at command time. Evidence is gathered and evaluated against the applicable policy version and jurisdiction.

A recommendation may be produced, but an authorized Decision remains separate. The Domain accepts an idempotent Command and records state plus an outbox entry atomically. When an external party participates, a timeout creates

Outcome Unknown; the workflow queries the external source or waits for a verified callback instead of repeating execution blindly.

On success or failure, the system records a completed fact, publishes a stable Event, updates downstream projections, and preserves source and knowledge cutoff. Later corrections append new facts and identify downstream impact rather than replacing history. AI assists through typed tools and cannot own the subject, grant authority, or invent an outcome.

19. Conformance Checklist

- ☐ Stable subject and version identities are explicit.
 - ☐ Proposition ownership and external authority are named.
 - ☐ Entry and exit criteria are testable.
 - ☐ State, Recommendation, Decision, Intent, execution, and outcome remain distinct.
 - ☐ Commands are authorized, version-guarded, and idempotent.
 - ☐ Events describe completed facts.
 - ☐ Evidence includes provenance, time, integrity, and correction links.
 - ☐ Portals show freshness, uncertainty, and Outcome Unknown.
 - ☐ AI uses typed tools and approved autonomy.
 - ☐ Security, privacy, tenant, facility, and jurisdiction boundaries are enforced.
 - ☐ Reconciliation precedes risky duplicate execution.
 - ☐ Corrections preserve history.
 - ☐ Tests prove invariants, security, failure, and recovery.
 - ☐ Implementation status is not upgraded without executable repository evidence.
-

20. Status Vocabulary

- VERIFIED_IMPLEMENTED: executable source and the relevant behavior were inspected.
- IMPLEMENTED_WITH_GAP: executable behavior exists with a material governance or completeness gap.
- PARTIAL_IMPLEMENTATION: only part of the target capability is evidenced.
- REFERENCE_ARCHITECTURE: normative target behavior; no claim that it exists today.
- PLANNED_NOT_IMPLEMENTED: explicitly planned but not evidenced as executable.
- DOCUMENTATION_ONLY_CLAIM: documentation or a client contract claims behavior without verified backend evidence.
- CONFLICT_REQUIRES_CANONICALIZATION: sources disagree and no final owner has been established.
- REQUIRES_REPOSITORY_REVIEW: the necessary executable evidence has not yet been inspected.

Publication Chapter 11 — Payment and Risk Approval

Metadata	Value
Publication Chapter	11
Canonical Lifecycle Stage	10
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Canon dependencies	CANON-001 through CANON-011
Constitutional dependencies	ART-001 through ART-016
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This stage exists to **decide whether a financial obligation may become an executable payment intent and preserve risk, approval, execution, provider outcome, settlement, and reconciliation as separate facts.**

Its output is not merely a screen state or document download. It is a governed set of identified subjects, Relationships, Decisions, Intent, Commands, completed facts, external observations, evidence, and correction history.

Reality or external outcome

- ≠ platform Observation
- ≠ derived Perspective
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1.1 Accountability

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- a testable lifecycle outcome;
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8. Any conflicting active process is resolved, suspended, or explicitly related.

2.2 Exit criteria

1. The authoritative Domain or external authority records the outcome.
2. State and version changes are durable.
3. A completed-fact Event or qualified observation is available.
4. Downstream consumers receive stable subject, correlation, and causation references.
5. Unknown, partial, disputed, or failed outcomes remain visible.
6. Corrections preserve the original record and reason.
7. Metrics, audit evidence, and operational signals are emitted.
8. A qualified reviewer can reconstruct who knew what, when, and why.

3. Ubiquitous Language and Subject Model

Primary subject families:

Subject	Required interpretation
PaymentApprovalCase	independent identified subject or governed subordinate entity; ownership must be declared
PaymentRiskAssessment	independent identified subject or governed subordinate entity; ownership must be declared
ApprovalDecision	independent identified subject or governed subordinate entity; ownership must be declared
PaymentIntent	independent identified subject or governed subordinate entity; ownership must be declared
ProviderExecutionReference	independent identified subject or governed subordinate entity; ownership must be declared
ReconciliationCase	independent identified subject or governed subordinate entity; ownership must be declared

Reference aggregate:

```
PaymentApprovalCase {
  aggregateId
  tenantId
  organizationContext
  canonicalSubjectRefs[]
  sourceVersionRefs[]
  lifecycleState
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
```

```

effectiveAt
updatedAt
version
correctionOf?
correlationId
}

```

3.1 Identity and relationship rules

- a display name, document number, provider reference, barcode, or URL is not automatically canonical identity;
- relationships such as custody, mandate, allocation, publication, reservation, or obligation are independent facts;
- cross-Domain references use stable IDs and qualified snapshots, not shared-table mutation;
- external identifiers retain issuer and namespace;
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3.2 Invariants

1. Submitted, approved, accepted, issued, or signed versions are immutable.
2. State changes occur through authorized Commands.
3. Completed-fact Events use past-tense semantics.
4. A Recommendation cannot self-promote to a Decision.
5. An external outcome retains its source authority.
6. Derived projections cannot become authoritative by repeated copying.
7. All writes are tenant-, organization-, facility-, and authority-scoped as applicable.
8. Retried writes are idempotent.
9. Outcome Unknown is not converted to success or failure by assumption.
10. Corrections reference the corrected fact and downstream impact.

4. Lifecycle State Model

4.1 State vocabulary

State	Meaning
REQUESTED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
RISK_ASSESSMENT	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
APPROVAL_PENDING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
APPROVED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
REJECTED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
EXECUTION_PENDING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
OUTCOME_UNKNOWN	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
AUTHORIZED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
CAPTURED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
FAILED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
CANCELLED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
RECONCILIATION_REQUIRED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification

4.2 Reference state machine

```

stateDiagram-v2
    [*] --> REQUESTED
    REQUESTED --> RISK_ASSESSMENT
    RISK_ASSESSMENT --> APPROVAL_PENDING
    APPROVAL_PENDING --> APPROVED
    APPROVED --> REJECTED

```

```

REJECTED --> EXECUTION_PENDING
EXECUTION_PENDING --> OUTCOME_UNKNOWN
OUTCOME_UNKNOWN --> AUTHORIZED
AUTHORIZED --> CAPTURED
CAPTURED --> FAILED
FAILED --> CANCELLED
CANCELLED --> RECONCILIATION_REQUIRED

```

The diagram shows a readable primary path. A conformant implementation may include parallel approvals, amendment, suspension, timeout, appeal, rejection, partial completion, reconciliation, and correction paths, but it must preserve the same semantic distinctions.

4.3 Transition matrix

From	To	Minimum guard
REQUESTED	RISK_ASSESSMENT	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
RISK_ASSESSMENT	APPROVAL_PENDING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
APPROVAL_PENDING	APPROVED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
APPROVED	REJECTED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
REJECTED	EXECUTION_PENDING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
EXECUTION_PENDING	OUTCOME_UNKNOWN	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
OUTCOME_UNKNOWN	AUTHORIZED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
AUTHORIZED	CAPTURED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
CAPTURED	FAILED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
FAILED	CANCELLED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
CANCELLED	RECONCILIATION_REQUIRED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard

4.4 Forbidden state behavior

- direct arbitrary assignment from a Portal request;
- using one status to represent lifecycle, approval, payment, document, provider, and dispute state simultaneously;
- deleting terminal history to reopen a process;
- inferring completion from elapsed time or an AI prediction;
- moving past a mandatory gate because a downstream system already acted.

5. Decision, Policy, and Authority

Material Decisions in this stage include readiness, eligibility, approval, exception, amendment, correction, and downstream activation decisions.

Reference Decision:

```

Decision {
  decisionId
  decisionType
  subjectRef
}

```

```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Decision rules

- the Decision references exact source versions;
 - authority and delegation are evaluated at Decision time;
 - a Policy version and jurisdiction context are recorded;
 - mandatory-gate failure blocks progression unless a lawful exception exists;
 - dissent, uncertainty, and missing Evidence remain visible;
 - Decision correction does not erase the original Decision;
 - AI may prepare a briefing, but cannot manufacture authority.
-

6. Evidence, Documents, and Knowledge Cutoff

Every binding or high-impact action requires an Evidence Set containing:

- evidence identity and integrity hash where required;
- issuing, observing, or signing party;
- observed time, received time, and effective-time interval;
- subject and source-version relationships;
- qualification, confidence, and limitations;
- contradiction, retraction, supersession, and correction links;
- jurisdiction, classification, retention, and residency;
- chain of custody for physical or high-integrity evidence;
- the knowledge cutoff used by the Decision.

A document is not automatically truth. It is an Artifact containing Claims. Its legal or operational effect depends on issuer, mandate, signature, version, applicable rules, and external authority.

7. Commands, Events, and Queries

7.1 Commands

- RequestPaymentApproval
- AssessPaymentRisk
- ApprovePayment
- RejectPayment
- CreatePaymentIntent
- ExecutePaymentIntent
- QueryProviderOutcome
- CancelPaymentIntent
- OpenReconciliationCase

Each Command shall contain an idempotency key, actor identity, represented principal, tenant and organization context, correlation ID, expected version where relevant, reason for privileged action, and exact target subject.

7.2 Events

- PaymentApprovalRequested
- PaymentRiskAssessed
- PaymentApproved
- PaymentRejected
- PaymentIntentCreated
- PaymentExecutionRequested
- PaymentOutcomeBecameUnknown

- PaymentAuthorized
- PaymentCaptured
- PaymentFailed
- ReconciliationRequired

Events describe completed facts or explicitly qualified observations. A request, proposed action, timer, or prediction is not a completed-fact Event.

7.3 Queries and perspectives

- GetPaymentApprovalCase
- ListPaymentApprovalCaseRecords
- GetPaymentApprovalCaseTimeline
- GetPaymentApprovalCaseEvidence
- GetPaymentApprovalCaseDecisionPackage
- GetPaymentApprovalCaseExceptions
- GetPaymentApprovalCaseAuditTrail
- GetPaymentApprovalCaseCurrentPerspective

Every query response states data freshness, source cutoff, version, and whether values are authoritative, observed, derived, predicted, or disputed.

8. Workflow, Timeout, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Evidence and policy gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Create Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and available compensation]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Trigger downstream stage]

```

Required workflow behavior:

- durable process and correlation identity;
- persisted approvals and timers;
- bounded retries and retry budgets;
- idempotent command submission;
- safe restart after deployment or outage;
- compensation that records new Intent instead of deleting history;
- visible stalled, orphaned, and manually intervened processes;
- operator access to source responses and downstream impact;
- reconciliation before repeating an external financial, legal, or physical action.

9. Target API Contract

Reference surface:

```

POST /api/v1/payment_and_risk_approval/commands
GET  /api/v1/payment_and_risk_approval/{id}
GET  /api/v1/payment_and_risk_approval/{id}/timeline
GET  /api/v1/payment_and_risk_approval/{id}/evidence
GET  /api/v1/payment_and_risk_approval/{id}/decisions
POST /api/v1/payment_and_risk_approval/{id}/exceptions
POST /api/v1/payment_and_risk_approval/{id}/corrections

```

These are reference endpoints, not claims that the exact routes exist.

Contract requirements:

- typed request and response schemas;
- action-level authorization;
- trusted authentication context for tenant identity;
- idempotency for writes;

- optimistic concurrency or equivalent aggregate guard;
- RFC problem-details or governed equivalent;
- stable pagination and sorting;
- versioning and deprecation policy;
- trace, correlation, and causation propagation;
- explicit Outcome Unknown response when external completion is unresolved.

10. Target Persistence and Data Ownership

Reference table families:

Table family	Purpose
stage_aggregate	authoritative lifecycle subject
stage_version	immutable approved/submitted/issued versions
stage_decision	binding Decisions and rationale
stage_evidence_link	evidence and provenance relationships
stage_event_outbox	transactional Event publication
stage_idempotency	duplicate-write protection
stage_exception	exception, claim, hold, or dispute
stage_correction	correction and supersession lineage
stage_audit	privileged action evidence

Data rules:

- the owning service controls its schema and migrations;
- foreign keys stay inside an ownership boundary;
- cross-Domain relationships use stable IDs or contracts;
- effective-time and recorded-time are separate when needed;
- binding versions are immutable;
- indexes support tenant, owner, state, correlation, external reference, and effective time;
- retention, legal hold, privacy, residency, backup, and restore are documented;
- analytics, search, reports, and vector stores are projections.

11. Portal and Experience Architecture

Required Portal behavior:

- display exact version, owner, state, and last-updated time;
- distinguish authoritative, observed, derived, predicted, and disputed values;
- expose required action and authority;
- show loading, empty, partial, stale, error, timeout, Outcome Unknown, and permission-denied states;
- provide accessible keyboard and screen-reader behavior;
- handle locale, currency, units, dates, and RTL/LTR correctly;
- preserve evidence and download audit;
- never trust a browser-provided tenant header as authority;
- prevent double submission while still supporting safe retry;
- show correction and supersession history.

Operator surfaces additionally expose correlation, source responses, retry budget, replay status, compensation options, and affected downstream subjects.

12. AI Participation and Tool Governance

Permitted AI tasks:

- summarize risk factors
- detect duplicate or anomalous payment requests
- recommend a provider route
- prioritize reconciliation cases

AI may also summarize evidence, detect gaps, recommend next actions, and draft human-readable explanations.

AI shall not:

- own or directly mutate Domain databases;

- fabricate external-provider, carrier, bank, customs, inspection, or delivery outcomes;
- grant itself permission or mandate;
- silently change an approved or issued version;
- suppress uncertainty, contradictions, policy failures, or human dissent;
- execute high-impact actions outside typed tools and approval policy;
- create shadow records for canonical subjects.

Reference automation:

Task	Maximum default
extraction and classification	A2, automatic with reviewable evidence
anomaly detection and recommendation	A2/A3 according to risk
binding commercial, legal, financial, quality, or custody Decision	A0/A1, human authority required
external execution	typed tool, policy, approval, idempotency, and outcome observation
correction of authoritative truth	governed human-authorized workflow

Every Agent Run records prompt/model/tool versions, context sources, policy evaluation, approvals, tool calls, structured outputs, uncertainty, final observed outcome, and recovery actions.

13. Security, Privacy, and Trust Boundaries

Minimum controls:

- authenticated human, service, organization, device, and AI identities;
- represented principal and effective mandate validation;
- least privilege, ABAC/ReBAC where appropriate, and separation of duties;
- tenant, organization, facility, channel, and jurisdiction isolation;
- integrity protection for binding documents and Evidence;
- external callback signature and replay verification;
- confidential handling of commercial, personal, financial, transport, and regulatory data;
- secret isolation and no sensitive values in logs or prompts;
- privileged-action audit and emergency-access review;
- threat models for impersonation, tampering, replay, confused deputy, fraud, exfiltration, and prompt injection.

14. Reliability, SLOs, and Operations

Reference objectives:

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by the business deadline of the stage

Operational requirements include dashboards, alerts, runbooks, ownership, dependency health, retry budgets, DLQ/replay controls, Event outbox/inbox, workflow restart, reconciliation, capacity, backup/restore, incident command, and post-incident correction.

15. Failure Modes and Recovery

Representative failures:

- stale or wrong source version;
- duplicate submission or replay;
- external timeout after possible success;
- contradictory source observations;

- lost Event after committed state;
- approval or mandate revoked during execution;
- downstream action performed before a mandatory gate;
- partial completion across multiple subjects;
- evidence later retracted or corrected;
- tenant, facility, or jurisdiction mismatch.

Recovery principles:

1. Query the authoritative source before repeating an external action.
2. Preserve partial success.
3. Use compensating Intent instead of destructive history rewrite.
4. Publish correction Events when distributed facts change.
5. Quarantine high-impact ambiguity.
6. Escalate when authority or legal effect is unclear.
7. Record business impact, affected subjects, and final reconciliation.

16. Testing and Conformance

Required tests:

- aggregate invariant and transition tests;
- authorization, mandate, separation-of-duties, and tenant-isolation tests;
- idempotency and optimistic-concurrency tests;
- migration, constraint, index, and rollback tests;
- outbox/inbox, ordering, duplicate, replay, and DLQ tests;
- workflow restart, timeout, compensation, and manual-intervention tests;
- provider callback verification and Outcome Unknown tests;
- Portal E2E for success, partial, stale, denied, and error journeys;
- accessibility, locale, currency, unit, and RTL tests;
- security adversarial and data-exfiltration tests;
- AI evaluation, prompt-injection, policy-bypass, and unauthorized-tool tests;
- historical reconstruction and correction tests;
- performance, capacity, disaster-recovery, and operational exercise evidence.

17. Repository Review Boundary and Previously Inspected Evidence

- Payment, wallet, billing, and finance-related modules are registered.
- The existing Payment Domain chapter set is the repository-grounded baseline for intent, provider, unknown-outcome, refund, settlement, and reconciliation semantics.
- No claim is made that every PI-to-payment integration path is already implemented.

Area	Classification for this chapter
Target aggregate, API, tables, state model, and workflow Registered supporting modules	REFERENCE_ARCHITECTURE DOCUMENTATION_ONLY_CLAIM until executable files are inspected
Concrete behavior already established in earlier repository-grounded chapters	retain the original classification and evidence
Exact integration paths not inspected for this stage	REQUIRES_REPOSITORY_REVIEW
External authority or provider outcome	qualified external observation, never platform-owned truth

No statement in this chapter upgrades an implementation status merely because a module name, frontend route, README, plan, or database catalog entry exists.

18. Worked Scenario

An upstream governed commitment enters the stage with exact subject and version references. The initiating principal is authenticated, and its mandate is checked at command time. Evidence is gathered and evaluated against the applicable policy version and jurisdiction.

A recommendation may be produced, but an authorized Decision remains separate. The Domain accepts an idempotent Command and records state plus an outbox entry atomically. When an external party participates, a timeout creates

Outcome Unknown; the workflow queries the external source or waits for a verified callback instead of repeating execution blindly.

On success or failure, the system records a completed fact, publishes a stable Event, updates downstream projections, and preserves source and knowledge cutoff. Later corrections append new facts and identify downstream impact rather than replacing history. AI assists through typed tools and cannot own the subject, grant authority, or invent an outcome.

19. Conformance Checklist

- ☐ Stable subject and version identities are explicit.
 - ☐ Proposition ownership and external authority are named.
 - ☐ Entry and exit criteria are testable.
 - ☐ State, Recommendation, Decision, Intent, execution, and outcome remain distinct.
 - ☐ Commands are authorized, version-guarded, and idempotent.
 - ☐ Events describe completed facts.
 - ☐ Evidence includes provenance, time, integrity, and correction links.
 - ☐ Portals show freshness, uncertainty, and Outcome Unknown.
 - ☐ AI uses typed tools and approved autonomy.
 - ☐ Security, privacy, tenant, facility, and jurisdiction boundaries are enforced.
 - ☐ Reconciliation precedes risky duplicate execution.
 - ☐ Corrections preserve history.
 - ☐ Tests prove invariants, security, failure, and recovery.
 - ☐ Implementation status is not upgraded without executable repository evidence.
-

20. Status Vocabulary

- VERIFIED_IMPLEMENTED: executable source and the relevant behavior were inspected.
- IMPLEMENTED_WITH_GAP: executable behavior exists with a material governance or completeness gap.
- PARTIAL_IMPLEMENTATION: only part of the target capability is evidenced.
- REFERENCE_ARCHITECTURE: normative target behavior; no claim that it exists today.
- PLANNED_NOT_IMPLEMENTED: explicitly planned but not evidenced as executable.
- DOCUMENTATION_ONLY_CLAIM: documentation or a client contract claims behavior without verified backend evidence.
- CONFLICT_REQUIRES_CANONICALIZATION: sources disagree and no final owner has been established.
- REQUIRES_REPOSITORY_REVIEW: the necessary executable evidence has not yet been inspected.

Publication Chapter 12 — Purchase Order Creation and Acknowledgement

Metadata	Value
Publication Chapter	12
Canonical Lifecycle Stage	11
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE with explicit repository notes
Accountable capability	Procurement / Purchase Order
Canon dependencies	CANON-001 through CANON-011
Constitutional dependencies	ART-001 through ART-016
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This stage exists to **create a buyer-controlled execution commitment and obtain explicit supplier acknowledgement without collapsing the PO into a sales order, contract, production order, or payment.**

Its output is not merely a screen state or document download. It is a governed set of identified subjects, Relationships, Decisions, Intent, Commands, completed facts, external observations, evidence, and correction history.

Reality or external outcome
≠ platform Observation
≠ derived Perspective
≠ AI Prediction
≠ Recommendation
≠ authorized Decision
≠ execution Intent

1.1 Accountability

The accountable capability is **Procurement / Purchase Order**. Supporting Portals, workflow engines, reports, search indexes, integration adapters, and AI Agents may coordinate or present the work, but they do not silently become owners of the underlying business truth.

1.2 Primary business outcomes

- a stable subject identity and exact source-version lineage;
- explicit authority and effective mandate;
- a testable lifecycle outcome;
- evidence sufficient for downstream use and later reconstruction;
- visible failure, partial completion, uncertainty, and correction;
- a completed-fact Event or qualified external observation;
- downstream activation without shared-database ownership transfer.

1.3 Non-goals

- treating a registered module as proof of complete implementation;
 - treating a frontend route as proof of backend behavior;
 - allowing an AI recommendation to become a binding Decision;
 - inventing external-provider or government-authority success;
 - overwriting history during amendment or correction;
 - combining unrelated states into one generic status field.
-

2. Entry and Exit Criteria

2.1 Entry criteria

1. Every upstream subject reference resolves to a stable identifier.
2. The exact approved, accepted, issued, or observed version is known.
3. The initiating human, service, organization, or AI Agent is authenticated.
4. The represented principal and mandate are valid at command time.
5. Required Evidence exists, or an explicit authorized exception is linked.
6. Applicable Policy and jurisdiction versions are fixed.
7. Duplicate, stale-version, and replay checks have run.
8. Any conflicting active process is resolved, suspended, or explicitly related.

2.2 Exit criteria

1. The authoritative Domain or external authority records the outcome.
 2. State and version changes are durable.
 3. A completed-fact Event or qualified observation is available.
 4. Downstream consumers receive stable subject, correlation, and causation references.
 5. Unknown, partial, disputed, or failed outcomes remain visible.
 6. Corrections preserve the original record and reason.
 7. Metrics, audit evidence, and operational signals are emitted.
 8. A qualified reviewer can reconstruct who knew what, when, and why.
-

3. Ubiquitous Language and Subject Model

Primary subject families:

Subject	Required interpretation
PurchaseOrder	independent identified subject or governed subordinate entity; ownership must be declared
PurchaseOrderVersion	independent identified subject or governed subordinate entity; ownership must be declared
PurchaseOrderLine	independent identified subject or governed subordinate entity; ownership must be declared
DeliverySchedule	independent identified subject or governed subordinate entity; ownership must be declared
SupplierAcknowledgement	independent identified subject or governed subordinate entity; ownership must be declared
POAmendment	independent identified subject or governed subordinate entity; ownership must be declared

Reference aggregate:

```
PurchaseOrder {
  aggregateId
  tenantId
  organizationContext
  canonicalSubjectRefs[]
  sourceVersionRefs[]
  lifecycleState
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
}
```

```

authorityRef
jurisdictionRefs[]
createdAt
effectiveAt
updatedAt
version
correctionOf?
correlationId
}

```

3.1 Identity and relationship rules

- a display name, document number, provider reference, barcode, or URL is not automatically canonical identity;
- relationships such as custody, mandate, allocation, publication, reservation, or obligation are independent facts;
- cross-Domain references use stable IDs and qualified snapshots, not shared-table mutation;
- external identifiers retain issuer and namespace;
- merge, split, correction, and supersession preserve lineage.

3.2 Invariants

1. Submitted, approved, accepted, issued, or signed versions are immutable.
2. State changes occur through authorized Commands.
3. Completed-fact Events use past-tense semantics.
4. A Recommendation cannot self-promote to a Decision.
5. An external outcome retains its source authority.
6. Derived projections cannot become authoritative by repeated copying.
7. All writes are tenant-, organization-, facility-, and authority-scoped as applicable.
8. Retried writes are idempotent.
9. Outcome Unknown is not converted to success or failure by assumption.
10. Corrections reference the corrected fact and downstream impact.

4. Lifecycle State Model

4.1 State vocabulary

State	Meaning
DRAFT	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
APPROVAL_PENDING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
APPROVED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
ISSUED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
ACKNOWLEDGEMENT_PENDING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
ACKNOWLEDGED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
REJECTED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
AMENDMENT_PENDING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
PARTIALLY_FULFILLED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
FULFILLED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
CANCELLED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
CLOSED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification

4.2 Reference state machine

```

stateDiagram-v2
    [*] --> DRAFT
    DRAFT --> APPROVAL_PENDING

```

```

APPROVAL_PENDING --> APPROVED
APPROVED --> ISSUED
ISSUED --> ACKNOWLEDGEMENT_PENDING
ACKNOWLEDGEMENT_PENDING --> ACKNOWLEDGED
ACKNOWLEDGED --> REJECTED
REJECTED --> AMENDMENT_PENDING
AMENDMENT_PENDING --> PARTIALLY_FULFILLED
PARTIALLY_FULFILLED --> FULFILLED
FULFILLED --> CANCELLED
CANCELLED --> CLOSED

```

The diagram shows a readable primary path. A conformant implementation may include parallel approvals, amendment, suspension, timeout, appeal, rejection, partial completion, reconciliation, and correction paths, but it must preserve the same semantic distinctions.

4.3 Transition matrix

From	To	Minimum guard
DRAFT	APPROVAL_PENDING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
APPROVAL_PENDING	APPROVED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
APPROVED	ISSUED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
ISSUED	ACKNOWLEDGEMENT_PENDING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
ACKNOWLEDGEMENT_PENDING	ACKNOWLEDGED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
ACKNOWLEDGED	REJECTED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
REJECTED	AMENDMENT_PENDING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
AMENDMENT_PENDING	PARTIALLY_FULFILLED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
PARTIALLY_FULFILLED	FULFILLED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
FULFILLED	CANCELLED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
CANCELLED	CLOSED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard

4.4 Forbidden state behavior

- direct arbitrary assignment from a Portal request;
- using one status to represent lifecycle, approval, payment, document, provider, and dispute state simultaneously;
- deleting terminal history to reopen a process;
- inferring completion from elapsed time or an AI prediction;
- moving past a mandatory gate because a downstream system already acted.

5. Decision, Policy, and Authority

Material Decisions in this stage include readiness, eligibility, approval, exception, amendment, correction, and down-stream activation decisions.

Reference Decision:

```

Decision {
  decisionId

```

```

decisionType
subjectRef
sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Decision rules

- the Decision references exact source versions;
 - authority and delegation are evaluated at Decision time;
 - a Policy version and jurisdiction context are recorded;
 - mandatory-gate failure blocks progression unless a lawful exception exists;
 - dissent, uncertainty, and missing Evidence remain visible;
 - Decision correction does not erase the original Decision;
 - AI may prepare a briefing, but cannot manufacture authority.
-

6. Evidence, Documents, and Knowledge Cutoff

Every binding or high-impact action requires an Evidence Set containing:

- evidence identity and integrity hash where required;
- issuing, observing, or signing party;
- observed time, received time, and effective-time interval;
- subject and source-version relationships;
- qualification, confidence, and limitations;
- contradiction, retraction, supersession, and correction links;
- jurisdiction, classification, retention, and residency;
- chain of custody for physical or high-integrity evidence;
- the knowledge cutoff used by the Decision.

A document is not automatically truth. It is an Artifact containing Claims. Its legal or operational effect depends on issuer, mandate, signature, version, applicable rules, and external authority.

7. Commands, Events, and Queries

7.1 Commands

- CreatePurchaseOrder
- ApprovePurchaseOrder
- IssuePurchaseOrder
- AcknowledgePurchaseOrder
- RejectPurchaseOrder
- ProposePurchaseOrderAmendment
- ApprovePurchaseOrderAmendment
- CancelPurchaseOrder

Each Command shall contain an idempotency key, actor identity, represented principal, tenant and organization context, correlation ID, expected version where relevant, reason for privileged action, and exact target subject.

7.2 Events

- PurchaseOrderCreated
- PurchaseOrderApproved
- PurchaseOrderIssued
- PurchaseOrderAcknowledged
- PurchaseOrderRejected
- PurchaseOrderAmended

- PurchaseOrderCancelled
- PurchaseOrderFulfilled

Events describe completed facts or explicitly qualified observations. A request, proposed action, timer, or prediction is not a completed-fact Event.

7.3 Queries and perspectives

- GetPurchaseOrder
- ListPurchaseOrderRecords
- GetPurchaseOrderTimeline
- GetPurchaseOrderEvidence
- GetPurchaseOrderDecisionPackage
- GetPurchaseOrderExceptions
- GetPurchaseOrderAuditTrail
- GetPurchaseOrderCurrentPerspective

Every query response states data freshness, source cutoff, version, and whether values are authoritative, observed, derived, predicted, or disputed.

8. Workflow, Timeout, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Evidence and policy gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Create Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and available compensation]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Trigger downstream stage]

```

Required workflow behavior:

- durable process and correlation identity;
- persisted approvals and timers;
- bounded retries and retry budgets;
- idempotent command submission;
- safe restart after deployment or outage;
- compensation that records new Intent instead of deleting history;
- visible stalled, orphaned, and manually intervened processes;
- operator access to source responses and downstream impact;
- reconciliation before repeating an external financial, legal, or physical action.

9. Target API Contract

Reference surface:

```

POST /api/v1/purchase_order_creation_and_acknowledgement/commands
GET  /api/v1/purchase_order_creation_and_acknowledgement/{id}
GET  /api/v1/purchase_order_creation_and_acknowledgement/{id}/timeline
GET  /api/v1/purchase_order_creation_and_acknowledgement/{id}/evidence
GET  /api/v1/purchase_order_creation_and_acknowledgement/{id}/decisions
POST /api/v1/purchase_order_creation_and_acknowledgement/{id}/exceptions
POST /api/v1/purchase_order_creation_and_acknowledgement/{id}/corrections

```

These are reference endpoints, not claims that the exact routes exist.

Contract requirements:

- typed request and response schemas;
- action-level authorization;
- trusted authentication context for tenant identity;
- idempotency for writes;
- optimistic concurrency or equivalent aggregate guard;
- RFC problem-details or governed equivalent;

- stable pagination and sorting;
- versioning and deprecation policy;
- trace, correlation, and causation propagation;
- explicit Outcome Unknown response when external completion is unresolved.

10. Target Persistence and Data Ownership

Reference table families:

Table family	Purpose
stage_aggregate	authoritative lifecycle subject
stage_version	immutable approved/submitted/issued versions
stage_decision	binding Decisions and rationale
stage_evidence_link	evidence and provenance relationships
stage_event_outbox	transactional Event publication
stage_idempotency	duplicate-write protection
stage_exception	exception, claim, hold, or dispute
stage_correction	correction and supersession lineage
stage_audit	privileged action evidence

Data rules:

- the owning service controls its schema and migrations;
- foreign keys stay inside an ownership boundary;
- cross-Domain relationships use stable IDs or contracts;
- effective-time and recorded-time are separate when needed;
- binding versions are immutable;
- indexes support tenant, owner, state, correlation, external reference, and effective time;
- retention, legal hold, privacy, residency, backup, and restore are documented;
- analytics, search, reports, and vector stores are projections.

11. Portal and Experience Architecture

Required Portal behavior:

- display exact version, owner, state, and last-updated time;
- distinguish authoritative, observed, derived, predicted, and disputed values;
- expose required action and authority;
- show loading, empty, partial, stale, error, timeout, Outcome Unknown, and permission-denied states;
- provide accessible keyboard and screen-reader behavior;
- handle locale, currency, units, dates, and RTL/LTR correctly;
- preserve evidence and download audit;
- never trust a browser-provided tenant header as authority;
- prevent double submission while still supporting safe retry;
- show correction and supersession history.

Operator surfaces additionally expose correlation, source responses, retry budget, replay status, compensation options, and affected downstream subjects.

12. AI Participation and Tool Governance

Permitted AI tasks:

- compare the PO with source terms
- detect missing clauses and schedule conflicts
- summarize supplier deviations
- predict acknowledgement delay

AI may also summarize evidence, detect gaps, recommend next actions, and draft human-readable explanations.

AI shall not:

- own or directly mutate Domain databases;
- fabricate external-provider, carrier, bank, customs, inspection, or delivery outcomes;
- grant itself permission or mandate;

- silently change an approved or issued version;
- suppress uncertainty, contradictions, policy failures, or human dissent;
- execute high-impact actions outside typed tools and approval policy;
- create shadow records for canonical subjects.

Reference automation:

Task	Maximum default
extraction and classification	A2, automatic with reviewable evidence
anomaly detection and recommendation	A2/A3 according to risk
binding commercial, legal, financial, quality, or custody Decision	A0/A1, human authority required
external execution	typed tool, policy, approval, idempotency, and outcome observation
correction of authoritative truth	governed human-authorized workflow

Every Agent Run records prompt/model/tool versions, context sources, policy evaluation, approvals, tool calls, structured outputs, uncertainty, final observed outcome, and recovery actions.

13. Security, Privacy, and Trust Boundaries

Minimum controls:

- authenticated human, service, organization, device, and AI identities;
- represented principal and effective mandate validation;
- least privilege, ABAC/ReBAC where appropriate, and separation of duties;
- tenant, organization, facility, channel, and jurisdiction isolation;
- integrity protection for binding documents and Evidence;
- external callback signature and replay verification;
- confidential handling of commercial, personal, financial, transport, and regulatory data;
- secret isolation and no sensitive values in logs or prompts;
- privileged-action audit and emergency-access review;
- threat models for impersonation, tampering, replay, confused deputy, fraud, exfiltration, and prompt injection.

14. Reliability, SLOs, and Operations

Reference objectives:

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by the business deadline of the stage

Operational requirements include dashboards, alerts, runbooks, ownership, dependency health, retry budgets, DLQ/replay controls, Event outbox/inbox, workflow restart, reconciliation, capacity, backup/restore, incident command, and post-incident correction.

15. Failure Modes and Recovery

Representative failures:

- stale or wrong source version;
- duplicate submission or replay;
- external timeout after possible success;
- contradictory source observations;
- lost Event after committed state;
- approval or mandate revoked during execution;

- downstream action performed before a mandatory gate;
- partial completion across multiple subjects;
- evidence later retracted or corrected;
- tenant, facility, or jurisdiction mismatch.

Recovery principles:

1. Query the authoritative source before repeating an external action.
2. Preserve partial success.
3. Use compensating Intent instead of destructive history rewrite.
4. Publish correction Events when distributed facts change.
5. Quarantine high-impact ambiguity.
6. Escalate when authority or legal effect is unclear.
7. Record business impact, affected subjects, and final reconciliation.

16. Testing and Conformance

Required tests:

- aggregate invariant and transition tests;
- authorization, mandate, separation-of-duties, and tenant-isolation tests;
- idempotency and optimistic-concurrency tests;
- migration, constraint, index, and rollback tests;
- outbox/inbox, ordering, duplicate, replay, and DLQ tests;
- workflow restart, timeout, compensation, and manual-intervention tests;
- provider callback verification and Outcome Unknown tests;
- Portal E2E for success, partial, stale, denied, and error journeys;
- accessibility, locale, currency, unit, and RTL tests;
- security adversarial and data-exfiltration tests;
- AI evaluation, prompt-injection, policy-bypass, and unauthorized-tool tests;
- historical reconstruction and correction tests;
- performance, capacity, disaster-recovery, and operational exercise evidence.

17. Repository Review Boundary and Previously Inspected Evidence

- Procurement, order, contract, and manufacturing capabilities are registered.
- Purchase Order truth must not be inferred from a generic customer sales-order aggregate.
- The dedicated PO aggregate and acknowledgement workflow remain `REQUIRES_REPOSITORY_REVIEW`.

Area	Classification for this chapter
Target aggregate, API, tables, state model, and workflow	<code>REFERENCE_ARCHITECTURE</code>
Registered supporting modules	<code>DOCUMENTATION_ONLY_CLAIM</code> until executable files are inspected
Concrete behavior already established in earlier repository-grounded chapters	retain the original classification and evidence
Exact integration paths not inspected for this stage	<code>REQUIRES_REPOSITORY_REVIEW</code>
External authority or provider outcome	qualified external observation, never platform-owned truth

No statement in this chapter upgrades an implementation status merely because a module name, frontend route, README, plan, or database catalog entry exists.

18. Worked Scenario

An upstream governed commitment enters the stage with exact subject and version references. The initiating principal is authenticated, and its mandate is checked at command time. Evidence is gathered and evaluated against the applicable policy version and jurisdiction.

A recommendation may be produced, but an authorized Decision remains separate. The Domain accepts an idempotent Command and records state plus an outbox entry atomically. When an external party participates, a timeout creates Outcome Unknown; the workflow queries the external source or waits for a verified callback instead of repeating execution blindly.

On success or failure, the system records a completed fact, publishes a stable Event, updates downstream projections, and preserves source and knowledge cutoff. Later corrections append new facts and identify downstream impact rather than replacing history. AI assists through typed tools and cannot own the subject, grant authority, or invent an outcome.

19. Conformance Checklist

- ☐ Stable subject and version identities are explicit.
 - ☐ Proposition ownership and external authority are named.
 - ☐ Entry and exit criteria are testable.
 - ☐ State, Recommendation, Decision, Intent, execution, and outcome remain distinct.
 - ☐ Commands are authorized, version-guarded, and idempotent.
 - ☐ Events describe completed facts.
 - ☐ Evidence includes provenance, time, integrity, and correction links.
 - ☐ Portals show freshness, uncertainty, and Outcome Unknown.
 - ☐ AI uses typed tools and approved autonomy.
 - ☐ Security, privacy, tenant, facility, and jurisdiction boundaries are enforced.
 - ☐ Reconciliation precedes risky duplicate execution.
 - ☐ Corrections preserve history.
 - ☐ Tests prove invariants, security, failure, and recovery.
 - ☐ Implementation status is not upgraded without executable repository evidence.
-

20. Status Vocabulary

- VERIFIED_IMPLEMENTED: executable source and the relevant behavior were inspected.
- IMPLEMENTED_WITH_GAP: executable behavior exists with a material governance or completeness gap.
- PARTIAL_IMPLEMENTATION: only part of the target capability is evidenced.
- REFERENCE_ARCHITECTURE: normative target behavior; no claim that it exists today.
- PLANNED_NOT_IMPLEMENTED: explicitly planned but not evidenced as executable.
- DOCUMENTATION_ONLY_CLAIM: documentation or a client contract claims behavior without verified backend evidence.
- CONFLICT_REQUIRES_CANONICALIZATION: sources disagree and no final owner has been established.
- REQUIRES_REPOSITORY_REVIEW: the necessary executable evidence has not yet been inspected.

Publication Chapter 13 — Production Planning and Capacity Commitment

Metadata	Value
Publication Chapter	13
Canonical Lifecycle Stage	12
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE with explicit repository notes
Accountable capability	Manufacturing / Supplier Factory
Canon dependencies	CANON-001 through CANON-011
Constitutional dependencies	ART-001 through ART-016
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This stage exists to **translate an acknowledged commitment into a feasible, approved, versioned production plan with capacity, material, lot, milestone, and quality dependencies.**

Its output is not merely a screen state or document download. It is a governed set of identified subjects, Relationships, Decisions, Intent, Commands, completed facts, external observations, evidence, and correction history.

Reality or external outcome
≠ platform Observation
≠ derived Perspective
≠ AI Prediction
≠ Recommendation
≠ authorized Decision
≠ execution Intent

1.1 Accountability

The accountable capability is **Manufacturing / Supplier Factory**. Supporting Portals, workflow engines, reports, search indexes, integration adapters, and AI Agents may coordinate or present the work, but they do not silently become owners of the underlying business truth.

1.2 Primary business outcomes

- a stable subject identity and exact source-version lineage;
- explicit authority and effective mandate;
- a testable lifecycle outcome;
- evidence sufficient for downstream use and later reconstruction;
- visible failure, partial completion, uncertainty, and correction;
- a completed-fact Event or qualified external observation;
- downstream activation without shared-database ownership transfer.

1.3 Non-goals

- treating a registered module as proof of complete implementation;

- treating a frontend route as proof of backend behavior;
- allowing an AI recommendation to become a binding Decision;
- inventing external-provider or government-authority success;
- overwriting history during amendment or correction;
- combining unrelated states into one generic status field.

2. Entry and Exit Criteria

2.1 Entry criteria

1. Every upstream subject reference resolves to a stable identifier.
2. The exact approved, accepted, issued, or observed version is known.
3. The initiating human, service, organization, or AI Agent is authenticated.
4. The represented principal and mandate are valid at command time.
5. Required Evidence exists, or an explicit authorized exception is linked.
6. Applicable Policy and jurisdiction versions are fixed.
7. Duplicate, stale-version, and replay checks have run.
8. Any conflicting active process is resolved, suspended, or explicitly related.

2.2 Exit criteria

1. The authoritative Domain or external authority records the outcome.
2. State and version changes are durable.
3. A completed-fact Event or qualified observation is available.
4. Downstream consumers receive stable subject, correlation, and causation references.
5. Unknown, partial, disputed, or failed outcomes remain visible.
6. Corrections preserve the original record and reason.
7. Metrics, audit evidence, and operational signals are emitted.
8. A qualified reviewer can reconstruct who knew what, when, and why.

3. Ubiquitous Language and Subject Model

Primary subject families:

Subject	Required interpretation
ProductionPlan	independent identified subject or governed subordinate entity; ownership must be declared
ProductionPlanVersion	independent identified subject or governed subordinate entity; ownership must be declared
ProductionOrder	independent identified subject or governed subordinate entity; ownership must be declared
BatchPlan	independent identified subject or governed subordinate entity; ownership must be declared
MaterialRequirement	independent identified subject or governed subordinate entity; ownership must be declared
CapacityCommitment	independent identified subject or governed subordinate entity; ownership must be declared
MilestoneBaseline	independent identified subject or governed subordinate entity; ownership must be declared
FactoryAssignment	independent identified subject or governed subordinate entity; ownership must be declared

Reference aggregate:

```
ProductionPlan {
  aggregateId
  tenantId
  organizationContext
  canonicalSubjectRefs[]
  sourceVersionRefs[]
  lifecycleState
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
```

```

policyRefs[]
authorityRef
jurisdictionRefs[]
createdAt
effectiveAt
updatedAt
version
correctionOf?
correlationId
}

```

3.1 Identity and relationship rules

- a display name, document number, provider reference, barcode, or URL is not automatically canonical identity;
- relationships such as custody, mandate, allocation, publication, reservation, or obligation are independent facts;
- cross-Domain references use stable IDs and qualified snapshots, not shared-table mutation;
- external identifiers retain issuer and namespace;
- merge, split, correction, and supersession preserve lineage.

3.2 Invariants

1. Submitted, approved, accepted, issued, or signed versions are immutable.
2. State changes occur through authorized Commands.
3. Completed-fact Events use past-tense semantics.
4. A Recommendation cannot self-promote to a Decision.
5. An external outcome retains its source authority.
6. Derived projections cannot become authoritative by repeated copying.
7. All writes are tenant-, organization-, facility-, and authority-scoped as applicable.
8. Retried writes are idempotent.
9. Outcome Unknown is not converted to success or failure by assumption.
10. Corrections reference the corrected fact and downstream impact.

4. Lifecycle State Model

4.1 State vocabulary

State	Meaning
PROPOSED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
FEASIBILITY_CHECK	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
MATERIAL_CHECK	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
CAPACITY_PENDING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
APPROVAL_PENDING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
APPROVED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
BASELINED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
REBASELINE_PENDING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
SUSPENDED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
CANCELLED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
COMPLETED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification

4.2 Reference state machine

```

stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> FEASIBILITY_CHECK
    FEASIBILITY_CHECK --> MATERIAL_CHECK

```

```

MATERIAL_CHECK --> CAPACITY_PENDING
CAPACITY_PENDING --> APPROVAL_PENDING
APPROVAL_PENDING --> APPROVED
APPROVED --> BASELINED
BASELINED --> REBASELINE_PENDING
REBASELINE_PENDING --> SUSPENDED
SUSPENDED --> CANCELLED
CANCELLED --> COMPLETED

```

The diagram shows a readable primary path. A conformant implementation may include parallel approvals, amendment, suspension, timeout, appeal, rejection, partial completion, reconciliation, and correction paths, but it must preserve the same semantic distinctions.

4.3 Transition matrix

From	To	Minimum guard
PROPOSED	FEASIBILITY_CHECK	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
FEASIBILITY_CHECK	MATERIAL_CHECK	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
MATERIAL_CHECK	CAPACITY_PENDING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
CAPACITY_PENDING	APPROVAL_PENDING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
APPROVAL_PENDING	APPROVED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
APPROVED	BASELINED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
BASELINED	REBASELINE_PENDING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
REBASELINE_PENDING	SUSPENDED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
SUSPENDED	CANCELLED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
CANCELLED	COMPLETED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard

4.4 Forbidden state behavior

- direct arbitrary assignment from a Portal request;
- using one status to represent lifecycle, approval, payment, document, provider, and dispute state simultaneously;
- deleting terminal history to reopen a process;
- inferring completion from elapsed time or an AI prediction;
- moving past a mandatory gate because a downstream system already acted.

5. Decision, Policy, and Authority

Material Decisions in this stage include readiness, eligibility, approval, exception, amendment, correction, and down-stream activation decisions.

Reference Decision:

```

Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
}

```

```

evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Decision rules

- the Decision references exact source versions;
 - authority and delegation are evaluated at Decision time;
 - a Policy version and jurisdiction context are recorded;
 - mandatory-gate failure blocks progression unless a lawful exception exists;
 - dissent, uncertainty, and missing Evidence remain visible;
 - Decision correction does not erase the original Decision;
 - AI may prepare a briefing, but cannot manufacture authority.
-

6. Evidence, Documents, and Knowledge Cutoff

Every binding or high-impact action requires an Evidence Set containing:

- evidence identity and integrity hash where required;
- issuing, observing, or signing party;
- observed time, received time, and effective-time interval;
- subject and source-version relationships;
- qualification, confidence, and limitations;
- contradiction, retraction, supersession, and correction links;
- jurisdiction, classification, retention, and residency;
- chain of custody for physical or high-integrity evidence;
- the knowledge cutoff used by the Decision.

A document is not automatically truth. It is an Artifact containing Claims. Its legal or operational effect depends on issuer, mandate, signature, version, applicable rules, and external authority.

7. Commands, Events, and Queries

7.1 Commands

- CreateProductionPlan
- AssessProductionFeasibility
- CommitCapacity
- ApproveProductionPlan
- BaselineMilestones
- RebaselineProductionPlan
- SuspendProductionPlan

Each Command shall contain an idempotency key, actor identity, represented principal, tenant and organization context, correlation ID, expected version where relevant, reason for privileged action, and exact target subject.

7.2 Events

- ProductionPlanCreated
- ProductionFeasibilityAssessed
- CapacityCommitted
- ProductionPlanApproved
- ProductionPlanBaselined
- ProductionPlanRebaselined
- ProductionPlanSuspended

Events describe completed facts or explicitly qualified observations. A request, proposed action, timer, or prediction is not a completed-fact Event.

7.3 Queries and perspectives

- GetProductionPlan
- ListProductionPlanRecords
- GetProductionPlanTimeline
- GetProductionPlanEvidence
- GetProductionPlanDecisionPackage
- GetProductionPlanExceptions
- GetProductionPlanAuditTrail
- GetProductionPlanCurrentPerspective

Every query response states data freshness, source cutoff, version, and whether values are authoritative, observed, derived, predicted, or disputed.

8. Workflow, Timeout, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Evidence and policy gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Create Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and available compensation]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Trigger downstream stage]
```

Required workflow behavior:

- durable process and correlation identity;
 - persisted approvals and timers;
 - bounded retries and retry budgets;
 - idempotent command submission;
 - safe restart after deployment or outage;
 - compensation that records new Intent instead of deleting history;
 - visible stalled, orphaned, and manually intervened processes;
 - operator access to source responses and downstream impact;
 - reconciliation before repeating an external financial, legal, or physical action.
-

9. Target API Contract

Reference surface:

```
POST /api/v1/production_planning_and_capacity_commitment/commands
GET /api/v1/production_planning_and_capacity_commitment/{id}
GET /api/v1/production_planning_and_capacity_commitment/{id}/timeline
GET /api/v1/production_planning_and_capacity_commitment/{id}/evidence
GET /api/v1/production_planning_and_capacity_commitment/{id}/decisions
POST /api/v1/production_planning_and_capacity_commitment/{id}/exceptions
POST /api/v1/production_planning_and_capacity_commitment/{id}/corrections
```

These are reference endpoints, not claims that the exact routes exist.

Contract requirements:

- typed request and response schemas;
 - action-level authorization;
 - trusted authentication context for tenant identity;
 - idempotency for writes;
 - optimistic concurrency or equivalent aggregate guard;
 - RFC problem-details or governed equivalent;
 - stable pagination and sorting;
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Reference table families:

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stage_audit	privileged action evidence

Data rules:

- the owning service controls its schema and migrations;
- foreign keys stay inside an ownership boundary;
- cross-Domain relationships use stable IDs or contracts;
- effective-time and recorded-time are separate when needed;
- binding versions are immutable;
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- display exact version, owner, state, and last-updated time;
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- show loading, empty, partial, stale, error, timeout, Outcome Unknown, and permission-denied states;
- provide accessible keyboard and screen-reader behavior;
- handle locale, currency, units, dates, and RTL/LTR correctly;
- preserve evidence and download audit;
- never trust a browser-provided tenant header as authority;
- prevent double submission while still supporting safe retry;
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Operator surfaces additionally expose correlation, source responses, retry budget, replay status, compensation options, and affected downstream subjects.

12. AI Participation and Tool Governance

Permitted AI tasks:

- optimize schedules
- predict material shortage
- score capacity risk
- compare alternative production plans

AI may also summarize evidence, detect gaps, recommend next actions, and draft human-readable explanations.

AI shall not:

- own or directly mutate Domain databases;
- fabricate external-provider, carrier, bank, customs, inspection, or delivery outcomes;
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- external callback signature and replay verification;
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binding Decision audit completeness	100%
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Operational requirements include dashboards, alerts, runbooks, ownership, dependency health, retry budgets, DLQ/replay controls, Event outbox/inbox, workflow restart, reconciliation, capacity, backup/restore, incident command, and post-incident correction.

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Representative failures:

- stale or wrong source version;
- duplicate submission or replay;
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1. Query the authoritative source before repeating an external action.

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Required tests:

- aggregate invariant and transition tests;
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 - accessibility, locale, currency, unit, and RTL tests;
 - security adversarial and data-exfiltration tests;
 - AI evaluation, prompt-injection, policy-bypass, and unauthorized-tool tests;
 - historical reconstruction and correction tests;
 - performance, capacity, disaster-recovery, and operational exercise evidence.
-

17. Repository Review Boundary and Previously Inspected Evidence

- Manufacturing, supplier/provider, workflow, and event capabilities are registered.
- Detailed production-plan and capacity-commitment behavior is not treated as implemented without direct source evidence.

Area	Classification for this chapter
Target aggregate, API, tables, state model, and workflow Registered supporting modules	REFERENCE_ARCHITECTURE DOCUMENTATION_ONLY_CLAIM until executable files are inspected
Concrete behavior already established in earlier repository-grounded chapters	retain the original classification and evidence
Exact integration paths not inspected for this stage External authority or provider outcome	REQUIRES_REPOSITORY_REVIEW qualified external observation, never platform-owned truth

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A recommendation may be produced, but an authorized Decision remains separate. The Domain accepts an idempotent Command and records state plus an outbox entry atomically. When an external party participates, a timeout creates Outcome Unknown; the workflow queries the external source or waits for a verified callback instead of repeating execution blindly.

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- ☐ Stable subject and version identities are explicit.
 - ☐ Proposition ownership and external authority are named.
 - ☐ Entry and exit criteria are testable.
 - ☐ State, Recommendation, Decision, Intent, execution, and outcome remain distinct.
 - ☐ Commands are authorized, version-guarded, and idempotent.
 - ☐ Events describe completed facts.
 - ☐ Evidence includes provenance, time, integrity, and correction links.
 - ☐ Portals show freshness, uncertainty, and Outcome Unknown.
 - ☐ AI uses typed tools and approved autonomy.
 - ☐ Security, privacy, tenant, facility, and jurisdiction boundaries are enforced.
 - ☐ Reconciliation precedes risky duplicate execution.
 - ☐ Corrections preserve history.
 - ☐ Tests prove invariants, security, failure, and recovery.
 - ☐ Implementation status is not upgraded without executable repository evidence.
-

20. Status Vocabulary

- VERIFIED_IMPLEMENTED: executable source and the relevant behavior were inspected.
- IMPLEMENTED_WITH_GAP: executable behavior exists with a material governance or completeness gap.
- PARTIAL_IMPLEMENTATION: only part of the target capability is evidenced.
- REFERENCE_ARCHITECTURE: normative target behavior; no claim that it exists today.
- PLANNED_NOT_IMPLEMENTED: explicitly planned but not evidenced as executable.
- DOCUMENTATION_ONLY_CLAIM: documentation or a client contract claims behavior without verified backend evidence.
- CONFLICT_REQUIRES_CANONICALIZATION: sources disagree and no final owner has been established.
- REQUIRES_REPOSITORY_REVIEW: the necessary executable evidence has not yet been inspected.

Publication Chapter 14 — Production Execution and Milestone Observation

Metadata	Value
Publication Chapter	14
Canonical Lifecycle Stage	13
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE with explicit repository notes
Accountable capability	Manufacturing
Canon dependencies	CANON-001 through CANON-011
Constitutional dependencies	ART-001 through ART-016
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This stage exists to **record qualified production progress, lots, quantities, delays, exceptions, and milestone evidence while distinguishing observed progress from approved quality release.**

Its output is not merely a screen state or document download. It is a governed set of identified subjects, Relationships, Decisions, Intent, Commands, completed facts, external observations, evidence, and correction history.

- Reality or external outcome
- ≠ platform Observation
 - ≠ derived Perspective
 - ≠ AI Prediction
 - ≠ Recommendation
 - ≠ authorized Decision
 - ≠ execution Intent

1.1 Accountability

The accountable capability is **Manufacturing**. Supporting Portals, workflow engines, reports, search indexes, integration adapters, and AI Agents may coordinate or present the work, but they do not silently become owners of the underlying business truth.

1.2 Primary business outcomes

- a stable subject identity and exact source-version lineage;
- explicit authority and effective mandate;
- a testable lifecycle outcome;
- evidence sufficient for downstream use and later reconstruction;
- visible failure, partial completion, uncertainty, and correction;
- a completed-fact Event or qualified external observation;
- downstream activation without shared-database ownership transfer.

1.3 Non-goals

- treating a registered module as proof of complete implementation;
- treating a frontend route as proof of backend behavior;

- allowing an AI recommendation to become a binding Decision;
- inventing external-provider or government-authority success;
- overwriting history during amendment or correction;
- combining unrelated states into one generic status field.

2. Entry and Exit Criteria

2.1 Entry criteria

1. Every upstream subject reference resolves to a stable identifier.
2. The exact approved, accepted, issued, or observed version is known.
3. The initiating human, service, organization, or AI Agent is authenticated.
4. The represented principal and mandate are valid at command time.
5. Required Evidence exists, or an explicit authorized exception is linked.
6. Applicable Policy and jurisdiction versions are fixed.
7. Duplicate, stale-version, and replay checks have run.
8. Any conflicting active process is resolved, suspended, or explicitly related.

2.2 Exit criteria

1. The authoritative Domain or external authority records the outcome.
2. State and version changes are durable.
3. A completed-fact Event or qualified observation is available.
4. Downstream consumers receive stable subject, correlation, and causation references.
5. Unknown, partial, disputed, or failed outcomes remain visible.
6. Corrections preserve the original record and reason.
7. Metrics, audit evidence, and operational signals are emitted.
8. A qualified reviewer can reconstruct who knew what, when, and why.

3. Ubiquitous Language and Subject Model

Primary subject families:

Subject	Required interpretation
ProductionExecution	independent identified subject or governed subordinate entity; ownership must be declared
ProductionBatch	independent identified subject or governed subordinate entity; ownership must be declared
ProductionLot	independent identified subject or governed subordinate entity; ownership must be declared
MilestoneObservation	independent identified subject or governed subordinate entity; ownership must be declared
QuantityProgress	independent identified subject or governed subordinate entity; ownership must be declared
ProductionException	independent identified subject or governed subordinate entity; ownership must be declared
ReworkInstruction	independent identified subject or governed subordinate entity; ownership must be declared

Reference aggregate:

```
ProductionExecution {
  aggregateId
  tenantId
  organizationContext
  canonicalSubjectRefs[]
  sourceVersionRefs[]
  lifecycleState
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
}
```

```

    createdAt
    effectiveAt
    updatedAt
    version
    correctionOf?
    correlationId
}

```

3.1 Identity and relationship rules

- a display name, document number, provider reference, barcode, or URL is not automatically canonical identity;
- relationships such as custody, mandate, allocation, publication, reservation, or obligation are independent facts;
- cross-Domain references use stable IDs and qualified snapshots, not shared-table mutation;
- external identifiers retain issuer and namespace;
- merge, split, correction, and supersession preserve lineage.

3.2 Invariants

1. Submitted, approved, accepted, issued, or signed versions are immutable.
2. State changes occur through authorized Commands.
3. Completed-fact Events use past-tense semantics.
4. A Recommendation cannot self-promote to a Decision.
5. An external outcome retains its source authority.
6. Derived projections cannot become authoritative by repeated copying.
7. All writes are tenant-, organization-, facility-, and authority-scoped as applicable.
8. Retried writes are idempotent.
9. Outcome Unknown is not converted to success or failure by assumption.
10. Corrections reference the corrected fact and downstream impact.

4. Lifecycle State Model

4.1 State vocabulary

State	Meaning
NOT_STARTED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
STARTED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
IN_PROGRESS	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
BLOCKED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
DELAYED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
PARTIALLY_COMPLETED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
COMPLETED_PENDING_QC	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
REWORK	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
CANCELLED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
CLOSED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification

4.2 Reference state machine

```

stateDiagram-v2
    [*] --> NOT_STARTED
    NOT_STARTED --> STARTED
    STARTED --> IN_PROGRESS
    IN_PROGRESS --> BLOCKED
    BLOCKED --> DELAYED
    DELAYED --> PARTIALLY_COMPLETED
    PARTIALLY_COMPLETED --> COMPLETED_PENDING_QC
    COMPLETED_PENDING_QC --> REWORK

```

REWORK --> CANCELLED
CANCELLED --> CLOSED

The diagram shows a readable primary path. A conformant implementation may include parallel approvals, amendment, suspension, timeout, appeal, rejection, partial completion, reconciliation, and correction paths, but it must preserve the same semantic distinctions.

4.3 Transition matrix

From	To	Minimum guard
NOT_STARTED	STARTED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
STARTED	IN_PROGRESS	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
IN_PROGRESS	BLOCKED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
BLOCKED	DELAYED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
DELAYED	PARTIALLY_COMPLETED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
PARTIALLY_COMPLETED	COMPLETED_PENDING_QC	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
COMPLETED_PENDING_QC	REWORK	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
REWORK	CANCELLED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
CANCELLED	CLOSED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard

4.4 Forbidden state behavior

- direct arbitrary assignment from a Portal request;
- using one status to represent lifecycle, approval, payment, document, provider, and dispute state simultaneously;
- deleting terminal history to reopen a process;
- inferring completion from elapsed time or an AI prediction;
- moving past a mandatory gate because a downstream system already acted.

5. Decision, Policy, and Authority

Material Decisions in this stage include readiness, eligibility, approval, exception, amendment, correction, and downstream activation decisions.

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
}
```

```
correctionOf?  
}
```

Decision rules

- the Decision references exact source versions;
 - authority and delegation are evaluated at Decision time;
 - a Policy version and jurisdiction context are recorded;
 - mandatory-gate failure blocks progression unless a lawful exception exists;
 - dissent, uncertainty, and missing Evidence remain visible;
 - Decision correction does not erase the original Decision;
 - AI may prepare a briefing, but cannot manufacture authority.
-

6. Evidence, Documents, and Knowledge Cutoff

Every binding or high-impact action requires an Evidence Set containing:

- evidence identity and integrity hash where required;
- issuing, observing, or signing party;
- observed time, received time, and effective-time interval;
- subject and source-version relationships;
- qualification, confidence, and limitations;
- contradiction, retraction, supersession, and correction links;
- jurisdiction, classification, retention, and residency;
- chain of custody for physical or high-integrity evidence;
- the knowledge cutoff used by the Decision.

A document is not automatically truth. It is an Artifact containing Claims. Its legal or operational effect depends on issuer, mandate, signature, version, applicable rules, and external authority.

7. Commands, Events, and Queries

7.1 Commands

- StartProduction
- RecordMilestone
- RecordQuantityProgress
- ReportProductionException
- RequestRework
- CompleteProduction
- CorrectProductionObservation

Each Command shall contain an idempotency key, actor identity, represented principal, tenant and organization context, correlation ID, expected version where relevant, reason for privileged action, and exact target subject.

7.2 Events

- ProductionStarted
- ProductionMilestoneObserved
- ProductionProgressRecorded
- ProductionDelayed
- ProductionBlocked
- ProductionCompleted
- ProductionObservationCorrected

Events describe completed facts or explicitly qualified observations. A request, proposed action, timer, or prediction is not a completed-fact Event.

7.3 Queries and perspectives

- GetProductionExecution
- ListProductionExecutionRecords
- GetProductionExecutionTimeline
- GetProductionExecutionEvidence
- GetProductionExecutionDecisionPackage
- GetProductionExecutionExceptions
- GetProductionExecutionAuditTrail
- GetProductionExecutionCurrentPerspective

Every query response states data freshness, source cutoff, version, and whether values are authoritative, observed, derived, predicted, or disputed.

8. Workflow, Timeout, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Evidence and policy gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Create Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and available compensation]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Trigger downstream stage]
```

Required workflow behavior:

- durable process and correlation identity;
 - persisted approvals and timers;
 - bounded retries and retry budgets;
 - idempotent command submission;
 - safe restart after deployment or outage;
 - compensation that records new Intent instead of deleting history;
 - visible stalled, orphaned, and manually intervened processes;
 - operator access to source responses and downstream impact;
 - reconciliation before repeating an external financial, legal, or physical action.
-

9. Target API Contract

Reference surface:

```
POST /api/v1/production_execution_and_milestone_observation/commands
GET /api/v1/production_execution_and_milestone_observation/{id}
GET /api/v1/production_execution_and_milestone_observation/{id}/timeline
GET /api/v1/production_execution_and_milestone_observation/{id}/evidence
GET /api/v1/production_execution_and_milestone_observation/{id}/decisions
POST /api/v1/production_execution_and_milestone_observation/{id}/exceptions
POST /api/v1/production_execution_and_milestone_observation/{id}/corrections
```

These are reference endpoints, not claims that the exact routes exist.

Contract requirements:

- typed request and response schemas;
 - action-level authorization;
 - trusted authentication context for tenant identity;
 - idempotency for writes;
 - optimistic concurrency or equivalent aggregate guard;
 - RFC problem-details or governed equivalent;
 - stable pagination and sorting;
 - versioning and deprecation policy;
 - trace, correlation, and causation propagation;
 - explicit Outcome Unknown response when external completion is unresolved.
-

10. Target Persistence and Data Ownership

Reference table families:

Table family	Purpose
stage_aggregate	authoritative lifecycle subject
stage_version	immutable approved/submitted/issued versions

Table family	Purpose
stage_decision	binding Decisions and rationale
stage_evidence_link	evidence and provenance relationships
stage_event_outbox	transactional Event publication
stage_idempotency	duplicate-write protection
stage_exception	exception, claim, hold, or dispute
stage_correction	correction and supersession lineage
stage_audit	privileged action evidence

Data rules:

- the owning service controls its schema and migrations;
- foreign keys stay inside an ownership boundary;
- cross-Domain relationships use stable IDs or contracts;
- effective-time and recorded-time are separate when needed;
- binding versions are immutable;
- indexes support tenant, owner, state, correlation, external reference, and effective time;
- retention, legal hold, privacy, residency, backup, and restore are documented;
- analytics, search, reports, and vector stores are projections.

11. Portal and Experience Architecture

Required Portal behavior:

- display exact version, owner, state, and last-updated time;
- distinguish authoritative, observed, derived, predicted, and disputed values;
- expose required action and authority;
- show loading, empty, partial, stale, error, timeout, Outcome Unknown, and permission-denied states;
- provide accessible keyboard and screen-reader behavior;
- handle locale, currency, units, dates, and RTL/LTR correctly;
- preserve evidence and download audit;
- never trust a browser-provided tenant header as authority;
- prevent double submission while still supporting safe retry;
- show correction and supersession history.

Operator surfaces additionally expose correlation, source responses, retry budget, replay status, compensation options, and affected downstream subjects.

12. AI Participation and Tool Governance

Permitted AI tasks:

- predict delay and completion
- detect inconsistent progress
- summarize production exceptions
- compare observed progress with baseline

AI may also summarize evidence, detect gaps, recommend next actions, and draft human-readable explanations.

AI shall not:

- own or directly mutate Domain databases;
- fabricate external-provider, carrier, bank, customs, inspection, or delivery outcomes;
- grant itself permission or mandate;
- silently change an approved or issued version;
- suppress uncertainty, contradictions, policy failures, or human dissent;
- execute high-impact actions outside typed tools and approval policy;
- create shadow records for canonical subjects.

Reference automation:

Task	Maximum default
extraction and classification	A2, automatic with reviewable evidence
anomaly detection and recommendation	A2/A3 according to risk
binding commercial, legal, financial, quality, or custody Decision	A0/A1, human authority required
external execution	typed tool, policy, approval, idempotency, and outcome observation

Task	Maximum default
correction of authoritative truth	governed human-authorized workflow

Every Agent Run records prompt/model/tool versions, context sources, policy evaluation, approvals, tool calls, structured outputs, uncertainty, final observed outcome, and recovery actions.

13. Security, Privacy, and Trust Boundaries

Minimum controls:

- authenticated human, service, organization, device, and AI identities;
- represented principal and effective mandate validation;
- least privilege, ABAC/ReBAC where appropriate, and separation of duties;
- tenant, organization, facility, channel, and jurisdiction isolation;
- integrity protection for binding documents and Evidence;
- external callback signature and replay verification;
- confidential handling of commercial, personal, financial, transport, and regulatory data;
- secret isolation and no sensitive values in logs or prompts;
- privileged-action audit and emergency-access review;
- threat models for impersonation, tampering, replay, confused deputy, fraud, exfiltration, and prompt injection.

14. Reliability, SLOs, and Operations

Reference objectives:

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by the business deadline of the stage

Operational requirements include dashboards, alerts, runbooks, ownership, dependency health, retry budgets, DLQ/replay controls, Event outbox/inbox, workflow restart, reconciliation, capacity, backup/restore, incident command, and post-incident correction.

15. Failure Modes and Recovery

Representative failures:

- stale or wrong source version;
- duplicate submission or replay;
- external timeout after possible success;
- contradictory source observations;
- lost Event after committed state;
- approval or mandate revoked during execution;
- downstream action performed before a mandatory gate;
- partial completion across multiple subjects;
- evidence later retracted or corrected;
- tenant, facility, or jurisdiction mismatch.

Recovery principles:

1. Query the authoritative source before repeating an external action.
2. Preserve partial success.
3. Use compensating Intent instead of destructive history rewrite.
4. Publish correction Events when distributed facts change.
5. Quarantine high-impact ambiguity.
6. Escalate when authority or legal effect is unclear.
7. Record business impact, affected subjects, and final reconciliation.

16. Testing and Conformance

Required tests:

- aggregate invariant and transition tests;
 - authorization, mandate, separation-of-duties, and tenant-isolation tests;
 - idempotency and optimistic-concurrency tests;
 - migration, constraint, index, and rollback tests;
 - outbox/inbox, ordering, duplicate, replay, and DLQ tests;
 - workflow restart, timeout, compensation, and manual-intervention tests;
 - provider callback verification and Outcome Unknown tests;
 - Portal E2E for success, partial, stale, denied, and error journeys;
 - accessibility, locale, currency, unit, and RTL tests;
 - security adversarial and data-exfiltration tests;
 - AI evaluation, prompt-injection, policy-bypass, and unauthorized-tool tests;
 - historical reconstruction and correction tests;
 - performance, capacity, disaster-recovery, and operational exercise evidence.
-

17. Repository Review Boundary and Previously Inspected Evidence

- Manufacturing and Event Bus capabilities are registered.
- Portal progress views, when present, do not by themselves prove a canonical production-event model.
- This chapter remains REFERENCE_ARCHITECTURE pending repository verification.

Area	Classification for this chapter
Target aggregate, API, tables, state model, and workflow	REFERENCE_ARCHITECTURE
Registered supporting modules	DOCUMENTATION_ONLY_CLAIM until executable files are inspected
Concrete behavior already established in earlier repository-grounded chapters	retain the original classification and evidence
Exact integration paths not inspected for this stage	REQUIRES_REPOSITORY_REVIEW
External authority or provider outcome	qualified external observation, never platform-owned truth

No statement in this chapter upgrades an implementation status merely because a module name, frontend route, README, plan, or database catalog entry exists.

18. Worked Scenario

An upstream governed commitment enters the stage with exact subject and version references. The initiating principal is authenticated, and its mandate is checked at command time. Evidence is gathered and evaluated against the applicable policy version and jurisdiction.

A recommendation may be produced, but an authorized Decision remains separate. The Domain accepts an idempotent Command and records state plus an outbox entry atomically. When an external party participates, a timeout creates Outcome Unknown; the workflow queries the external source or waits for a verified callback instead of repeating execution blindly.

On success or failure, the system records a completed fact, publishes a stable Event, updates downstream projections, and preserves source and knowledge cutoff. Later corrections append new facts and identify downstream impact rather than replacing history. AI assists through typed tools and cannot own the subject, grant authority, or invent an outcome.

19. Conformance Checklist

- ☐ Stable subject and version identities are explicit.
- ☐ Proposition ownership and external authority are named.
- ☐ Entry and exit criteria are testable.
- ☐ State, Recommendation, Decision, Intent, execution, and outcome remain distinct.
- ☐ Commands are authorized, version-guarded, and idempotent.
- ☐ Events describe completed facts.

- ☐ Evidence includes provenance, time, integrity, and correction links.
 - ☐ Portals show freshness, uncertainty, and Outcome Unknown.
 - ☐ AI uses typed tools and approved autonomy.
 - ☐ Security, privacy, tenant, facility, and jurisdiction boundaries are enforced.
 - ☐ Reconciliation precedes risky duplicate execution.
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- PLANNED_NOT_IMPLEMENTED: explicitly planned but not evidenced as executable.
- DOCUMENTATION_ONLY_CLAIM: documentation or a client contract claims behavior without verified backend evidence.
- CONFLICT_REQUIRES_CANONICALIZATION: sources disagree and no final owner has been established.
- REQUIRES_REPOSITORY_REVIEW: the necessary executable evidence has not yet been inspected.

Publication Chapter 15 — Quality Inspection, Nonconformance, and Release

Metadata	Value
Publication Chapter	15
Canonical Lifecycle Stage	14
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE with explicit repository notes
Accountable capability	Quality / external inspector or laboratory
Canon dependencies	CANON-001 through CANON-011
Constitutional dependencies	ART-001 through ART-016
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This stage exists to **plan and execute inspection, record measurements and defects, manage nonconformance, and make explicit release, hold, rework, or rejection decisions.**

Its output is not merely a screen state or document download. It is a governed set of identified subjects, Relationships, Decisions, Intent, Commands, completed facts, external observations, evidence, and correction history.

- Reality or external outcome
- ≠ platform Observation
 - ≠ derived Perspective
 - ≠ AI Prediction
 - ≠ Recommendation
 - ≠ authorized Decision
 - ≠ execution Intent

1.1 Accountability

The accountable capability is **Quality / external inspector or laboratory**. Supporting Portals, workflow engines, reports, search indexes, integration adapters, and AI Agents may coordinate or present the work, but they do not silently become owners of the underlying business truth.

1.2 Primary business outcomes

- a stable subject identity and exact source-version lineage;
- explicit authority and effective mandate;
- a testable lifecycle outcome;
- evidence sufficient for downstream use and later reconstruction;
- visible failure, partial completion, uncertainty, and correction;
- a completed-fact Event or qualified external observation;
- downstream activation without shared-database ownership transfer.

1.3 Non-goals

- treating a registered module as proof of complete implementation;
 - treating a frontend route as proof of backend behavior;
 - allowing an AI recommendation to become a binding Decision;
 - inventing external-provider or government-authority success;
 - overwriting history during amendment or correction;
 - combining unrelated states into one generic status field.
-

2. Entry and Exit Criteria

2.1 Entry criteria

1. Every upstream subject reference resolves to a stable identifier.
2. The exact approved, accepted, issued, or observed version is known.
3. The initiating human, service, organization, or AI Agent is authenticated.
4. The represented principal and mandate are valid at command time.
5. Required Evidence exists, or an explicit authorized exception is linked.
6. Applicable Policy and jurisdiction versions are fixed.
7. Duplicate, stale-version, and replay checks have run.
8. Any conflicting active process is resolved, suspended, or explicitly related.

2.2 Exit criteria

1. The authoritative Domain or external authority records the outcome.
 2. State and version changes are durable.
 3. A completed-fact Event or qualified observation is available.
 4. Downstream consumers receive stable subject, correlation, and causation references.
 5. Unknown, partial, disputed, or failed outcomes remain visible.
 6. Corrections preserve the original record and reason.
 7. Metrics, audit evidence, and operational signals are emitted.
 8. A qualified reviewer can reconstruct who knew what, when, and why.
-

3. Ubiquitous Language and Subject Model

Primary subject families:

Subject	Required interpretation
InspectionPlan	independent identified subject or governed subordinate entity; ownership must be declared
Inspection	independent identified subject or governed subordinate entity; ownership must be declared
Sample	independent identified subject or governed subordinate entity; ownership must be declared
Measurement	independent identified subject or governed subordinate entity; ownership must be declared
Defect	independent identified subject or governed subordinate entity; ownership must be declared
Nonconformance	independent identified subject or governed subordinate entity; ownership must be declared
CorrectiveAction	independent identified subject or governed subordinate entity; ownership must be declared
QualityDecision	independent identified subject or governed subordinate entity; ownership must be declared
ReleaseCertificate	independent identified subject or governed subordinate entity; ownership must be declared

Reference aggregate:

```
InspectionPlan {
  aggregateId
  tenantId
  organizationContext
  canonicalSubjectRefs[]
  sourceVersionRefs[]
}
```

```

lifecycleState
decisionRefs[]
intentRefs[]
evidenceSetRef
knowledgeCutoff
policyRefs[]
authorityRef
jurisdictionRefs[]
createdAt
effectiveAt
updatedAt
version
correctionOf?
correlationId
}

```

3.1 Identity and relationship rules

- a display name, document number, provider reference, barcode, or URL is not automatically canonical identity;
- relationships such as custody, mandate, allocation, publication, reservation, or obligation are independent facts;
- cross-Domain references use stable IDs and qualified snapshots, not shared-table mutation;
- external identifiers retain issuer and namespace;
- merge, split, correction, and supersession preserve lineage.

3.2 Invariants

1. Submitted, approved, accepted, issued, or signed versions are immutable.
2. State changes occur through authorized Commands.
3. Completed-fact Events use past-tense semantics.
4. A Recommendation cannot self-promote to a Decision.
5. An external outcome retains its source authority.
6. Derived projections cannot become authoritative by repeated copying.
7. All writes are tenant-, organization-, facility-, and authority-scoped as applicable.
8. Retried writes are idempotent.
9. Outcome Unknown is not converted to success or failure by assumption.
10. Corrections reference the corrected fact and downstream impact.

4. Lifecycle State Model

4.1 State vocabulary

State	Meaning
PLANNED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
READY	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
IN_PROGRESS	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
EVIDENCE_PENDING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
DECISION_PENDING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
RELEASED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
CONDITIONALLY_RELEASED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
HELD	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
REWORK_REQUIRED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
REJECTED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
REVOKED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
CLOSED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification

4.2 Reference state machine

```
stateDiagram-v2
    [*] --> PLANNED
    PLANNED --> READY
    READY --> IN_PROGRESS
    IN_PROGRESS --> EVIDENCE_PENDING
    EVIDENCE_PENDING --> DECISION_PENDING
    DECISION_PENDING --> RELEASED
    RELEASED --> CONDITIONALLY_RELEASED
    CONDITIONALLY_RELEASED --> HELD
    HELD --> REWORK_REQUIRED
    REWORK_REQUIRED --> REJECTED
    REJECTED --> REVOKED
    REVOKED --> CLOSED
```

The diagram shows a readable primary path. A conformant implementation may include parallel approvals, amendment, suspension, timeout, appeal, rejection, partial completion, reconciliation, and correction paths, but it must preserve the same semantic distinctions.

4.3 Transition matrix

From	To	Minimum guard
PLANNED	READY	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
READY	IN_PROGRESS	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
IN_PROGRESS	EVIDENCE_PENDING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
EVIDENCE_PENDING	DECISION_PENDING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
DECISION_PENDING	RELEASED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
RELEASED	CONDITIONALLY_RELEASED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
CONDITIONALLY_RELEASED	HELD	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
HELD	REWORK_REQUIRED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
REWORK_REQUIRED	REJECTED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
REJECTED	REVOKED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
REVOKED	CLOSED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard

4.4 Forbidden state behavior

- direct arbitrary assignment from a Portal request;
- using one status to represent lifecycle, approval, payment, document, provider, and dispute state simultaneously;
- deleting terminal history to reopen a process;
- inferring completion from elapsed time or an AI prediction;
- moving past a mandatory gate because a downstream system already acted.

5. Decision, Policy, and Authority

Material Decisions in this stage include readiness, eligibility, approval, exception, amendment, correction, and downstream activation decisions.

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Decision rules

- the Decision references exact source versions;
 - authority and delegation are evaluated at Decision time;
 - a Policy version and jurisdiction context are recorded;
 - mandatory-gate failure blocks progression unless a lawful exception exists;
 - dissent, uncertainty, and missing Evidence remain visible;
 - Decision correction does not erase the original Decision;
 - AI may prepare a briefing, but cannot manufacture authority.
-

6. Evidence, Documents, and Knowledge Cutoff

Every binding or high-impact action requires an Evidence Set containing:

- evidence identity and integrity hash where required;
- issuing, observing, or signing party;
- observed time, received time, and effective-time interval;
- subject and source-version relationships;
- qualification, confidence, and limitations;
- contradiction, retraction, supersession, and correction links;
- jurisdiction, classification, retention, and residency;
- chain of custody for physical or high-integrity evidence;
- the knowledge cutoff used by the Decision.

A document is not automatically truth. It is an Artifact containing Claims. Its legal or operational effect depends on issuer, mandate, signature, version, applicable rules, and external authority.

7. Commands, Events, and Queries

7.1 Commands

- CreateInspectionPlan
- ScheduleInspection
- RecordMeasurement
- RecordDefect
- OpenNonconformance
- RequestCorrectiveAction
- DecideQualityRelease
- RevokeQualityRelease

Each Command shall contain an idempotency key, actor identity, represented principal, tenant and organization context, correlation ID, expected version where relevant, reason for privileged action, and exact target subject.

7.2 Events

- InspectionPlanned
- InspectionStarted

- MeasurementRecorded
- DefectObserved
- NonconformanceOpened
- CorrectiveActionRequested
- QualityReleased
- QualityHeld
- QualityRejected
- QualityReleaseRevoked

Events describe completed facts or explicitly qualified observations. A request, proposed action, timer, or prediction is not a completed-fact Event.

7.3 Queries and perspectives

- GetInspectionPlan
- ListInspectionPlanRecords
- GetInspectionPlanTimeline
- GetInspectionPlanEvidence
- GetInspectionPlanDecisionPackage
- GetInspectionPlanExceptions
- GetInspectionPlanAuditTrail
- GetInspectionPlanCurrentPerspective

Every query response states data freshness, source cutoff, version, and whether values are authoritative, observed, derived, predicted, or disputed.

8. Workflow, Timeout, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Evidence and policy gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Create Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and available compensation]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Trigger downstream stage]

```

Required workflow behavior:

- durable process and correlation identity;
- persisted approvals and timers;
- bounded retries and retry budgets;
- idempotent command submission;
- safe restart after deployment or outage;
- compensation that records new Intent instead of deleting history;
- visible stalled, orphaned, and manually intervened processes;
- operator access to source responses and downstream impact;
- reconciliation before repeating an external financial, legal, or physical action.

9. Target API Contract

Reference surface:

```

POST /api/v1/quality_inspection_nonconformance_and_release/commands
GET  /api/v1/quality_inspection_nonconformance_and_release/{id}
GET  /api/v1/quality_inspection_nonconformance_and_release/{id}/timeline
GET  /api/v1/quality_inspection_nonconformance_and_release/{id}/evidence
GET  /api/v1/quality_inspection_nonconformance_and_release/{id}/decisions
POST /api/v1/quality_inspection_nonconformance_and_release/{id}/exceptions
POST /api/v1/quality_inspection_nonconformance_and_release/{id}/corrections

```

These are reference endpoints, not claims that the exact routes exist.

Contract requirements:

- typed request and response schemas;
- action-level authorization;
- trusted authentication context for tenant identity;
- idempotency for writes;
- optimistic concurrency or equivalent aggregate guard;
- RFC problem-details or governed equivalent;
- stable pagination and sorting;
- versioning and deprecation policy;
- trace, correlation, and causation propagation;
- explicit Outcome Unknown response when external completion is unresolved.

10. Target Persistence and Data Ownership

Reference table families:

Table family	Purpose
stage_aggregate	authoritative lifecycle subject
stage_version	immutable approved/submitted/issued versions
stage_decision	binding Decisions and rationale
stage_evidence_link	evidence and provenance relationships
stage_event_outbox	transactional Event publication
stage_idempotency	duplicate-write protection
stage_exception	exception, claim, hold, or dispute
stage_correction	correction and supersession lineage
stage_audit	privileged action evidence

Data rules:

- the owning service controls its schema and migrations;
- foreign keys stay inside an ownership boundary;
- cross-Domain relationships use stable IDs or contracts;
- effective-time and recorded-time are separate when needed;
- binding versions are immutable;
- indexes support tenant, owner, state, correlation, external reference, and effective time;
- retention, legal hold, privacy, residency, backup, and restore are documented;
- analytics, search, reports, and vector stores are projections.

11. Portal and Experience Architecture

Required Portal behavior:

- display exact version, owner, state, and last-updated time;
- distinguish authoritative, observed, derived, predicted, and disputed values;
- expose required action and authority;
- show loading, empty, partial, stale, error, timeout, Outcome Unknown, and permission-denied states;
- provide accessible keyboard and screen-reader behavior;
- handle locale, currency, units, dates, and RTL/LTR correctly;
- preserve evidence and download audit;
- never trust a browser-provided tenant header as authority;
- prevent double submission while still supporting safe retry;
- show correction and supersession history.

Operator surfaces additionally expose correlation, source responses, retry budget, replay status, compensation options, and affected downstream subjects.

12. AI Participation and Tool Governance

Permitted AI tasks:

- assist defect detection
- detect measurement anomalies
- classify nonconformance
- prepare a release-decision briefing

AI may also summarize evidence, detect gaps, recommend next actions, and draft human-readable explanations.

AI shall not:

- own or directly mutate Domain databases;
- fabricate external-provider, carrier, bank, customs, inspection, or delivery outcomes;
- grant itself permission or mandate;
- silently change an approved or issued version;
- suppress uncertainty, contradictions, policy failures, or human dissent;
- execute high-impact actions outside typed tools and approval policy;
- create shadow records for canonical subjects.

Reference automation:

Task	Maximum default
extraction and classification	A2, automatic with reviewable evidence
anomaly detection and recommendation	A2/A3 according to risk
binding commercial, legal, financial, quality, or custody Decision	A0/A1, human authority required
external execution	typed tool, policy, approval, idempotency, and outcome observation
correction of authoritative truth	governed human-authorized workflow

Every Agent Run records prompt/model/tool versions, context sources, policy evaluation, approvals, tool calls, structured outputs, uncertainty, final observed outcome, and recovery actions.

13. Security, Privacy, and Trust Boundaries

Minimum controls:

- authenticated human, service, organization, device, and AI identities;
- represented principal and effective mandate validation;
- least privilege, ABAC/ReBAC where appropriate, and separation of duties;
- tenant, organization, facility, channel, and jurisdiction isolation;
- integrity protection for binding documents and Evidence;
- external callback signature and replay verification;
- confidential handling of commercial, personal, financial, transport, and regulatory data;
- secret isolation and no sensitive values in logs or prompts;
- privileged-action audit and emergency-access review;
- threat models for impersonation, tampering, replay, confused deputy, fraud, exfiltration, and prompt injection.

14. Reliability, SLOs, and Operations

Reference objectives:

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by the business deadline of the stage

Operational requirements include dashboards, alerts, runbooks, ownership, dependency health, retry budgets, DLQ/replay controls, Event outbox/inbox, workflow restart, reconciliation, capacity, backup/restore, incident command, and post-incident correction.

15. Failure Modes and Recovery

Representative failures:

- stale or wrong source version;
- duplicate submission or replay;
- external timeout after possible success;
- contradictory source observations;
- lost Event after committed state;
- approval or mandate revoked during execution;
- downstream action performed before a mandatory gate;
- partial completion across multiple subjects;
- evidence later retracted or corrected;
- tenant, facility, or jurisdiction mismatch.

Recovery principles:

1. Query the authoritative source before repeating an external action.
2. Preserve partial success.
3. Use compensating Intent instead of destructive history rewrite.
4. Publish correction Events when distributed facts change.
5. Quarantine high-impact ambiguity.
6. Escalate when authority or legal effect is unclear.
7. Record business impact, affected subjects, and final reconciliation.

16. Testing and Conformance

Required tests:

- aggregate invariant and transition tests;
- authorization, mandate, separation-of-duties, and tenant-isolation tests;
- idempotency and optimistic-concurrency tests;
- migration, constraint, index, and rollback tests;
- outbox/inbox, ordering, duplicate, replay, and DLQ tests;
- workflow restart, timeout, compensation, and manual-intervention tests;
- provider callback verification and Outcome Unknown tests;
- Portal E2E for success, partial, stale, denied, and error journeys;
- accessibility, locale, currency, unit, and RTL tests;
- security adversarial and data-exfiltration tests;
- AI evaluation, prompt-injection, policy-bypass, and unauthorized-tool tests;
- historical reconstruction and correction tests;
- performance, capacity, disaster-recovery, and operational exercise evidence.

17. Repository Review Boundary and Previously Inspected Evidence

- Quality, regulatory, document, and AI vision capabilities are registered or referenced.
- A complete end-to-end quality aggregate and release authority require targeted repository inspection.

Area	Classification for this chapter
Target aggregate, API, tables, state model, and workflow Registered supporting modules	REFERENCE_ARCHITECTURE DOCUMENTATION_ONLY_CLAIM until executable files are inspected
Concrete behavior already established in earlier repository-grounded chapters	retain the original classification and evidence
Exact integration paths not inspected for this stage External authority or provider outcome	REQUIRES_REPOSITORY_REVIEW qualified external observation, never platform-owned truth

No statement in this chapter upgrades an implementation status merely because a module name, frontend route, README, plan, or database catalog entry exists.

18. Worked Scenario

An upstream governed commitment enters the stage with exact subject and version references. The initiating principal is authenticated, and its mandate is checked at command time. Evidence is gathered and evaluated against the applicable policy version and jurisdiction.

A recommendation may be produced, but an authorized Decision remains separate. The Domain accepts an idempotent Command and records state plus an outbox entry atomically. When an external party participates, a timeout creates Outcome Unknown; the workflow queries the external source or waits for a verified callback instead of repeating execution blindly.

On success or failure, the system records a completed fact, publishes a stable Event, updates downstream projections, and preserves source and knowledge cutoff. Later corrections append new facts and identify downstream impact rather than replacing history. AI assists through typed tools and cannot own the subject, grant authority, or invent an outcome.

19. Conformance Checklist

- ☐ Stable subject and version identities are explicit.
 - ☐ Proposition ownership and external authority are named.
 - ☐ Entry and exit criteria are testable.
 - ☐ State, Recommendation, Decision, Intent, execution, and outcome remain distinct.
 - ☐ Commands are authorized, version-guarded, and idempotent.
 - ☐ Events describe completed facts.
 - ☐ Evidence includes provenance, time, integrity, and correction links.
 - ☐ Portals show freshness, uncertainty, and Outcome Unknown.
 - ☐ AI uses typed tools and approved autonomy.
 - ☐ Security, privacy, tenant, facility, and jurisdiction boundaries are enforced.
 - ☐ Reconciliation precedes risky duplicate execution.
 - ☐ Corrections preserve history.
 - ☐ Tests prove invariants, security, failure, and recovery.
 - ☐ Implementation status is not upgraded without executable repository evidence.
-

20. Status Vocabulary

- VERIFIED_IMPLEMENTED: executable source and the relevant behavior were inspected.
- IMPLEMENTED_WITH_GAP: executable behavior exists with a material governance or completeness gap.
- PARTIAL_IMPLEMENTATION: only part of the target capability is evidenced.
- REFERENCE_ARCHITECTURE: normative target behavior; no claim that it exists today.
- PLANNED_NOT_IMPLEMENTED: explicitly planned but not evidenced as executable.
- DOCUMENTATION_ONLY_CLAIM: documentation or a client contract claims behavior without verified backend evidence.
- CONFLICT_REQUIRES_CANONICALIZATION: sources disagree and no final owner has been established.
- REQUIRES_REPOSITORY_REVIEW: the necessary executable evidence has not yet been inspected.

Publication Chapter 16 — Packing List and Commercial Invoice

Metadata	Value
Publication Chapter	16
Canonical Lifecycle Stage	15
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE with explicit repository notes
Accountable capability	Trade Documents / Finance
Canon dependencies	CANON-001 through CANON-011
Constitutional dependencies	ART-001 through ART-016
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This stage exists to **produce shipment-ready, versioned commercial documents that reconcile products, packages, quantities, values, weights, and terms with production and quality evidence.**

Its output is not merely a screen state or document download. It is a governed set of identified subjects, Relationships, Decisions, Intent, Commands, completed facts, external observations, evidence, and correction history.

Reality or external outcome

- ≠ platform Observation
- ≠ derived Perspective
- ≠ AI Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountability

The accountable capability is **Trade Documents / Finance**. Supporting Portals, workflow engines, reports, search indexes, integration adapters, and AI Agents may coordinate or present the work, but they do not silently become owners of the underlying business truth.

1.2 Primary business outcomes

- a stable subject identity and exact source-version lineage;
- explicit authority and effective mandate;
- a testable lifecycle outcome;
- evidence sufficient for downstream use and later reconstruction;
- visible failure, partial completion, uncertainty, and correction;
- a completed-fact Event or qualified external observation;
- downstream activation without shared-database ownership transfer.

1.3 Non-goals

- treating a registered module as proof of complete implementation;
- treating a frontend route as proof of backend behavior;

- allowing an AI recommendation to become a binding Decision;
- inventing external-provider or government-authority success;
- overwriting history during amendment or correction;
- combining unrelated states into one generic status field.

2. Entry and Exit Criteria

2.1 Entry criteria

1. Every upstream subject reference resolves to a stable identifier.
2. The exact approved, accepted, issued, or observed version is known.
3. The initiating human, service, organization, or AI Agent is authenticated.
4. The represented principal and mandate are valid at command time.
5. Required Evidence exists, or an explicit authorized exception is linked.
6. Applicable Policy and jurisdiction versions are fixed.
7. Duplicate, stale-version, and replay checks have run.
8. Any conflicting active process is resolved, suspended, or explicitly related.

2.2 Exit criteria

1. The authoritative Domain or external authority records the outcome.
2. State and version changes are durable.
3. A completed-fact Event or qualified observation is available.
4. Downstream consumers receive stable subject, correlation, and causation references.
5. Unknown, partial, disputed, or failed outcomes remain visible.
6. Corrections preserve the original record and reason.
7. Metrics, audit evidence, and operational signals are emitted.
8. A qualified reviewer can reconstruct who knew what, when, and why.

3. Ubiquitous Language and Subject Model

Primary subject families:

Subject	Required interpretation
PackingList	independent identified subject or governed subordinate entity; ownership must be declared
PackingListVersion	independent identified subject or governed subordinate entity; ownership must be declared
Package	independent identified subject or governed subordinate entity; ownership must be declared
CargoUnit	independent identified subject or governed subordinate entity; ownership must be declared
CommercialInvoice	independent identified subject or governed subordinate entity; ownership must be declared
CommercialInvoiceLine	independent identified subject or governed subordinate entity; ownership must be declared
DocumentReconciliation	independent identified subject or governed subordinate entity; ownership must be declared
DocumentSignature	independent identified subject or governed subordinate entity; ownership must be declared

Reference aggregate:

```
PackingList {
  aggregateId
  tenantId
  organizationContext
  canonicalSubjectRefs[]
  sourceVersionRefs[]
  lifecycleState
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
}
```

```

authorityRef
jurisdictionRefs[]
createdAt
effectiveAt
updatedAt
version
correctionOf?
correlationId
}

```

3.1 Identity and relationship rules

- a display name, document number, provider reference, barcode, or URL is not automatically canonical identity;
- relationships such as custody, mandate, allocation, publication, reservation, or obligation are independent facts;
- cross-Domain references use stable IDs and qualified snapshots, not shared-table mutation;
- external identifiers retain issuer and namespace;
- merge, split, correction, and supersession preserve lineage.

3.2 Invariants

1. Submitted, approved, accepted, issued, or signed versions are immutable.
2. State changes occur through authorized Commands.
3. Completed-fact Events use past-tense semantics.
4. A Recommendation cannot self-promote to a Decision.
5. An external outcome retains its source authority.
6. Derived projections cannot become authoritative by repeated copying.
7. All writes are tenant-, organization-, facility-, and authority-scoped as applicable.
8. Retried writes are idempotent.
9. Outcome Unknown is not converted to success or failure by assumption.
10. Corrections reference the corrected fact and downstream impact.

4. Lifecycle State Model

4.1 State vocabulary

State	Meaning
DRAFT	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
VALIDATION_PENDING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
APPROVAL_PENDING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
ISSUED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
AMENDED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
ACCEPTED_FOR_SHIPMENT	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
REJECTED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
SUPERSEDED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
CANCELLED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification

4.2 Reference state machine

```

stateDiagram-v2
    [*] --> DRAFT
    DRAFT --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ISSUED
    ISSUED --> AMENDED
    AMENDED --> ACCEPTED_FOR_SHIPMENT
    ACCEPTED_FOR_SHIPMENT --> REJECTED
    REJECTED --> SUPERSEDED
    SUPERSEDED --> CANCELLED

```

The diagram shows a readable primary path. A conformant implementation may include parallel approvals, amendment, suspension, timeout, appeal, rejection, partial completion, reconciliation, and correction paths, but it must preserve the same semantic distinctions.

4.3 Transition matrix

From	To	Minimum guard
DRAFT	VALIDATION_PENDING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
VALIDATION_PENDING	APPROVAL_PENDING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
APPROVAL_PENDING	ISSUED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
ISSUED	AMENDED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
AMENDED	ACCEPTED_FOR_SHIPMENT	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
ACCEPTED_FOR_SHIPMENT	REJECTED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
REJECTED	SUPERSEDED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
SUPERSEDED	CANCELLED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard

4.4 Forbidden state behavior

- direct arbitrary assignment from a Portal request;
- using one status to represent lifecycle, approval, payment, document, provider, and dispute state simultaneously;
- deleting terminal history to reopen a process;
- inferring completion from elapsed time or an AI prediction;
- moving past a mandatory gate because a downstream system already acted.

5. Decision, Policy, and Authority

Material Decisions in this stage include readiness, eligibility, approval, exception, amendment, correction, and downstream activation decisions.

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
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  effectiveAt
  correctionOf?
}
```

Decision rules

- the Decision references exact source versions;
 - authority and delegation are evaluated at Decision time;
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Every binding or high-impact action requires an Evidence Set containing:

- evidence identity and integrity hash where required;
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- observed time, received time, and effective-time interval;
- subject and source-version relationships;
- qualification, confidence, and limitations;
- contradiction, retraction, supersession, and correction links;
- jurisdiction, classification, retention, and residency;
- chain of custody for physical or high-integrity evidence;
- the knowledge cutoff used by the Decision.

A document is not automatically truth. It is an Artifact containing Claims. Its legal or operational effect depends on issuer, mandate, signature, version, applicable rules, and external authority.

7. Commands, Events, and Queries

7.1 Commands

- CreatePackingList
- CreateCommercialInvoice
- ValidateShipmentDocuments
- ApproveShipmentDocuments
- IssueShipmentDocuments
- AmendShipmentDocuments
- ReconcileShipmentDocuments

Each Command shall contain an idempotency key, actor identity, represented principal, tenant and organization context, correlation ID, expected version where relevant, reason for privileged action, and exact target subject.

7.2 Events

- PackingListCreated
- CommercialInvoiceCreated
- ShipmentDocumentsValidated
- ShipmentDocumentsIssued
- ShipmentDocumentsAmended
- DocumentDiscrepancyDetected

Events describe completed facts or explicitly qualified observations. A request, proposed action, timer, or prediction is not a completed-fact Event.

7.3 Queries and perspectives

- GetPackingList
- ListPackingListRecords
- GetPackingListTimeline
- GetPackingListEvidence
- GetPackingListDecisionPackage
- GetPackingListExceptions
- GetPackingListAuditTrail
- GetPackingListCurrentPerspective

Every query response states data freshness, source cutoff, version, and whether values are authoritative, observed, derived, predicted, or disputed.

8. Workflow, Timeout, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Evidence and policy gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Create Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and available compensation]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Trigger downstream stage]
```

Required workflow behavior:

- durable process and correlation identity;
 - persisted approvals and timers;
 - bounded retries and retry budgets;
 - idempotent command submission;
 - safe restart after deployment or outage;
 - compensation that records new Intent instead of deleting history;
 - visible stalled, orphaned, and manually intervened processes;
 - operator access to source responses and downstream impact;
 - reconciliation before repeating an external financial, legal, or physical action.
-

9. Target API Contract

Reference surface:

```
POST /api/v1/packing_list_and_commercial_invoice/commands
GET  /api/v1/packing_list_and_commercial_invoice/{id}
GET  /api/v1/packing_list_and_commercial_invoice/{id}/timeline
GET  /api/v1/packing_list_and_commercial_invoice/{id}/evidence
GET  /api/v1/packing_list_and_commercial_invoice/{id}/decisions
POST /api/v1/packing_list_and_commercial_invoice/{id}/exceptions
POST /api/v1/packing_list_and_commercial_invoice/{id}/corrections
```

These are reference endpoints, not claims that the exact routes exist.

Contract requirements:

- typed request and response schemas;
 - action-level authorization;
 - trusted authentication context for tenant identity;
 - idempotency for writes;
 - optimistic concurrency or equivalent aggregate guard;
 - RFC problem-details or governed equivalent;
 - stable pagination and sorting;
 - versioning and deprecation policy;
 - trace, correlation, and causation propagation;
 - explicit Outcome Unknown response when external completion is unresolved.
-

10. Target Persistence and Data Ownership

Reference table families:

Table family	Purpose
stage_aggregate	authoritative lifecycle subject
stage_version	immutable approved/submitted/issued versions
stage_decision	binding Decisions and rationale
stage_evidence_link	evidence and provenance relationships
stage_event_outbox	transactional Event publication
stage_idempotency	duplicate-write protection
stage_exception	exception, claim, hold, or dispute
stage_correction	correction and supersession lineage

Table family	Purpose
stage_audit	privileged action evidence

Data rules:

- the owning service controls its schema and migrations;
- foreign keys stay inside an ownership boundary;
- cross-Domain relationships use stable IDs or contracts;
- effective-time and recorded-time are separate when needed;
- binding versions are immutable;
- indexes support tenant, owner, state, correlation, external reference, and effective time;
- retention, legal hold, privacy, residency, backup, and restore are documented;
- analytics, search, reports, and vector stores are projections.

11. Portal and Experience Architecture

Required Portal behavior:

- display exact version, owner, state, and last-updated time;
- distinguish authoritative, observed, derived, predicted, and disputed values;
- expose required action and authority;
- show loading, empty, partial, stale, error, timeout, Outcome Unknown, and permission-denied states;
- provide accessible keyboard and screen-reader behavior;
- handle locale, currency, units, dates, and RTL/LTR correctly;
- preserve evidence and download audit;
- never trust a browser-provided tenant header as authority;
- prevent double submission while still supporting safe retry;
- show correction and supersession history.

Operator surfaces additionally expose correlation, source responses, retry budget, replay status, compensation options, and affected downstream subjects.

12. AI Participation and Tool Governance

Permitted AI tasks:

- extract document data
- reconcile documents
- suggest HS codes with confidence
- detect weight and value anomalies

AI may also summarize evidence, detect gaps, recommend next actions, and draft human-readable explanations.

AI shall not:

- own or directly mutate Domain databases;
- fabricate external-provider, carrier, bank, customs, inspection, or delivery outcomes;
- grant itself permission or mandate;
- silently change an approved or issued version;
- suppress uncertainty, contradictions, policy failures, or human dissent;
- execute high-impact actions outside typed tools and approval policy;
- create shadow records for canonical subjects.

Reference automation:

Task	Maximum default
extraction and classification	A2, automatic with reviewable evidence
anomaly detection and recommendation	A2/A3 according to risk
binding commercial, legal, financial, quality, or custody Decision	A0/A1, human authority required
external execution	typed tool, policy, approval, idempotency, and outcome observation
correction of authoritative truth	governed human-authorized workflow

Every Agent Run records prompt/model/tool versions, context sources, policy evaluation, approvals, tool calls, structured outputs, uncertainty, final observed outcome, and recovery actions.

13. Security, Privacy, and Trust Boundaries

Minimum controls:

- authenticated human, service, organization, device, and AI identities;
- represented principal and effective mandate validation;
- least privilege, ABAC/ReBAC where appropriate, and separation of duties;
- tenant, organization, facility, channel, and jurisdiction isolation;
- integrity protection for binding documents and Evidence;
- external callback signature and replay verification;
- confidential handling of commercial, personal, financial, transport, and regulatory data;
- secret isolation and no sensitive values in logs or prompts;
- privileged-action audit and emergency-access review;
- threat models for impersonation, tampering, replay, confused deputy, fraud, exfiltration, and prompt injection.

14. Reliability, SLOs, and Operations

Reference objectives:

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by the business deadline of the stage

Operational requirements include dashboards, alerts, runbooks, ownership, dependency health, retry budgets, DLQ/replay controls, Event outbox/inbox, workflow restart, reconciliation, capacity, backup/restore, incident command, and post-incident correction.

15. Failure Modes and Recovery

Representative failures:

- stale or wrong source version;
- duplicate submission or replay;
- external timeout after possible success;
- contradictory source observations;
- lost Event after committed state;
- approval or mandate revoked during execution;
- downstream action performed before a mandatory gate;
- partial completion across multiple subjects;
- evidence later retracted or corrected;
- tenant, facility, or jurisdiction mismatch.

Recovery principles:

1. Query the authoritative source before repeating an external action.
2. Preserve partial success.
3. Use compensating Intent instead of destructive history rewrite.
4. Publish correction Events when distributed facts change.
5. Quarantine high-impact ambiguity.
6. Escalate when authority or legal effect is unclear.
7. Record business impact, affected subjects, and final reconciliation.

16. Testing and Conformance

Required tests:

- aggregate invariant and transition tests;
- authorization, mandate, separation-of-duties, and tenant-isolation tests;
- idempotency and optimistic-concurrency tests;

- migration, constraint, index, and rollback tests;
- outbox/inbox, ordering, duplicate, replay, and DLQ tests;
- workflow restart, timeout, compensation, and manual-intervention tests;
- provider callback verification and Outcome Unknown tests;
- Portal E2E for success, partial, stale, denied, and error journeys;
- accessibility, locale, currency, unit, and RTL tests;
- security adversarial and data-exfiltration tests;
- AI evaluation, prompt-injection, policy-bypass, and unauthorized-tool tests;
- historical reconstruction and correction tests;
- performance, capacity, disaster-recovery, and operational exercise evidence.

17. Repository Review Boundary and Previously Inspected Evidence

- Document, finance, trade, and shipping capabilities are registered.
- Exact document schemas and generation behavior are not asserted implemented without executable evidence.

Area	Classification for this chapter
Target aggregate, API, tables, state model, and workflow Registered supporting modules	REFERENCE_ARCHITECTURE DOCUMENTATION_ONLY_CLAIM until executable files are inspected
Concrete behavior already established in earlier repository-grounded chapters	retain the original classification and evidence
Exact integration paths not inspected for this stage	REQUIRES_REPOSITORY_REVIEW
External authority or provider outcome	qualified external observation, never platform-owned truth

No statement in this chapter upgrades an implementation status merely because a module name, frontend route, README, plan, or database catalog entry exists.

18. Worked Scenario

An upstream governed commitment enters the stage with exact subject and version references. The initiating principal is authenticated, and its mandate is checked at command time. Evidence is gathered and evaluated against the applicable policy version and jurisdiction.

A recommendation may be produced, but an authorized Decision remains separate. The Domain accepts an idempotent Command and records state plus an outbox entry atomically. When an external party participates, a timeout creates Outcome Unknown; the workflow queries the external source or waits for a verified callback instead of repeating execution blindly.

On success or failure, the system records a completed fact, publishes a stable Event, updates downstream projections, and preserves source and knowledge cutoff. Later corrections append new facts and identify downstream impact rather than replacing history. AI assists through typed tools and cannot own the subject, grant authority, or invent an outcome.

19. Conformance Checklist

- ☐ Stable subject and version identities are explicit.
- ☐ Proposition ownership and external authority are named.
- ☐ Entry and exit criteria are testable.
- ☐ State, Recommendation, Decision, Intent, execution, and outcome remain distinct.
- ☐ Commands are authorized, version-guarded, and idempotent.
- ☐ Events describe completed facts.
- ☐ Evidence includes provenance, time, integrity, and correction links.
- ☐ Portals show freshness, uncertainty, and Outcome Unknown.
- ☐ AI uses typed tools and approved autonomy.
- ☐ Security, privacy, tenant, facility, and jurisdiction boundaries are enforced.
- ☐ Reconciliation precedes risky duplicate execution.
- ☐ Corrections preserve history.
- ☐ Tests prove invariants, security, failure, and recovery.
- ☐ Implementation status is not upgraded without executable repository evidence.

20. Status Vocabulary

- VERIFIED_IMPLEMENTED: executable source and the relevant behavior were inspected.
- IMPLEMENTED_WITH_GAP: executable behavior exists with a material governance or completeness gap.
- PARTIAL_IMPLEMENTATION: only part of the target capability is evidenced.
- REFERENCE_ARCHITECTURE: normative target behavior; no claim that it exists today.
- PLANNED_NOT_IMPLEMENTED: explicitly planned but not evidenced as executable.
- DOCUMENTATION_ONLY_CLAIM: documentation or a client contract claims behavior without verified backend evidence.
- CONFLICT_REQUIRES_CANONICALIZATION: sources disagree and no final owner has been established.
- REQUIRES_REPOSITORY_REVIEW: the necessary executable evidence has not yet been inspected.

Publication Chapter 17 — Export Customs and Legal Release

Metadata	Value
Publication Chapter	17
Canonical Lifecycle Stage	16
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE with explicit repository notes
Accountable capability	Regulatory / Compliance / external customs authority
Canon dependencies	CANON-001 through CANON-011
Constitutional dependencies	ART-001 through ART-016
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This stage exists to **prepare, submit, amend, and track export declarations and authoritative legal-release outcomes without treating platform readiness as customs approval.**

Its output is not merely a screen state or document download. It is a governed set of identified subjects, Relationships, Decisions, Intent, Commands, completed facts, external observations, evidence, and correction history.

- Reality or external outcome
- ≠ platform Observation
 - ≠ derived Perspective
 - ≠ AI Prediction
 - ≠ Recommendation
 - ≠ authorized Decision
 - ≠ execution Intent

1.1 Accountability

The accountable capability is **Regulatory / Compliance / external customs authority**. Supporting Portals, workflow engines, reports, search indexes, integration adapters, and AI Agents may coordinate or present the work, but they do not silently become owners of the underlying business truth.

1.2 Primary business outcomes

- a stable subject identity and exact source-version lineage;
- explicit authority and effective mandate;
- a testable lifecycle outcome;
- evidence sufficient for downstream use and later reconstruction;
- visible failure, partial completion, uncertainty, and correction;
- a completed-fact Event or qualified external observation;
- downstream activation without shared-database ownership transfer.

1.3 Non-goals

- treating a registered module as proof of complete implementation;

- treating a frontend route as proof of backend behavior;
- allowing an AI recommendation to become a binding Decision;
- inventing external-provider or government-authority success;
- overwriting history during amendment or correction;
- combining unrelated states into one generic status field.

2. Entry and Exit Criteria

2.1 Entry criteria

1. Every upstream subject reference resolves to a stable identifier.
2. The exact approved, accepted, issued, or observed version is known.
3. The initiating human, service, organization, or AI Agent is authenticated.
4. The represented principal and mandate are valid at command time.
5. Required Evidence exists, or an explicit authorized exception is linked.
6. Applicable Policy and jurisdiction versions are fixed.
7. Duplicate, stale-version, and replay checks have run.
8. Any conflicting active process is resolved, suspended, or explicitly related.

2.2 Exit criteria

1. The authoritative Domain or external authority records the outcome.
2. State and version changes are durable.
3. A completed-fact Event or qualified observation is available.
4. Downstream consumers receive stable subject, correlation, and causation references.
5. Unknown, partial, disputed, or failed outcomes remain visible.
6. Corrections preserve the original record and reason.
7. Metrics, audit evidence, and operational signals are emitted.
8. A qualified reviewer can reconstruct who knew what, when, and why.

3. Ubiquitous Language and Subject Model

Primary subject families:

Subject	Required interpretation
ExportDeclaration	independent identified subject or governed subordinate entity; ownership must be declared
DeclarationVersion	independent identified subject or governed subordinate entity; ownership must be declared
CustomsDocumentSet	independent identified subject or governed subordinate entity; ownership must be declared
BrokerMandate	independent identified subject or governed subordinate entity; ownership must be declared
CustomsSubmission	independent identified subject or governed subordinate entity; ownership must be declared
CustomsHold	independent identified subject or governed subordinate entity; ownership must be declared
CustomsInspection	independent identified subject or governed subordinate entity; ownership must be declared
LegalReleaseObservation	independent identified subject or governed subordinate entity; ownership must be declared

Reference aggregate:

```
ExportDeclaration {
  aggregateId
  tenantId
  organizationContext
  canonicalSubjectRefs[]
  sourceVersionRefs[]
  lifecycleState
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
```

```

policyRefs[]
authorityRef
jurisdictionRefs[]
createdAt
effectiveAt
updatedAt
version
correctionOf?
correlationId
}

```

3.1 Identity and relationship rules

- a display name, document number, provider reference, barcode, or URL is not automatically canonical identity;
- relationships such as custody, mandate, allocation, publication, reservation, or obligation are independent facts;
- cross-Domain references use stable IDs and qualified snapshots, not shared-table mutation;
- external identifiers retain issuer and namespace;
- merge, split, correction, and supersession preserve lineage.

3.2 Invariants

1. Submitted, approved, accepted, issued, or signed versions are immutable.
2. State changes occur through authorized Commands.
3. Completed-fact Events use past-tense semantics.
4. A Recommendation cannot self-promote to a Decision.
5. An external outcome retains its source authority.
6. Derived projections cannot become authoritative by repeated copying.
7. All writes are tenant-, organization-, facility-, and authority-scoped as applicable.
8. Retried writes are idempotent.
9. Outcome Unknown is not converted to success or failure by assumption.
10. Corrections reference the corrected fact and downstream impact.

4. Lifecycle State Model

4.1 State vocabulary

State	Meaning
PREPARATION	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
VALIDATION_PENDING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
READY_TO_SUBMIT	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
SUBMITTED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
ACCEPTED_FOR_PROCESSING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
HOLD	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
INSPECTION	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
AMENDMENT_REQUIRED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
RELEASED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
REJECTED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
CANCELLED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification

4.2 Reference state machine

```

stateDiagram-v2
    [*] --> PREPARATION
    PREPARATION --> VALIDATION_PENDING
    VALIDATION_PENDING --> READY_TO_SUBMIT

```

```

READY_TO_SUBMIT --> SUBMITTED
SUBMITTED --> ACCEPTED_FOR_PROCESSING
ACCEPTED_FOR_PROCESSING --> HOLD
HOLD --> INSPECTION
INSPECTION --> AMENDMENT_REQUIRED
AMENDMENT_REQUIRED --> RELEASED
RELEASED --> REJECTED
REJECTED --> CANCELLED

```

The diagram shows a readable primary path. A conformant implementation may include parallel approvals, amendment, suspension, timeout, appeal, rejection, partial completion, reconciliation, and correction paths, but it must preserve the same semantic distinctions.

4.3 Transition matrix

From	To	Minimum guard
PREPARATION	VALIDATION_PENDING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
VALIDATION_PENDING	READY_TO_SUBMIT	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
READY_TO_SUBMIT	SUBMITTED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
SUBMITTED	ACCEPTED_FOR_PROCESSING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
ACCEPTED_FOR_PROCESSING	HOLD	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
HOLD	INSPECTION	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
INSPECTION	AMENDMENT_REQUIRED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
AMENDMENT_REQUIRED	RELEASED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
RELEASED	REJECTED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
REJECTED	CANCELLED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard

4.4 Forbidden state behavior

- direct arbitrary assignment from a Portal request;
- using one status to represent lifecycle, approval, payment, document, provider, and dispute state simultaneously;
- deleting terminal history to reopen a process;
- inferring completion from elapsed time or an AI prediction;
- moving past a mandatory gate because a downstream system already acted.

5. Decision, Policy, and Authority

Material Decisions in this stage include readiness, eligibility, approval, exception, amendment, correction, and downstream activation decisions.

Reference Decision:

```

Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
}

```

```

evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Decision rules

- the Decision references exact source versions;
 - authority and delegation are evaluated at Decision time;
 - a Policy version and jurisdiction context are recorded;
 - mandatory-gate failure blocks progression unless a lawful exception exists;
 - dissent, uncertainty, and missing Evidence remain visible;
 - Decision correction does not erase the original Decision;
 - AI may prepare a briefing, but cannot manufacture authority.
-

6. Evidence, Documents, and Knowledge Cutoff

Every binding or high-impact action requires an Evidence Set containing:

- evidence identity and integrity hash where required;
- issuing, observing, or signing party;
- observed time, received time, and effective-time interval;
- subject and source-version relationships;
- qualification, confidence, and limitations;
- contradiction, retraction, supersession, and correction links;
- jurisdiction, classification, retention, and residency;
- chain of custody for physical or high-integrity evidence;
- the knowledge cutoff used by the Decision.

A document is not automatically truth. It is an Artifact containing Claims. Its legal or operational effect depends on issuer, mandate, signature, version, applicable rules, and external authority.

7. Commands, Events, and Queries

7.1 Commands

- PrepareExportDeclaration
- ValidateExportDeclaration
- SubmitExportDeclaration
- RecordCustomsResponse
- AmendExportDeclaration
- RecordCustomsHold
- RecordExportRelease

Each Command shall contain an idempotency key, actor identity, represented principal, tenant and organization context, correlation ID, expected version where relevant, reason for privileged action, and exact target subject.

7.2 Events

- ExportDeclarationPrepared
- ExportDeclarationSubmitted
- CustomsHoldRecorded
- CustomsInspectionRequested
- ExportDeclarationAmended
- ExportReleased
- ExportRejected

Events describe completed facts or explicitly qualified observations. A request, proposed action, timer, or prediction is not a completed-fact Event.

7.3 Queries and perspectives

- `GetExportDeclaration`
- `ListExportDeclarationRecords`
- `GetExportDeclarationTimeline`
- `GetExportDeclarationEvidence`
- `GetExportDeclarationDecisionPackage`
- `GetExportDeclarationExceptions`
- `GetExportDeclarationAuditTrail`
- `GetExportDeclarationCurrentPerspective`

Every query response states data freshness, source cutoff, version, and whether values are authoritative, observed, derived, predicted, or disputed.

8. Workflow, Timeout, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Evidence and policy gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Create Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and available compensation]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Trigger downstream stage]
```

Required workflow behavior:

- durable process and correlation identity;
 - persisted approvals and timers;
 - bounded retries and retry budgets;
 - idempotent command submission;
 - safe restart after deployment or outage;
 - compensation that records new Intent instead of deleting history;
 - visible stalled, orphaned, and manually intervened processes;
 - operator access to source responses and downstream impact;
 - reconciliation before repeating an external financial, legal, or physical action.
-

9. Target API Contract

Reference surface:

```
POST /api/v1/export_customs_and_legal_release/commands
GET  /api/v1/export_customs_and_legal_release/{id}
GET  /api/v1/export_customs_and_legal_release/{id}/timeline
GET  /api/v1/export_customs_and_legal_release/{id}/evidence
GET  /api/v1/export_customs_and_legal_release/{id}/decisions
POST /api/v1/export_customs_and_legal_release/{id}/exceptions
POST /api/v1/export_customs_and_legal_release/{id}/corrections
```

These are reference endpoints, not claims that the exact routes exist.

Contract requirements:

- typed request and response schemas;
 - action-level authorization;
 - trusted authentication context for tenant identity;
 - idempotency for writes;
 - optimistic concurrency or equivalent aggregate guard;
 - RFC problem-details or governed equivalent;
 - stable pagination and sorting;
 - versioning and deprecation policy;
 - trace, correlation, and causation propagation;
 - explicit Outcome Unknown response when external completion is unresolved.
-

10. Target Persistence and Data Ownership

Reference table families:

Table family	Purpose
stage_aggregate	authoritative lifecycle subject
stage_version	immutable approved/submitted/issued versions
stage_decision	binding Decisions and rationale
stage_evidence_link	evidence and provenance relationships
stage_event_outbox	transactional Event publication
stage_idempotency	duplicate-write protection
stage_exception	exception, claim, hold, or dispute
stage_correction	correction and supersession lineage
stage_audit	privileged action evidence

Data rules:

- the owning service controls its schema and migrations;
- foreign keys stay inside an ownership boundary;
- cross-Domain relationships use stable IDs or contracts;
- effective-time and recorded-time are separate when needed;
- binding versions are immutable;
- indexes support tenant, owner, state, correlation, external reference, and effective time;
- retention, legal hold, privacy, residency, backup, and restore are documented;
- analytics, search, reports, and vector stores are projections.

11. Portal and Experience Architecture

Required Portal behavior:

- display exact version, owner, state, and last-updated time;
- distinguish authoritative, observed, derived, predicted, and disputed values;
- expose required action and authority;
- show loading, empty, partial, stale, error, timeout, Outcome Unknown, and permission-denied states;
- provide accessible keyboard and screen-reader behavior;
- handle locale, currency, units, dates, and RTL/LTR correctly;
- preserve evidence and download audit;
- never trust a browser-provided tenant header as authority;
- prevent double submission while still supporting safe retry;
- show correction and supersession history.

Operator surfaces additionally expose correlation, source responses, retry budget, replay status, compensation options, and affected downstream subjects.

12. AI Participation and Tool Governance

Permitted AI tasks:

- check document completeness
- assist classification
- triage customs risk
- summarize authority messages

AI may also summarize evidence, detect gaps, recommend next actions, and draft human-readable explanations.

AI shall not:

- own or directly mutate Domain databases;
- fabricate external-provider, carrier, bank, customs, inspection, or delivery outcomes;
- grant itself permission or mandate;
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Every Agent Run records prompt/model/tool versions, context sources, policy evaluation, approvals, tool calls, structured outputs, uncertainty, final observed outcome, and recovery actions.

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15. Failure Modes and Recovery

Representative failures:

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- duplicate submission or replay;
- external timeout after possible success;
- contradictory source observations;
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1. Query the authoritative source before repeating an external action.

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 4. Publish correction Events when distributed facts change.
 5. Quarantine high-impact ambiguity.
 6. Escalate when authority or legal effect is unclear.
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-

16. Testing and Conformance

Required tests:

- aggregate invariant and transition tests;
 - authorization, mandate, separation-of-duties, and tenant-isolation tests;
 - idempotency and optimistic-concurrency tests;
 - migration, constraint, index, and rollback tests;
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 - provider callback verification and Outcome Unknown tests;
 - Portal E2E for success, partial, stale, denied, and error journeys;
 - accessibility, locale, currency, unit, and RTL tests;
 - security adversarial and data-exfiltration tests;
 - AI evaluation, prompt-injection, policy-bypass, and unauthorized-tool tests;
 - historical reconstruction and correction tests;
 - performance, capacity, disaster-recovery, and operational exercise evidence.
-

17. Repository Review Boundary and Previously Inspected Evidence

- Regulatory, provider, and shipping capabilities are registered.
- The external customs authority owns legal release; platform records are qualified observations.
- Adapters and route behavior require repository verification.

Area	Classification for this chapter
Target aggregate, API, tables, state model, and workflow Registered supporting modules	REFERENCE_ARCHITECTURE DOCUMENTATION_ONLY_CLAIM until executable files are inspected
Concrete behavior already established in earlier repository-grounded chapters	retain the original classification and evidence
Exact integration paths not inspected for this stage	REQUIRES_REPOSITORY_REVIEW
External authority or provider outcome	qualified external observation, never platform-owned truth

No statement in this chapter upgrades an implementation status merely because a module name, frontend route, README, plan, or database catalog entry exists.

18. Worked Scenario

An upstream governed commitment enters the stage with exact subject and version references. The initiating principal is authenticated, and its mandate is checked at command time. Evidence is gathered and evaluated against the applicable policy version and jurisdiction.

A recommendation may be produced, but an authorized Decision remains separate. The Domain accepts an idempotent Command and records state plus an outbox entry atomically. When an external party participates, a timeout creates Outcome Unknown; the workflow queries the external source or waits for a verified callback instead of repeating execution blindly.

On success or failure, the system records a completed fact, publishes a stable Event, updates downstream projections, and preserves source and knowledge cutoff. Later corrections append new facts and identify downstream impact rather than replacing history. AI assists through typed tools and cannot own the subject, grant authority, or invent an outcome.

19. Conformance Checklist

- ☐ Stable subject and version identities are explicit.
 - ☐ Proposition ownership and external authority are named.
 - ☐ Entry and exit criteria are testable.
 - ☐ State, Recommendation, Decision, Intent, execution, and outcome remain distinct.
 - ☐ Commands are authorized, version-guarded, and idempotent.
 - ☐ Events describe completed facts.
 - ☐ Evidence includes provenance, time, integrity, and correction links.
 - ☐ Portals show freshness, uncertainty, and Outcome Unknown.
 - ☐ AI uses typed tools and approved autonomy.
 - ☐ Security, privacy, tenant, facility, and jurisdiction boundaries are enforced.
 - ☐ Reconciliation precedes risky duplicate execution.
 - ☐ Corrections preserve history.
 - ☐ Tests prove invariants, security, failure, and recovery.
 - ☐ Implementation status is not upgraded without executable repository evidence.
-

20. Status Vocabulary

- VERIFIED_IMPLEMENTED: executable source and the relevant behavior were inspected.
- IMPLEMENTED_WITH_GAP: executable behavior exists with a material governance or completeness gap.
- PARTIAL_IMPLEMENTATION: only part of the target capability is evidenced.
- REFERENCE_ARCHITECTURE: normative target behavior; no claim that it exists today.
- PLANNED_NOT_IMPLEMENTED: explicitly planned but not evidenced as executable.
- DOCUMENTATION_ONLY_CLAIM: documentation or a client contract claims behavior without verified backend evidence.
- CONFLICT_REQUIRES_CANONICALIZATION: sources disagree and no final owner has been established.
- REQUIRES_REPOSITORY_REVIEW: the necessary executable evidence has not yet been inspected.

Publication Chapter 18 — Transport Planning, Booking, and Capacity

Metadata	Value
Publication Chapter	18
Canonical Lifecycle Stage	17
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE with explicit repository notes
Accountable capability	Shipping / Provider / external carrier
Canon dependencies	CANON-001 through CANON-011
Constitutional dependencies	ART-001 through ART-016
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This stage exists to **convert a shipment requirement into a confirmed transport arrangement with route, mode, provider, carrier, capacity, cost, and schedule.**

Its output is not merely a screen state or document download. It is a governed set of identified subjects, Relationships, Decisions, Intent, Commands, completed facts, external observations, evidence, and correction history.

- Reality or external outcome
- ≠ platform Observation
 - ≠ derived Perspective
 - ≠ AI Prediction
 - ≠ Recommendation
 - ≠ authorized Decision
 - ≠ execution Intent

1.1 Accountability

The accountable capability is **Shipping / Provider / external carrier**. Supporting Portals, workflow engines, reports, search indexes, integration adapters, and AI Agents may coordinate or present the work, but they do not silently become owners of the underlying business truth.

1.2 Primary business outcomes

- a stable subject identity and exact source-version lineage;
- explicit authority and effective mandate;
- a testable lifecycle outcome;
- evidence sufficient for downstream use and later reconstruction;
- visible failure, partial completion, uncertainty, and correction;
- a completed-fact Event or qualified external observation;
- downstream activation without shared-database ownership transfer.

1.3 Non-goals

- treating a registered module as proof of complete implementation;

- treating a frontend route as proof of backend behavior;
- allowing an AI recommendation to become a binding Decision;
- inventing external-provider or government-authority success;
- overwriting history during amendment or correction;
- combining unrelated states into one generic status field.

2. Entry and Exit Criteria

2.1 Entry criteria

1. Every upstream subject reference resolves to a stable identifier.
2. The exact approved, accepted, issued, or observed version is known.
3. The initiating human, service, organization, or AI Agent is authenticated.
4. The represented principal and mandate are valid at command time.
5. Required Evidence exists, or an explicit authorized exception is linked.
6. Applicable Policy and jurisdiction versions are fixed.
7. Duplicate, stale-version, and replay checks have run.
8. Any conflicting active process is resolved, suspended, or explicitly related.

2.2 Exit criteria

1. The authoritative Domain or external authority records the outcome.
2. State and version changes are durable.
3. A completed-fact Event or qualified observation is available.
4. Downstream consumers receive stable subject, correlation, and causation references.
5. Unknown, partial, disputed, or failed outcomes remain visible.
6. Corrections preserve the original record and reason.
7. Metrics, audit evidence, and operational signals are emitted.
8. A qualified reviewer can reconstruct who knew what, when, and why.

3. Ubiquitous Language and Subject Model

Primary subject families:

Subject	Required interpretation
TransportRequirement	independent identified subject or governed subordinate entity; ownership must be declared
RouteOption	independent identified subject or governed subordinate entity; ownership must be declared
TransportQuote	independent identified subject or governed subordinate entity; ownership must be declared
CarrierSelectionDecision	independent identified subject or governed subordinate entity; ownership must be declared
Booking	independent identified subject or governed subordinate entity; ownership must be declared
BookingVersion	independent identified subject or governed subordinate entity; ownership must be declared
CapacityConfirmation	independent identified subject or governed subordinate entity; ownership must be declared
ShipmentPlan	independent identified subject or governed subordinate entity; ownership must be declared

Reference aggregate:

```
TransportRequirement {
  aggregateId
  tenantId
  organizationContext
  canonicalSubjectRefs[]
  sourceVersionRefs[]
  lifecycleState
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
```

```

policyRefs[]
authorityRef
jurisdictionRefs[]
createdAt
effectiveAt
updatedAt
version
correctionOf?
correlationId
}

```

3.1 Identity and relationship rules

- a display name, document number, provider reference, barcode, or URL is not automatically canonical identity;
- relationships such as custody, mandate, allocation, publication, reservation, or obligation are independent facts;
- cross-Domain references use stable IDs and qualified snapshots, not shared-table mutation;
- external identifiers retain issuer and namespace;
- merge, split, correction, and supersession preserve lineage.

3.2 Invariants

1. Submitted, approved, accepted, issued, or signed versions are immutable.
2. State changes occur through authorized Commands.
3. Completed-fact Events use past-tense semantics.
4. A Recommendation cannot self-promote to a Decision.
5. An external outcome retains its source authority.
6. Derived projections cannot become authoritative by repeated copying.
7. All writes are tenant-, organization-, facility-, and authority-scoped as applicable.
8. Retried writes are idempotent.
9. Outcome Unknown is not converted to success or failure by assumption.
10. Corrections reference the corrected fact and downstream impact.

4. Lifecycle State Model

4.1 State vocabulary

State	Meaning
REQUIREMENT_READY	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
OPTIONS_REQUESTED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
OPTIONS_RECEIVED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
SELECTION_PENDING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
BOOKING_REQUESTED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
CONFIRMATION_PENDING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
CONFIRMED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
AMENDMENT_PENDING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
CANCELLED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
FAILED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification

4.2 Reference state machine

```

stateDiagram-v2
    [*] --> REQUIREMENT_READY
    REQUIREMENT_READY --> OPTIONS_REQUESTED
    OPTIONS_REQUESTED --> OPTIONS_RECEIVED
    OPTIONS_RECEIVED --> SELECTION_PENDING
    SELECTION_PENDING --> BOOKING_REQUESTED

```

```

BOOKING_REQUESTED --> CONFIRMATION_PENDING
CONFIRMATION_PENDING --> CONFIRMED
CONFIRMED --> AMENDMENT_PENDING
AMENDMENT_PENDING --> CANCELLED
CANCELLED --> FAILED

```

The diagram shows a readable primary path. A conformant implementation may include parallel approvals, amendment, suspension, timeout, appeal, rejection, partial completion, reconciliation, and correction paths, but it must preserve the same semantic distinctions.

4.3 Transition matrix

From	To	Minimum guard
REQUIREMENT_READY	OPTIONS_REQUESTED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
OPTIONS_REQUESTED	OPTIONS_RECEIVED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
OPTIONS_RECEIVED	SELECTION_PENDING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
SELECTION_PENDING	BOOKING_REQUESTED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
BOOKING_REQUESTED	CONFIRMATION_PENDING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
CONFIRMATION_PENDING	CONFIRMED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
CONFIRMED	AMENDMENT_PENDING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
AMENDMENT_PENDING	CANCELLED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
CANCELLED	FAILED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard

4.4 Forbidden state behavior

- direct arbitrary assignment from a Portal request;
- using one status to represent lifecycle, approval, payment, document, provider, and dispute state simultaneously;
- deleting terminal history to reopen a process;
- inferring completion from elapsed time or an AI prediction;
- moving past a mandatory gate because a downstream system already acted.

5. Decision, Policy, and Authority

Material Decisions in this stage include readiness, eligibility, approval, exception, amendment, correction, and downstream activation decisions.

Reference Decision:

```

Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
}

```

```

outcome
decidedAt
effectiveAt
correctionOf?
}

```

Decision rules

- the Decision references exact source versions;
 - authority and delegation are evaluated at Decision time;
 - a Policy version and jurisdiction context are recorded;
 - mandatory-gate failure blocks progression unless a lawful exception exists;
 - dissent, uncertainty, and missing Evidence remain visible;
 - Decision correction does not erase the original Decision;
 - AI may prepare a briefing, but cannot manufacture authority.
-

6. Evidence, Documents, and Knowledge Cutoff

Every binding or high-impact action requires an Evidence Set containing:

- evidence identity and integrity hash where required;
- issuing, observing, or signing party;
- observed time, received time, and effective-time interval;
- subject and source-version relationships;
- qualification, confidence, and limitations;
- contradiction, retraction, supersession, and correction links;
- jurisdiction, classification, retention, and residency;
- chain of custody for physical or high-integrity evidence;
- the knowledge cutoff used by the Decision.

A document is not automatically truth. It is an Artifact containing Claims. Its legal or operational effect depends on issuer, mandate, signature, version, applicable rules, and external authority.

7. Commands, Events, and Queries

7.1 Commands

- CreateTransportRequirement
- RequestTransportOptions
- SelectTransportOption
- RequestBooking
- ConfirmBooking
- AmendBooking
- CancelBooking

Each Command shall contain an idempotency key, actor identity, represented principal, tenant and organization context, correlation ID, expected version where relevant, reason for privileged action, and exact target subject.

7.2 Events

- TransportRequirementCreated
- TransportOptionsReceived
- TransportOptionSelected
- BookingRequested
- BookingConfirmed
- BookingAmended
- BookingCancelled
- BookingFailed

Events describe completed facts or explicitly qualified observations. A request, proposed action, timer, or prediction is not a completed-fact Event.

7.3 Queries and perspectives

- GetTransportRequirement
- ListTransportRequirementRecords
- GetTransportRequirementTimeline
- GetTransportRequirementEvidence

- GetTransportRequirementDecisionPackage
- GetTransportRequirementExceptions
- GetTransportRequirementAuditTrail
- GetTransportRequirementCurrentPerspective

Every query response states data freshness, source cutoff, version, and whether values are authoritative, observed, derived, predicted, or disputed.

8. Workflow, Timeout, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C[Evidence and policy gates pass?]
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Create Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G[Outcome known?]
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and available compensation]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Trigger downstream stage]

```

Required workflow behavior:

- durable process and correlation identity;
 - persisted approvals and timers;
 - bounded retries and retry budgets;
 - idempotent command submission;
 - safe restart after deployment or outage;
 - compensation that records new Intent instead of deleting history;
 - visible stalled, orphaned, and manually intervened processes;
 - operator access to source responses and downstream impact;
 - reconciliation before repeating an external financial, legal, or physical action.
-

9. Target API Contract

Reference surface:

```

POST /api/v1/transport_planning_booking_and_capacity/commands
GET  /api/v1/transport_planning_booking_and_capacity/{id}
GET  /api/v1/transport_planning_booking_and_capacity/{id}/timeline
GET  /api/v1/transport_planning_booking_and_capacity/{id}/evidence
GET  /api/v1/transport_planning_booking_and_capacity/{id}/decisions
POST /api/v1/transport_planning_booking_and_capacity/{id}/exceptions
POST /api/v1/transport_planning_booking_and_capacity/{id}/corrections

```

These are reference endpoints, not claims that the exact routes exist.

Contract requirements:

- typed request and response schemas;
 - action-level authorization;
 - trusted authentication context for tenant identity;
 - idempotency for writes;
 - optimistic concurrency or equivalent aggregate guard;
 - RFC problem-details or governed equivalent;
 - stable pagination and sorting;
 - versioning and deprecation policy;
 - trace, correlation, and causation propagation;
 - explicit Outcome Unknown response when external completion is unresolved.
-

10. Target Persistence and Data Ownership

Reference table families:

Table family	Purpose
stage_aggregate	authoritative lifecycle subject
stage_version	immutable approved/submitted/issued versions
stage_decision	binding Decisions and rationale
stage_evidence_link	evidence and provenance relationships
stage_event_outbox	transactional Event publication
stage_idempotency	duplicate-write protection
stage_exception	exception, claim, hold, or dispute
stage_correction	correction and supersession lineage
stage_audit	privileged action evidence

Data rules:

- the owning service controls its schema and migrations;
- foreign keys stay inside an ownership boundary;
- cross-Domain relationships use stable IDs or contracts;
- effective-time and recorded-time are separate when needed;
- binding versions are immutable;
- indexes support tenant, owner, state, correlation, external reference, and effective time;
- retention, legal hold, privacy, residency, backup, and restore are documented;
- analytics, search, reports, and vector stores are projections.

11. Portal and Experience Architecture

Required Portal behavior:

- display exact version, owner, state, and last-updated time;
- distinguish authoritative, observed, derived, predicted, and disputed values;
- expose required action and authority;
- show loading, empty, partial, stale, error, timeout, Outcome Unknown, and permission-denied states;
- provide accessible keyboard and screen-reader behavior;
- handle locale, currency, units, dates, and RTL/LTR correctly;
- preserve evidence and download audit;
- never trust a browser-provided tenant header as authority;
- prevent double submission while still supporting safe retry;
- show correction and supersession history.

Operator surfaces additionally expose correlation, source responses, retry budget, replay status, compensation options, and affected downstream subjects.

12. AI Participation and Tool Governance

Permitted AI tasks:

- compare routes
- predict ETA and delay
- normalize transport cost
- recommend capacity options

AI may also summarize evidence, detect gaps, recommend next actions, and draft human-readable explanations.

AI shall not:

- own or directly mutate Domain databases;
- fabricate external-provider, carrier, bank, customs, inspection, or delivery outcomes;
- grant itself permission or mandate;
- silently change an approved or issued version;
- suppress uncertainty, contradictions, policy failures, or human dissent;
- execute high-impact actions outside typed tools and approval policy;
- create shadow records for canonical subjects.

Reference automation:

Task	Maximum default
extraction and classification	A2, automatic with reviewable evidence
anomaly detection and recommendation	A2/A3 according to risk
binding commercial, legal, financial, quality, or custody Decision	A0/A1, human authority required

Task	Maximum default
external execution	typed tool, policy, approval, idempotency, and outcome observation
correction of authoritative truth	governed human-authorized workflow

Every Agent Run records prompt/model/tool versions, context sources, policy evaluation, approvals, tool calls, structured outputs, uncertainty, final observed outcome, and recovery actions.

13. Security, Privacy, and Trust Boundaries

Minimum controls:

- authenticated human, service, organization, device, and AI identities;
- represented principal and effective mandate validation;
- least privilege, ABAC/ReBAC where appropriate, and separation of duties;
- tenant, organization, facility, channel, and jurisdiction isolation;
- integrity protection for binding documents and Evidence;
- external callback signature and replay verification;
- confidential handling of commercial, personal, financial, transport, and regulatory data;
- secret isolation and no sensitive values in logs or prompts;
- privileged-action audit and emergency-access review;
- threat models for impersonation, tampering, replay, confused deputy, fraud, exfiltration, and prompt injection.

14. Reliability, SLOs, and Operations

Reference objectives:

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by the business deadline of the stage

Operational requirements include dashboards, alerts, runbooks, ownership, dependency health, retry budgets, DLQ/replay controls, Event outbox/inbox, workflow restart, reconciliation, capacity, backup/restore, incident command, and post-incident correction.

15. Failure Modes and Recovery

Representative failures:

- stale or wrong source version;
- duplicate submission or replay;
- external timeout after possible success;
- contradictory source observations;
- lost Event after committed state;
- approval or mandate revoked during execution;
- downstream action performed before a mandatory gate;
- partial completion across multiple subjects;
- evidence later retracted or corrected;
- tenant, facility, or jurisdiction mismatch.

Recovery principles:

1. Query the authoritative source before repeating an external action.
2. Preserve partial success.
3. Use compensating Intent instead of destructive history rewrite.
4. Publish correction Events when distributed facts change.
5. Quarantine high-impact ambiguity.

- 6. Escalate when authority or legal effect is unclear.
- 7. Record business impact, affected subjects, and final reconciliation.

16. Testing and Conformance

Required tests:

- aggregate invariant and transition tests;
- authorization, mandate, separation-of-duties, and tenant-isolation tests;
- idempotency and optimistic-concurrency tests;
- migration, constraint, index, and rollback tests;
- outbox/inbox, ordering, duplicate, replay, and DLQ tests;
- workflow restart, timeout, compensation, and manual-intervention tests;
- provider callback verification and Outcome Unknown tests;
- Portal E2E for success, partial, stale, denied, and error journeys;
- accessibility, locale, currency, unit, and RTL tests;
- security adversarial and data-exfiltration tests;
- AI evaluation, prompt-injection, policy-bypass, and unauthorized-tool tests;
- historical reconstruction and correction tests;
- performance, capacity, disaster-recovery, and operational exercise evidence.

17. Repository Review Boundary and Previously Inspected Evidence

- Shipping, provider, workflow, and event capabilities are registered.
- Carrier integrations and booking schemas require direct evidence before implementation status can be upgraded.

Area	Classification for this chapter
Target aggregate, API, tables, state model, and workflow Registered supporting modules	REFERENCE_ARCHITECTURE DOCUMENTATION_ONLY_CLAIM until executable files are inspected
Concrete behavior already established in earlier repository-grounded chapters	retain the original classification and evidence
Exact integration paths not inspected for this stage External authority or provider outcome	REQUIRES_REPOSITORY_REVIEW qualified external observation, never platform-owned truth

No statement in this chapter upgrades an implementation status merely because a module name, frontend route, README, plan, or database catalog entry exists.

18. Worked Scenario

An upstream governed commitment enters the stage with exact subject and version references. The initiating principal is authenticated, and its mandate is checked at command time. Evidence is gathered and evaluated against the applicable policy version and jurisdiction.

A recommendation may be produced, but an authorized Decision remains separate. The Domain accepts an idempotent Command and records state plus an outbox entry atomically. When an external party participates, a timeout creates Outcome Unknown; the workflow queries the external source or waits for a verified callback instead of repeating execution blindly.

On success or failure, the system records a completed fact, publishes a stable Event, updates downstream projections, and preserves source and knowledge cutoff. Later corrections append new facts and identify downstream impact rather than replacing history. AI assists through typed tools and cannot own the subject, grant authority, or invent an outcome.

19. Conformance Checklist

- ☐ Stable subject and version identities are explicit.
- ☐ Proposition ownership and external authority are named.
- ☐ Entry and exit criteria are testable.
- ☐ State, Recommendation, Decision, Intent, execution, and outcome remain distinct.

- ☐ Commands are authorized, version-guarded, and idempotent.
 - ☐ Events describe completed facts.
 - ☐ Evidence includes provenance, time, integrity, and correction links.
 - ☐ Portals show freshness, uncertainty, and Outcome Unknown.
 - ☐ AI uses typed tools and approved autonomy.
 - ☐ Security, privacy, tenant, facility, and jurisdiction boundaries are enforced.
 - ☐ Reconciliation precedes risky duplicate execution.
 - ☐ Corrections preserve history.
 - ☐ Tests prove invariants, security, failure, and recovery.
 - ☐ Implementation status is not upgraded without executable repository evidence.
-

20. Status Vocabulary

- VERIFIED_IMPLEMENTED: executable source and the relevant behavior were inspected.
- IMPLEMENTED_WITH_GAP: executable behavior exists with a material governance or completeness gap.
- PARTIAL_IMPLEMENTATION: only part of the target capability is evidenced.
- REFERENCE_ARCHITECTURE: normative target behavior; no claim that it exists today.
- PLANNED_NOT_IMPLEMENTED: explicitly planned but not evidenced as executable.
- DOCUMENTATION_ONLY_CLAIM: documentation or a client contract claims behavior without verified backend evidence.
- CONFLICT_REQUIRES_CANONICALIZATION: sources disagree and no final owner has been established.
- REQUIRES_REPOSITORY_REVIEW: the necessary executable evidence has not yet been inspected.

Publication Chapter 19 — Pickup, Custody Transfer, and Handover

Metadata	Value
Publication Chapter	19
Canonical Lifecycle Stage	18
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE with explicit repository notes
Accountable capability	Shipping / Provider / handing and receiving parties
Canon dependencies	CANON-001 through CANON-011
Constitutional dependencies	ART-001 through ART-016
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This stage exists to **evidence physical handover of identified cargo and establish custody without conflating custody, possession, risk, and ownership.**

Its output is not merely a screen state or document download. It is a governed set of identified subjects, Relationships, Decisions, Intent, Commands, completed facts, external observations, evidence, and correction history.

- Reality or external outcome
- ≠ platform Observation
 - ≠ derived Perspective
 - ≠ AI Prediction
 - ≠ Recommendation
 - ≠ authorized Decision
 - ≠ execution Intent

1.1 Accountability

The accountable capability is **Shipping / Provider / handing and receiving parties.** Supporting Portals, workflow engines, reports, search indexes, integration adapters, and AI Agents may coordinate or present the work, but they do not silently become owners of the underlying business truth.

1.2 Primary business outcomes

- a stable subject identity and exact source-version lineage;
- explicit authority and effective mandate;
- a testable lifecycle outcome;
- evidence sufficient for downstream use and later reconstruction;
- visible failure, partial completion, uncertainty, and correction;
- a completed-fact Event or qualified external observation;
- downstream activation without shared-database ownership transfer.

1.3 Non-goals

- treating a registered module as proof of complete implementation;

- treating a frontend route as proof of backend behavior;
- allowing an AI recommendation to become a binding Decision;
- inventing external-provider or government-authority success;
- overwriting history during amendment or correction;
- combining unrelated states into one generic status field.

2. Entry and Exit Criteria

2.1 Entry criteria

1. Every upstream subject reference resolves to a stable identifier.
2. The exact approved, accepted, issued, or observed version is known.
3. The initiating human, service, organization, or AI Agent is authenticated.
4. The represented principal and mandate are valid at command time.
5. Required Evidence exists, or an explicit authorized exception is linked.
6. Applicable Policy and jurisdiction versions are fixed.
7. Duplicate, stale-version, and replay checks have run.
8. Any conflicting active process is resolved, suspended, or explicitly related.

2.2 Exit criteria

1. The authoritative Domain or external authority records the outcome.
2. State and version changes are durable.
3. A completed-fact Event or qualified observation is available.
4. Downstream consumers receive stable subject, correlation, and causation references.
5. Unknown, partial, disputed, or failed outcomes remain visible.
6. Corrections preserve the original record and reason.
7. Metrics, audit evidence, and operational signals are emitted.
8. A qualified reviewer can reconstruct who knew what, when, and why.

3. Ubiquitous Language and Subject Model

Primary subject families:

Subject	Required interpretation
PickupAppointment	independent identified subject or governed subordinate entity; ownership must be declared
PickupInstruction	independent identified subject or governed subordinate entity; ownership must be declared
CargoHandover	independent identified subject or governed subordinate entity; ownership must be declared
CustodyRelationship	independent identified subject or governed subordinate entity; ownership must be declared
SealRecord	independent identified subject or governed subordinate entity; ownership must be declared
PackageCountObservation	independent identified subject or governed subordinate entity; ownership must be declared
PickupDiscrepancy	independent identified subject or governed subordinate entity; ownership must be declared
ProofOfPickup	independent identified subject or governed subordinate entity; ownership must be declared

Reference aggregate:

```
PickupAppointment {
  aggregateId
  tenantId
  organizationContext
  canonicalSubjectRefs[]
  sourceVersionRefs[]
  lifecycleState
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
```

```

policyRefs[]
authorityRef
jurisdictionRefs[]
createdAt
effectiveAt
updatedAt
version
correctionOf?
correlationId
}

```

3.1 Identity and relationship rules

- a display name, document number, provider reference, barcode, or URL is not automatically canonical identity;
- relationships such as custody, mandate, allocation, publication, reservation, or obligation are independent facts;
- cross-Domain references use stable IDs and qualified snapshots, not shared-table mutation;
- external identifiers retain issuer and namespace;
- merge, split, correction, and supersession preserve lineage.

3.2 Invariants

1. Submitted, approved, accepted, issued, or signed versions are immutable.
2. State changes occur through authorized Commands.
3. Completed-fact Events use past-tense semantics.
4. A Recommendation cannot self-promote to a Decision.
5. An external outcome retains its source authority.
6. Derived projections cannot become authoritative by repeated copying.
7. All writes are tenant-, organization-, facility-, and authority-scoped as applicable.
8. Retried writes are idempotent.
9. Outcome Unknown is not converted to success or failure by assumption.
10. Corrections reference the corrected fact and downstream impact.

4. Lifecycle State Model

4.1 State vocabulary

State	Meaning
SCHEDULED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
ARRIVED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
IDENTITY_CHECK	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
LOADING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
DISCREPANCY	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
HANDED_OVER	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
REFUSED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
FAILED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
CORRECTED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
CLOSED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification

4.2 Reference state machine

```

stateDiagram-v2
    [*] --> SCHEDULED
    SCHEDULED --> ARRIVED
    ARRIVED --> IDENTITY_CHECK
    IDENTITY_CHECK --> LOADING
    LOADING --> DISCREPANCY

```

```

DISCREPANCY --> HANDED_OVER
HANDED_OVER --> REFUSED
REFUSED --> FAILED
FAILED --> CORRECTED
CORRECTED --> CLOSED

```

The diagram shows a readable primary path. A conformant implementation may include parallel approvals, amendment, suspension, timeout, appeal, rejection, partial completion, reconciliation, and correction paths, but it must preserve the same semantic distinctions.

4.3 Transition matrix

From	To	Minimum guard
SCHEDULED	ARRIVED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
ARRIVED	IDENTITY_CHECK	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
IDENTITY_CHECK	LOADING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
LOADING	DISCREPANCY	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
DISCREPANCY	HANDED_OVER	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
HANDED_OVER	REFUSED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
REFUSED	FAILED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
FAILED	CORRECTED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
CORRECTED	CLOSED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard

4.4 Forbidden state behavior

- direct arbitrary assignment from a Portal request;
- using one status to represent lifecycle, approval, payment, document, provider, and dispute state simultaneously;
- deleting terminal history to reopen a process;
- inferring completion from elapsed time or an AI prediction;
- moving past a mandatory gate because a downstream system already acted.

5. Decision, Policy, and Authority

Material Decisions in this stage include readiness, eligibility, approval, exception, amendment, correction, and downstream activation decisions.

Reference Decision:

```

Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
}

```

```

outcome
decidedAt
effectiveAt
correctionOf?
}

```

Decision rules

- the Decision references exact source versions;
 - authority and delegation are evaluated at Decision time;
 - a Policy version and jurisdiction context are recorded;
 - mandatory-gate failure blocks progression unless a lawful exception exists;
 - dissent, uncertainty, and missing Evidence remain visible;
 - Decision correction does not erase the original Decision;
 - AI may prepare a briefing, but cannot manufacture authority.
-

6. Evidence, Documents, and Knowledge Cutoff

Every binding or high-impact action requires an Evidence Set containing:

- evidence identity and integrity hash where required;
- issuing, observing, or signing party;
- observed time, received time, and effective-time interval;
- subject and source-version relationships;
- qualification, confidence, and limitations;
- contradiction, retraction, supersession, and correction links;
- jurisdiction, classification, retention, and residency;
- chain of custody for physical or high-integrity evidence;
- the knowledge cutoff used by the Decision.

A document is not automatically truth. It is an Artifact containing Claims. Its legal or operational effect depends on issuer, mandate, signature, version, applicable rules, and external authority.

7. Commands, Events, and Queries

7.1 Commands

- SchedulePickup
- ConfirmPickupArrival
- VerifyCargoIdentity
- RecordPackageCount
- RecordSeal
- AcceptCustody
- RefusePickup
- CorrectHandoverRecord

Each Command shall contain an idempotency key, actor identity, represented principal, tenant and organization context, correlation ID, expected version where relevant, reason for privileged action, and exact target subject.

7.2 Events

- PickupScheduled
- PickupVehicleArrived
- CargoIdentityVerified
- PickupDiscrepancyObserved
- CustodyTransferred
- PickupRefused
- ProofOfPickupIssued

Events describe completed facts or explicitly qualified observations. A request, proposed action, timer, or prediction is not a completed-fact Event.

7.3 Queries and perspectives

- GetPickupAppointment
- ListPickupAppointmentRecords
- GetPickupAppointmentTimeline
- GetPickupAppointmentEvidence

- GetPickupAppointmentDecisionPackage
- GetPickupAppointmentExceptions
- GetPickupAppointmentAuditTrail
- GetPickupAppointmentCurrentPerspective

Every query response states data freshness, source cutoff, version, and whether values are authoritative, observed, derived, predicted, or disputed.

8. Workflow, Timeout, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Evidence and policy gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Create Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and available compensation]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Trigger downstream stage]

```

Required workflow behavior:

- durable process and correlation identity;
 - persisted approvals and timers;
 - bounded retries and retry budgets;
 - idempotent command submission;
 - safe restart after deployment or outage;
 - compensation that records new Intent instead of deleting history;
 - visible stalled, orphaned, and manually intervened processes;
 - operator access to source responses and downstream impact;
 - reconciliation before repeating an external financial, legal, or physical action.
-

9. Target API Contract

Reference surface:

```

POST /api/v1/pickup_custody_and_handover/commands
GET  /api/v1/pickup_custody_and_handover/{id}
GET  /api/v1/pickup_custody_and_handover/{id}/timeline
GET  /api/v1/pickup_custody_and_handover/{id}/evidence
GET  /api/v1/pickup_custody_and_handover/{id}/decisions
POST /api/v1/pickup_custody_and_handover/{id}/exceptions
POST /api/v1/pickup_custody_and_handover/{id}/corrections

```

These are reference endpoints, not claims that the exact routes exist.

Contract requirements:

- typed request and response schemas;
 - action-level authorization;
 - trusted authentication context for tenant identity;
 - idempotency for writes;
 - optimistic concurrency or equivalent aggregate guard;
 - RFC problem-details or governed equivalent;
 - stable pagination and sorting;
 - versioning and deprecation policy;
 - trace, correlation, and causation propagation;
 - explicit Outcome Unknown response when external completion is unresolved.
-

10. Target Persistence and Data Ownership

Reference table families:

Table family	Purpose
stage_aggregate	authoritative lifecycle subject
stage_version	immutable approved/submitted/issued versions
stage_decision	binding Decisions and rationale
stage_evidence_link	evidence and provenance relationships
stage_event_outbox	transactional Event publication
stage_idempotency	duplicate-write protection
stage_exception	exception, claim, hold, or dispute
stage_correction	correction and supersession lineage
stage_audit	privileged action evidence

Data rules:

- the owning service controls its schema and migrations;
- foreign keys stay inside an ownership boundary;
- cross-Domain relationships use stable IDs or contracts;
- effective-time and recorded-time are separate when needed;
- binding versions are immutable;
- indexes support tenant, owner, state, correlation, external reference, and effective time;
- retention, legal hold, privacy, residency, backup, and restore are documented;
- analytics, search, reports, and vector stores are projections.

11. Portal and Experience Architecture

Required Portal behavior:

- display exact version, owner, state, and last-updated time;
- distinguish authoritative, observed, derived, predicted, and disputed values;
- expose required action and authority;
- show loading, empty, partial, stale, error, timeout, Outcome Unknown, and permission-denied states;
- provide accessible keyboard and screen-reader behavior;
- handle locale, currency, units, dates, and RTL/LTR correctly;
- preserve evidence and download audit;
- never trust a browser-provided tenant header as authority;
- prevent double submission while still supporting safe retry;
- show correction and supersession history.

Operator surfaces additionally expose correlation, source responses, retry budget, replay status, compensation options, and affected downstream subjects.

12. AI Participation and Tool Governance

Permitted AI tasks:

- read package and seal identifiers
- detect count or weight anomalies
- summarize discrepancy
- assist pickup fraud detection

AI may also summarize evidence, detect gaps, recommend next actions, and draft human-readable explanations.

AI shall not:

- own or directly mutate Domain databases;
- fabricate external-provider, carrier, bank, customs, inspection, or delivery outcomes;
- grant itself permission or mandate;
- silently change an approved or issued version;
- suppress uncertainty, contradictions, policy failures, or human dissent;
- execute high-impact actions outside typed tools and approval policy;
- create shadow records for canonical subjects.

Reference automation:

Task	Maximum default
extraction and classification	A2, automatic with reviewable evidence
anomaly detection and recommendation	A2/A3 according to risk
binding commercial, legal, financial, quality, or custody Decision	A0/A1, human authority required

Task	Maximum default
external execution	typed tool, policy, approval, idempotency, and outcome observation
correction of authoritative truth	governed human-authorized workflow

Every Agent Run records prompt/model/tool versions, context sources, policy evaluation, approvals, tool calls, structured outputs, uncertainty, final observed outcome, and recovery actions.

13. Security, Privacy, and Trust Boundaries

Minimum controls:

- authenticated human, service, organization, device, and AI identities;
- represented principal and effective mandate validation;
- least privilege, ABAC/ReBAC where appropriate, and separation of duties;
- tenant, organization, facility, channel, and jurisdiction isolation;
- integrity protection for binding documents and Evidence;
- external callback signature and replay verification;
- confidential handling of commercial, personal, financial, transport, and regulatory data;
- secret isolation and no sensitive values in logs or prompts;
- privileged-action audit and emergency-access review;
- threat models for impersonation, tampering, replay, confused deputy, fraud, exfiltration, and prompt injection.

14. Reliability, SLOs, and Operations

Reference objectives:

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by the business deadline of the stage

Operational requirements include dashboards, alerts, runbooks, ownership, dependency health, retry budgets, DLQ/replay controls, Event outbox/inbox, workflow restart, reconciliation, capacity, backup/restore, incident command, and post-incident correction.

15. Failure Modes and Recovery

Representative failures:

- stale or wrong source version;
- duplicate submission or replay;
- external timeout after possible success;
- contradictory source observations;
- lost Event after committed state;
- approval or mandate revoked during execution;
- downstream action performed before a mandatory gate;
- partial completion across multiple subjects;
- evidence later retracted or corrected;
- tenant, facility, or jurisdiction mismatch.

Recovery principles:

1. Query the authoritative source before repeating an external action.
2. Preserve partial success.
3. Use compensating Intent instead of destructive history rewrite.
4. Publish correction Events when distributed facts change.
5. Quarantine high-impact ambiguity.

6. Escalate when authority or legal effect is unclear.
7. Record business impact, affected subjects, and final reconciliation.

16. Testing and Conformance

Required tests:

- aggregate invariant and transition tests;
- authorization, mandate, separation-of-duties, and tenant-isolation tests;
- idempotency and optimistic-concurrency tests;
- migration, constraint, index, and rollback tests;
- outbox/inbox, ordering, duplicate, replay, and DLQ tests;
- workflow restart, timeout, compensation, and manual-intervention tests;
- provider callback verification and Outcome Unknown tests;
- Portal E2E for success, partial, stale, denied, and error journeys;
- accessibility, locale, currency, unit, and RTL tests;
- security adversarial and data-exfiltration tests;
- AI evaluation, prompt-injection, policy-bypass, and unauthorized-tool tests;
- historical reconstruction and correction tests;
- performance, capacity, disaster-recovery, and operational exercise evidence.

17. Repository Review Boundary and Previously Inspected Evidence

- Shipping, provider, warehouse, and Event Bus capabilities are registered.
- A canonical custody aggregate is `REFERENCE_ARCHITECTURE` until verified.

Area	Classification for this chapter
Target aggregate, API, tables, state model, and workflow Registered supporting modules	<code>REFERENCE_ARCHITECTURE</code> <code>DOCUMENTATION_ONLY_CLAIM</code> until executable files are inspected
Concrete behavior already established in earlier repository-grounded chapters	retain the original classification and evidence
Exact integration paths not inspected for this stage	<code>REQUIRES_REPOSITORY_REVIEW</code>
External authority or provider outcome	qualified external observation, never platform-owned truth

No statement in this chapter upgrades an implementation status merely because a module name, frontend route, README, plan, or database catalog entry exists.

18. Worked Scenario

An upstream governed commitment enters the stage with exact subject and version references. The initiating principal is authenticated, and its mandate is checked at command time. Evidence is gathered and evaluated against the applicable policy version and jurisdiction.

A recommendation may be produced, but an authorized Decision remains separate. The Domain accepts an idempotent Command and records state plus an outbox entry atomically. When an external party participates, a timeout creates Outcome Unknown; the workflow queries the external source or waits for a verified callback instead of repeating execution blindly.

On success or failure, the system records a completed fact, publishes a stable Event, updates downstream projections, and preserves source and knowledge cutoff. Later corrections append new facts and identify downstream impact rather than replacing history. AI assists through typed tools and cannot own the subject, grant authority, or invent an outcome.

19. Conformance Checklist

- ☐ Stable subject and version identities are explicit.
- ☐ Proposition ownership and external authority are named.
- ☐ Entry and exit criteria are testable.
- ☐ State, Recommendation, Decision, Intent, execution, and outcome remain distinct.

- ☐ Commands are authorized, version-guarded, and idempotent.
 - ☐ Events describe completed facts.
 - ☐ Evidence includes provenance, time, integrity, and correction links.
 - ☐ Portals show freshness, uncertainty, and Outcome Unknown.
 - ☐ AI uses typed tools and approved autonomy.
 - ☐ Security, privacy, tenant, facility, and jurisdiction boundaries are enforced.
 - ☐ Reconciliation precedes risky duplicate execution.
 - ☐ Corrections preserve history.
 - ☐ Tests prove invariants, security, failure, and recovery.
 - ☐ Implementation status is not upgraded without executable repository evidence.
-

20. Status Vocabulary

- VERIFIED_IMPLEMENTED: executable source and the relevant behavior were inspected.
- IMPLEMENTED_WITH_GAP: executable behavior exists with a material governance or completeness gap.
- PARTIAL_IMPLEMENTATION: only part of the target capability is evidenced.
- REFERENCE_ARCHITECTURE: normative target behavior; no claim that it exists today.
- PLANNED_NOT_IMPLEMENTED: explicitly planned but not evidenced as executable.
- DOCUMENTATION_ONLY_CLAIM: documentation or a client contract claims behavior without verified backend evidence.
- CONFLICT_REQUIRES_CANONICALIZATION: sources disagree and no final owner has been established.
- REQUIRES_REPOSITORY_REVIEW: the necessary executable evidence has not yet been inspected.

Publication Chapter 20 — International Transport, Tracking, and Exceptions

Metadata	Value
Publication Chapter	20
Canonical Lifecycle Stage	19
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE with explicit repository notes
Accountable capability	Shipping / Provider / external carrier
Canon dependencies	CANON-001 through CANON-011
Constitutional dependencies	ART-001 through ART-016
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This stage exists to **assemble a qualified transport perspective from provider observations while preserving source, confidence, contradictions, corrections, and route-leg ownership.**

Its output is not merely a screen state or document download. It is a governed set of identified subjects, Relationships, Decisions, Intent, Commands, completed facts, external observations, evidence, and correction history.

- Reality or external outcome
- ≠ platform Observation
 - ≠ derived Perspective
 - ≠ AI Prediction
 - ≠ Recommendation
 - ≠ authorized Decision
 - ≠ execution Intent

1.1 Accountability

The accountable capability is **Shipping / Provider / external carrier**. Supporting Portals, workflow engines, reports, search indexes, integration adapters, and AI Agents may coordinate or present the work, but they do not silently become owners of the underlying business truth.

1.2 Primary business outcomes

- a stable subject identity and exact source-version lineage;
- explicit authority and effective mandate;
- a testable lifecycle outcome;
- evidence sufficient for downstream use and later reconstruction;
- visible failure, partial completion, uncertainty, and correction;
- a completed-fact Event or qualified external observation;
- downstream activation without shared-database ownership transfer.

1.3 Non-goals

- treating a registered module as proof of complete implementation;
 - treating a frontend route as proof of backend behavior;
 - allowing an AI recommendation to become a binding Decision;
 - inventing external-provider or government-authority success;
 - overwriting history during amendment or correction;
 - combining unrelated states into one generic status field.
-

2. Entry and Exit Criteria

2.1 Entry criteria

1. Every upstream subject reference resolves to a stable identifier.
2. The exact approved, accepted, issued, or observed version is known.
3. The initiating human, service, organization, or AI Agent is authenticated.
4. The represented principal and mandate are valid at command time.
5. Required Evidence exists, or an explicit authorized exception is linked.
6. Applicable Policy and jurisdiction versions are fixed.
7. Duplicate, stale-version, and replay checks have run.
8. Any conflicting active process is resolved, suspended, or explicitly related.

2.2 Exit criteria

1. The authoritative Domain or external authority records the outcome.
 2. State and version changes are durable.
 3. A completed-fact Event or qualified observation is available.
 4. Downstream consumers receive stable subject, correlation, and causation references.
 5. Unknown, partial, disputed, or failed outcomes remain visible.
 6. Corrections preserve the original record and reason.
 7. Metrics, audit evidence, and operational signals are emitted.
 8. A qualified reviewer can reconstruct who knew what, when, and why.
-

3. Ubiquitous Language and Subject Model

Primary subject families:

Subject	Required interpretation
Shipment	independent identified subject or governed subordinate entity; ownership must be declared
ShipmentLeg	independent identified subject or governed subordinate entity; ownership must be declared
TransportMilestone	independent identified subject or governed subordinate entity; ownership must be declared
TrackingObservation	independent identified subject or governed subordinate entity; ownership must be declared
EstimatedArrival	independent identified subject or governed subordinate entity; ownership must be declared
TransportException	independent identified subject or governed subordinate entity; ownership must be declared
RouteDeviation	independent identified subject or governed subordinate entity; ownership must be declared
TrackingCorrection	independent identified subject or governed subordinate entity; ownership must be declared

Reference aggregate:

```
Shipment {
  aggregateId
  tenantId
  organizationContext
  canonicalSubjectRefs[]
  sourceVersionRefs[]
  lifecycleState
  decisionRefs[]
}
```

```

intentRefs[]
evidenceSetRef
knowledgeCutoff
policyRefs[]
authorityRef
jurisdictionRefs[]
createdAt
effectiveAt
updatedAt
version
correctionOf?
correlationId
}

```

3.1 Identity and relationship rules

- a display name, document number, provider reference, barcode, or URL is not automatically canonical identity;
- relationships such as custody, mandate, allocation, publication, reservation, or obligation are independent facts;
- cross-Domain references use stable IDs and qualified snapshots, not shared-table mutation;
- external identifiers retain issuer and namespace;
- merge, split, correction, and supersession preserve lineage.

3.2 Invariants

1. Submitted, approved, accepted, issued, or signed versions are immutable.
2. State changes occur through authorized Commands.
3. Completed-fact Events use past-tense semantics.
4. A Recommendation cannot self-promote to a Decision.
5. An external outcome retains its source authority.
6. Derived projections cannot become authoritative by repeated copying.
7. All writes are tenant-, organization-, facility-, and authority-scoped as applicable.
8. Retried writes are idempotent.
9. Outcome Unknown is not converted to success or failure by assumption.
10. Corrections reference the corrected fact and downstream impact.

4. Lifecycle State Model

4.1 State vocabulary

State	Meaning
BOOKED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
PICKED_UP	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
DEPARTED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
IN_TRANSIT	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
TRANSSHIPMENT	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
DELAYED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
EXCEPTION	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
ARRIVED_AT_PORT	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
CUSTOMS_PENDING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
DELIVERED_TO_DESTINATION	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
LOST	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
CANCELLED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification

4.2 Reference state machine

```
stateDiagram-v2
    [*] --> BOOKED
    BOOKED --> PICKED_UP
    PICKED_UP --> DEPARTED
    DEPARTED --> IN_TRANSIT
    IN_TRANSIT --> TRANSSHIPMENT
    TRANSSHIPMENT --> DELAYED
    DELAYED --> EXCEPTION
    EXCEPTION --> ARRIVED_AT_PORT
    ARRIVED_AT_PORT --> CUSTOMS_PENDING
    CUSTOMS_PENDING --> DELIVERED_TO_DESTINATION
    DELIVERED_TO_DESTINATION --> LOST
    LOST --> CANCELLED
```

The diagram shows a readable primary path. A conformant implementation may include parallel approvals, amendment, suspension, timeout, appeal, rejection, partial completion, reconciliation, and correction paths, but it must preserve the same semantic distinctions.

4.3 Transition matrix

From	To	Minimum guard
BOOKED	PICKED_UP	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
PICKED_UP	DEPARTED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
DEPARTED	IN_TRANSIT	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
IN_TRANSIT	TRANSSHIPMENT	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
TRANSSHIPMENT	DELAYED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
DELAYED	EXCEPTION	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
EXCEPTION	ARRIVED_AT_PORT	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
ARRIVED_AT_PORT	CUSTOMS_PENDING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
CUSTOMS_PENDING	DELIVERED_TO_DESTINATION	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
DELIVERED_TO_DESTINATION	LOST	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
LOST	CANCELLED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard

4.4 Forbidden state behavior

- direct arbitrary assignment from a Portal request;
- using one status to represent lifecycle, approval, payment, document, provider, and dispute state simultaneously;
- deleting terminal history to reopen a process;
- inferring completion from elapsed time or an AI prediction;
- moving past a mandatory gate because a downstream system already acted.

5. Decision, Policy, and Authority

Material Decisions in this stage include readiness, eligibility, approval, exception, amendment, correction, and downstream activation decisions.

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Decision rules

- the Decision references exact source versions;
 - authority and delegation are evaluated at Decision time;
 - a Policy version and jurisdiction context are recorded;
 - mandatory-gate failure blocks progression unless a lawful exception exists;
 - dissent, uncertainty, and missing Evidence remain visible;
 - Decision correction does not erase the original Decision;
 - AI may prepare a briefing, but cannot manufacture authority.
-

6. Evidence, Documents, and Knowledge Cutoff

Every binding or high-impact action requires an Evidence Set containing:

- evidence identity and integrity hash where required;
- issuing, observing, or signing party;
- observed time, received time, and effective-time interval;
- subject and source-version relationships;
- qualification, confidence, and limitations;
- contradiction, retraction, supersession, and correction links;
- jurisdiction, classification, retention, and residency;
- chain of custody for physical or high-integrity evidence;
- the knowledge cutoff used by the Decision.

A document is not automatically truth. It is an Artifact containing Claims. Its legal or operational effect depends on issuer, mandate, signature, version, applicable rules, and external authority.

7. Commands, Events, and Queries

7.1 Commands

- RecordTrackingObservation
- QualifyTransportMilestone
- UpdateEstimatedArrival
- OpenTransportException
- CorrectTrackingObservation
- CloseTransportException

Each Command shall contain an idempotency key, actor identity, represented principal, tenant and organization context, correlation ID, expected version where relevant, reason for privileged action, and exact target subject.

7.2 Events

- ShipmentDeparted
- ShipmentMilestoneObserved
- ShipmentDelayed
- EstimatedArrivalChanged

- `TransportExceptionOpened`
- `RouteDeviationObserved`
- `ShipmentArrived`

Events describe completed facts or explicitly qualified observations. A request, proposed action, timer, or prediction is not a completed-fact Event.

7.3 Queries and perspectives

- `GetShipment`
- `ListShipmentRecords`
- `GetShipmentTimeline`
- `GetShipmentEvidence`
- `GetShipmentDecisionPackage`
- `GetShipmentExceptions`
- `GetShipmentAuditTrail`
- `GetShipmentCurrentPerspective`

Every query response states data freshness, source cutoff, version, and whether values are authoritative, observed, derived, predicted, or disputed.

8. Workflow, Timeout, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Evidence and policy gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Create Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and available compensation]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Trigger downstream stage]

```

Required workflow behavior:

- durable process and correlation identity;
- persisted approvals and timers;
- bounded retries and retry budgets;
- idempotent command submission;
- safe restart after deployment or outage;
- compensation that records new Intent instead of deleting history;
- visible stalled, orphaned, and manually intervened processes;
- operator access to source responses and downstream impact;
- reconciliation before repeating an external financial, legal, or physical action.

9. Target API Contract

Reference surface:

```

POST /api/v1/international_transport_tracking_and_exceptions/commands
GET  /api/v1/international_transport_tracking_and_exceptions/{id}
GET  /api/v1/international_transport_tracking_and_exceptions/{id}/timeline
GET  /api/v1/international_transport_tracking_and_exceptions/{id}/evidence
GET  /api/v1/international_transport_tracking_and_exceptions/{id}/decisions
POST /api/v1/international_transport_tracking_and_exceptions/{id}/exceptions
POST /api/v1/international_transport_tracking_and_exceptions/{id}/corrections

```

These are reference endpoints, not claims that the exact routes exist.

Contract requirements:

- typed request and response schemas;
- action-level authorization;
- trusted authentication context for tenant identity;
- idempotency for writes;
- optimistic concurrency or equivalent aggregate guard;

- RFC problem-details or governed equivalent;
- stable pagination and sorting;
- versioning and deprecation policy;
- trace, correlation, and causation propagation;
- explicit Outcome Unknown response when external completion is unresolved.

10. Target Persistence and Data Ownership

Reference table families:

Table family	Purpose
stage_aggregate	authoritative lifecycle subject
stage_version	immutable approved/submitted/issued versions
stage_decision	binding Decisions and rationale
stage_evidence_link	evidence and provenance relationships
stage_event_outbox	transactional Event publication
stage_idempotency	duplicate-write protection
stage_exception	exception, claim, hold, or dispute
stage_correction	correction and supersession lineage
stage_audit	privileged action evidence

Data rules:

- the owning service controls its schema and migrations;
- foreign keys stay inside an ownership boundary;
- cross-Domain relationships use stable IDs or contracts;
- effective-time and recorded-time are separate when needed;
- binding versions are immutable;
- indexes support tenant, owner, state, correlation, external reference, and effective time;
- retention, legal hold, privacy, residency, backup, and restore are documented;
- analytics, search, reports, and vector stores are projections.

11. Portal and Experience Architecture

Required Portal behavior:

- display exact version, owner, state, and last-updated time;
- distinguish authoritative, observed, derived, predicted, and disputed values;
- expose required action and authority;
- show loading, empty, partial, stale, error, timeout, Outcome Unknown, and permission-denied states;
- provide accessible keyboard and screen-reader behavior;
- handle locale, currency, units, dates, and RTL/LTR correctly;
- preserve evidence and download audit;
- never trust a browser-provided tenant header as authority;
- prevent double submission while still supporting safe retry;
- show correction and supersession history.

Operator surfaces additionally expose correlation, source responses, retry budget, replay status, compensation options, and affected downstream subjects.

12. AI Participation and Tool Governance

Permitted AI tasks:

- predict ETA
- detect delay and route deviation
- reconcile multi-source observations
- summarize transport exceptions

AI may also summarize evidence, detect gaps, recommend next actions, and draft human-readable explanations.

AI shall not:

- own or directly mutate Domain databases;
- fabricate external-provider, carrier, bank, customs, inspection, or delivery outcomes;

- grant itself permission or mandate;
- silently change an approved or issued version;
- suppress uncertainty, contradictions, policy failures, or human dissent;
- execute high-impact actions outside typed tools and approval policy;
- create shadow records for canonical subjects.

Reference automation:

Task	Maximum default
extraction and classification	A2, automatic with reviewable evidence
anomaly detection and recommendation	A2/A3 according to risk
binding commercial, legal, financial, quality, or custody Decision	A0/A1, human authority required
external execution	typed tool, policy, approval, idempotency, and outcome observation
correction of authoritative truth	governed human-authorized workflow

Every Agent Run records prompt/model/tool versions, context sources, policy evaluation, approvals, tool calls, structured outputs, uncertainty, final observed outcome, and recovery actions.

13. Security, Privacy, and Trust Boundaries

Minimum controls:

- authenticated human, service, organization, device, and AI identities;
- represented principal and effective mandate validation;
- least privilege, ABAC/ReBAC where appropriate, and separation of duties;
- tenant, organization, facility, channel, and jurisdiction isolation;
- integrity protection for binding documents and Evidence;
- external callback signature and replay verification;
- confidential handling of commercial, personal, financial, transport, and regulatory data;
- secret isolation and no sensitive values in logs or prompts;
- privileged-action audit and emergency-access review;
- threat models for impersonation, tampering, replay, confused deputy, fraud, exfiltration, and prompt injection.

14. Reliability, SLOs, and Operations

Reference objectives:

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by the business deadline of the stage

Operational requirements include dashboards, alerts, runbooks, ownership, dependency health, retry budgets, DLQ/replay controls, Event outbox/inbox, workflow restart, reconciliation, capacity, backup/restore, incident command, and post-incident correction.

15. Failure Modes and Recovery

Representative failures:

- stale or wrong source version;
- duplicate submission or replay;
- external timeout after possible success;
- contradictory source observations;
- lost Event after committed state;

- approval or mandate revoked during execution;
- downstream action performed before a mandatory gate;
- partial completion across multiple subjects;
- evidence later retracted or corrected;
- tenant, facility, or jurisdiction mismatch.

Recovery principles:

1. Query the authoritative source before repeating an external action.
2. Preserve partial success.
3. Use compensating Intent instead of destructive history rewrite.
4. Publish correction Events when distributed facts change.
5. Quarantine high-impact ambiguity.
6. Escalate when authority or legal effect is unclear.
7. Record business impact, affected subjects, and final reconciliation.

16. Testing and Conformance

Required tests:

- aggregate invariant and transition tests;
- authorization, mandate, separation-of-duties, and tenant-isolation tests;
- idempotency and optimistic-concurrency tests;
- migration, constraint, index, and rollback tests;
- outbox/inbox, ordering, duplicate, replay, and DLQ tests;
- workflow restart, timeout, compensation, and manual-intervention tests;
- provider callback verification and Outcome Unknown tests;
- Portal E2E for success, partial, stale, denied, and error journeys;
- accessibility, locale, currency, unit, and RTL tests;
- security adversarial and data-exfiltration tests;
- AI evaluation, prompt-injection, policy-bypass, and unauthorized-tool tests;
- historical reconstruction and correction tests;
- performance, capacity, disaster-recovery, and operational exercise evidence.

17. Repository Review Boundary and Previously Inspected Evidence

- Shipping, provider, Event Bus, and WebSocket capabilities are registered.
- External provider adapters and topic contracts require per-file verification.

Area	Classification for this chapter
Target aggregate, API, tables, state model, and workflow	REFERENCE_ARCHITECTURE
Registered supporting modules	DOCUMENTATION_ONLY_CLAIM until executable files are inspected
Concrete behavior already established in earlier repository-grounded chapters	retain the original classification and evidence
Exact integration paths not inspected for this stage	REQUIRES_REPOSITORY_REVIEW
External authority or provider outcome	qualified external observation, never platform-owned truth

No statement in this chapter upgrades an implementation status merely because a module name, frontend route, README, plan, or database catalog entry exists.

18. Worked Scenario

An upstream governed commitment enters the stage with exact subject and version references. The initiating principal is authenticated, and its mandate is checked at command time. Evidence is gathered and evaluated against the applicable policy version and jurisdiction.

A recommendation may be produced, but an authorized Decision remains separate. The Domain accepts an idempotent Command and records state plus an outbox entry atomically. When an external party participates, a timeout creates Outcome Unknown; the workflow queries the external source or waits for a verified callback instead of repeating execution blindly.

On success or failure, the system records a completed fact, publishes a stable Event, updates downstream projections, and preserves source and knowledge cutoff. Later corrections append new facts and identify downstream impact rather than replacing history. AI assists through typed tools and cannot own the subject, grant authority, or invent an outcome.

19. Conformance Checklist

- ☐ Stable subject and version identities are explicit.
 - ☐ Proposition ownership and external authority are named.
 - ☐ Entry and exit criteria are testable.
 - ☐ State, Recommendation, Decision, Intent, execution, and outcome remain distinct.
 - ☐ Commands are authorized, version-guarded, and idempotent.
 - ☐ Events describe completed facts.
 - ☐ Evidence includes provenance, time, integrity, and correction links.
 - ☐ Portals show freshness, uncertainty, and Outcome Unknown.
 - ☐ AI uses typed tools and approved autonomy.
 - ☐ Security, privacy, tenant, facility, and jurisdiction boundaries are enforced.
 - ☐ Reconciliation precedes risky duplicate execution.
 - ☐ Corrections preserve history.
 - ☐ Tests prove invariants, security, failure, and recovery.
 - ☐ Implementation status is not upgraded without executable repository evidence.
-

20. Status Vocabulary

- VERIFIED_IMPLEMENTED: executable source and the relevant behavior were inspected.
- IMPLEMENTED_WITH_GAP: executable behavior exists with a material governance or completeness gap.
- PARTIAL_IMPLEMENTATION: only part of the target capability is evidenced.
- REFERENCE_ARCHITECTURE: normative target behavior; no claim that it exists today.
- PLANNED_NOT_IMPLEMENTED: explicitly planned but not evidenced as executable.
- DOCUMENTATION_ONLY_CLAIM: documentation or a client contract claims behavior without verified backend evidence.
- CONFLICT_REQUIRES_CANONICALIZATION: sources disagree and no final owner has been established.
- REQUIRES_REPOSITORY_REVIEW: the necessary executable evidence has not yet been inspected.

Publication Chapter 21 — Import Customs, Duties, and Legal Release

Metadata	Value
Publication Chapter	21
Canonical Lifecycle Stage	20
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE with explicit repository notes
Accountable capability	Regulatory / Compliance / external customs authority
Canon dependencies	CANON-001 through CANON-011
Constitutional dependencies	ART-001 through ART-016
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This stage exists to **process destination declarations, duties, inspections, holds, and authoritative release while preserving legal-authority and financial boundaries.**

Its output is not merely a screen state or document download. It is a governed set of identified subjects, Relationships, Decisions, Intent, Commands, completed facts, external observations, evidence, and correction history.

Reality or external outcome
≠ platform Observation
≠ derived Perspective
≠ AI Prediction
≠ Recommendation
≠ authorized Decision
≠ execution Intent

1.1 Accountability

The accountable capability is **Regulatory / Compliance / external customs authority**. Supporting Portals, workflow engines, reports, search indexes, integration adapters, and AI Agents may coordinate or present the work, but they do not silently become owners of the underlying business truth.

1.2 Primary business outcomes

- a stable subject identity and exact source-version lineage;
- explicit authority and effective mandate;
- a testable lifecycle outcome;
- evidence sufficient for downstream use and later reconstruction;
- visible failure, partial completion, uncertainty, and correction;
- a completed-fact Event or qualified external observation;
- downstream activation without shared-database ownership transfer.

1.3 Non-goals

- treating a registered module as proof of complete implementation;

- treating a frontend route as proof of backend behavior;
 - allowing an AI recommendation to become a binding Decision;
 - inventing external-provider or government-authority success;
 - overwriting history during amendment or correction;
 - combining unrelated states into one generic status field.
-

2. Entry and Exit Criteria

2.1 Entry criteria

1. Every upstream subject reference resolves to a stable identifier.
2. The exact approved, accepted, issued, or observed version is known.
3. The initiating human, service, organization, or AI Agent is authenticated.
4. The represented principal and mandate are valid at command time.
5. Required Evidence exists, or an explicit authorized exception is linked.
6. Applicable Policy and jurisdiction versions are fixed.
7. Duplicate, stale-version, and replay checks have run.
8. Any conflicting active process is resolved, suspended, or explicitly related.

2.2 Exit criteria

1. The authoritative Domain or external authority records the outcome.
 2. State and version changes are durable.
 3. A completed-fact Event or qualified observation is available.
 4. Downstream consumers receive stable subject, correlation, and causation references.
 5. Unknown, partial, disputed, or failed outcomes remain visible.
 6. Corrections preserve the original record and reason.
 7. Metrics, audit evidence, and operational signals are emitted.
 8. A qualified reviewer can reconstruct who knew what, when, and why.
-

3. Ubiquitous Language and Subject Model

Primary subject families:

Subject	Required interpretation
ImportDeclaration	independent identified subject or governed subordinate entity; ownership must be declared
ImportDeclarationVersion	independent identified subject or governed subordinate entity; ownership must be declared
DutyAssessment	independent identified subject or governed subordinate entity; ownership must be declared
TaxAssessment	independent identified subject or governed subordinate entity; ownership must be declared
ImportLicense	independent identified subject or governed subordinate entity; ownership must be declared
CustomsHold	independent identified subject or governed subordinate entity; ownership must be declared
CustomsInspection	independent identified subject or governed subordinate entity; ownership must be declared
ImportRelease	independent identified subject or governed subordinate entity; ownership must be declared

Reference aggregate:

```
ImportDeclaration {
  aggregateId
  tenantId
  organizationContext
  canonicalSubjectRefs[]
  sourceVersionRefs[]
  lifecycleState
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
```

```

policyRefs[]
authorityRef
jurisdictionRefs[]
createdAt
effectiveAt
updatedAt
version
correctionOf?
correlationId
}

```

3.1 Identity and relationship rules

- a display name, document number, provider reference, barcode, or URL is not automatically canonical identity;
- relationships such as custody, mandate, allocation, publication, reservation, or obligation are independent facts;
- cross-Domain references use stable IDs and qualified snapshots, not shared-table mutation;
- external identifiers retain issuer and namespace;
- merge, split, correction, and supersession preserve lineage.

3.2 Invariants

1. Submitted, approved, accepted, issued, or signed versions are immutable.
2. State changes occur through authorized Commands.
3. Completed-fact Events use past-tense semantics.
4. A Recommendation cannot self-promote to a Decision.
5. An external outcome retains its source authority.
6. Derived projections cannot become authoritative by repeated copying.
7. All writes are tenant-, organization-, facility-, and authority-scoped as applicable.
8. Retried writes are idempotent.
9. Outcome Unknown is not converted to success or failure by assumption.
10. Corrections reference the corrected fact and downstream impact.

4. Lifecycle State Model

4.1 State vocabulary

State	Meaning
PREPARATION	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
DOCUMENT_PENDING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
SUBMITTED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
ACCEPTED_FOR_PROCESSING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
DUTY_PENDING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
PAYMENT_PENDING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
HOLD	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
INSPECTION	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
AMENDMENT_REQUIRED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
RELEASED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
REJECTED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification

4.2 Reference state machine

```

stateDiagram-v2
    [*] --> PREPARATION
    PREPARATION --> DOCUMENT_PENDING
    DOCUMENT_PENDING --> SUBMITTED

```

```

SUBMITTED --> ACCEPTED_FOR_PROCESSING
ACCEPTED_FOR_PROCESSING --> DUTY_PENDING
DUTY_PENDING --> PAYMENT_PENDING
PAYMENT_PENDING --> HOLD
HOLD --> INSPECTION
INSPECTION --> AMENDMENT_REQUIRED
AMENDMENT_REQUIRED --> RELEASED
RELEASED --> REJECTED

```

The diagram shows a readable primary path. A conformant implementation may include parallel approvals, amendment, suspension, timeout, appeal, rejection, partial completion, reconciliation, and correction paths, but it must preserve the same semantic distinctions.

4.3 Transition matrix

From	To	Minimum guard
PREPARATION	DOCUMENT_PENDING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
DOCUMENT_PENDING	SUBMITTED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
SUBMITTED	ACCEPTED_FOR_PROCESSING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
ACCEPTED_FOR_PROCESSING	DUTY_PENDING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
DUTY_PENDING	PAYMENT_PENDING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
PAYMENT_PENDING	HOLD	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
HOLD	INSPECTION	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
INSPECTION	AMENDMENT_REQUIRED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
AMENDMENT_REQUIRED	RELEASED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
RELEASED	REJECTED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard

4.4 Forbidden state behavior

- direct arbitrary assignment from a Portal request;
- using one status to represent lifecycle, approval, payment, document, provider, and dispute state simultaneously;
- deleting terminal history to reopen a process;
- inferring completion from elapsed time or an AI prediction;
- moving past a mandatory gate because a downstream system already acted.

5. Decision, Policy, and Authority

Material Decisions in this stage include readiness, eligibility, approval, exception, amendment, correction, and downstream activation decisions.

Reference Decision:

```

Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
}

```

```

evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Decision rules

- the Decision references exact source versions;
 - authority and delegation are evaluated at Decision time;
 - a Policy version and jurisdiction context are recorded;
 - mandatory-gate failure blocks progression unless a lawful exception exists;
 - dissent, uncertainty, and missing Evidence remain visible;
 - Decision correction does not erase the original Decision;
 - AI may prepare a briefing, but cannot manufacture authority.
-

6. Evidence, Documents, and Knowledge Cutoff

Every binding or high-impact action requires an Evidence Set containing:

- evidence identity and integrity hash where required;
- issuing, observing, or signing party;
- observed time, received time, and effective-time interval;
- subject and source-version relationships;
- qualification, confidence, and limitations;
- contradiction, retraction, supersession, and correction links;
- jurisdiction, classification, retention, and residency;
- chain of custody for physical or high-integrity evidence;
- the knowledge cutoff used by the Decision.

A document is not automatically truth. It is an Artifact containing Claims. Its legal or operational effect depends on issuer, mandate, signature, version, applicable rules, and external authority.

7. Commands, Events, and Queries

7.1 Commands

- PrepareImportDeclaration
- SubmitImportDeclaration
- RecordDutyAssessment
- RecordDutyPayment
- RecordImportHold
- AmendImportDeclaration
- RecordImportRelease

Each Command shall contain an idempotency key, actor identity, represented principal, tenant and organization context, correlation ID, expected version where relevant, reason for privileged action, and exact target subject.

7.2 Events

- ImportDeclarationSubmitted
- DutyAssessed
- ImportDutyPaid
- ImportHoldRecorded
- ImportInspectionRequested
- ImportDeclarationAmended
- ImportReleased

Events describe completed facts or explicitly qualified observations. A request, proposed action, timer, or prediction is not a completed-fact Event.

7.3 Queries and perspectives

- `GetImportDeclaration`
- `ListImportDeclarationRecords`
- `GetImportDeclarationTimeline`
- `GetImportDeclarationEvidence`
- `GetImportDeclarationDecisionPackage`
- `GetImportDeclarationExceptions`
- `GetImportDeclarationAuditTrail`
- `GetImportDeclarationCurrentPerspective`

Every query response states data freshness, source cutoff, version, and whether values are authoritative, observed, derived, predicted, or disputed.

8. Workflow, Timeout, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Evidence and policy gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Create Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and available compensation]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Trigger downstream stage]
```

Required workflow behavior:

- durable process and correlation identity;
 - persisted approvals and timers;
 - bounded retries and retry budgets;
 - idempotent command submission;
 - safe restart after deployment or outage;
 - compensation that records new Intent instead of deleting history;
 - visible stalled, orphaned, and manually intervened processes;
 - operator access to source responses and downstream impact;
 - reconciliation before repeating an external financial, legal, or physical action.
-

9. Target API Contract

Reference surface:

```
POST /api/v1/import_customs_duties_and_legal_release/commands
GET  /api/v1/import_customs_duties_and_legal_release/{id}
GET  /api/v1/import_customs_duties_and_legal_release/{id}/timeline
GET  /api/v1/import_customs_duties_and_legal_release/{id}/evidence
GET  /api/v1/import_customs_duties_and_legal_release/{id}/decisions
POST /api/v1/import_customs_duties_and_legal_release/{id}/exceptions
POST /api/v1/import_customs_duties_and_legal_release/{id}/corrections
```

These are reference endpoints, not claims that the exact routes exist.

Contract requirements:

- typed request and response schemas;
 - action-level authorization;
 - trusted authentication context for tenant identity;
 - idempotency for writes;
 - optimistic concurrency or equivalent aggregate guard;
 - RFC problem-details or governed equivalent;
 - stable pagination and sorting;
 - versioning and deprecation policy;
 - trace, correlation, and causation propagation;
 - explicit Outcome Unknown response when external completion is unresolved.
-

10. Target Persistence and Data Ownership

Reference table families:

Table family	Purpose
stage_aggregate	authoritative lifecycle subject
stage_version	immutable approved/submitted/issued versions
stage_decision	binding Decisions and rationale
stage_evidence_link	evidence and provenance relationships
stage_event_outbox	transactional Event publication
stage_idempotency	duplicate-write protection
stage_exception	exception, claim, hold, or dispute
stage_correction	correction and supersession lineage
stage_audit	privileged action evidence

Data rules:

- the owning service controls its schema and migrations;
- foreign keys stay inside an ownership boundary;
- cross-Domain relationships use stable IDs or contracts;
- effective-time and recorded-time are separate when needed;
- binding versions are immutable;
- indexes support tenant, owner, state, correlation, external reference, and effective time;
- retention, legal hold, privacy, residency, backup, and restore are documented;
- analytics, search, reports, and vector stores are projections.

11. Portal and Experience Architecture

Required Portal behavior:

- display exact version, owner, state, and last-updated time;
- distinguish authoritative, observed, derived, predicted, and disputed values;
- expose required action and authority;
- show loading, empty, partial, stale, error, timeout, Outcome Unknown, and permission-denied states;
- provide accessible keyboard and screen-reader behavior;
- handle locale, currency, units, dates, and RTL/LTR correctly;
- preserve evidence and download audit;
- never trust a browser-provided tenant header as authority;
- prevent double submission while still supporting safe retry;
- show correction and supersession history.

Operator surfaces additionally expose correlation, source responses, retry budget, replay status, compensation options, and affected downstream subjects.

12. AI Participation and Tool Governance

Permitted AI tasks:

- detect document gaps
- triage valuation anomalies
- assist classification
- predict demurrage risk

AI may also summarize evidence, detect gaps, recommend next actions, and draft human-readable explanations.

AI shall not:

- own or directly mutate Domain databases;
- fabricate external-provider, carrier, bank, customs, inspection, or delivery outcomes;
- grant itself permission or mandate;
- silently change an approved or issued version;
- suppress uncertainty, contradictions, policy failures, or human dissent;
- execute high-impact actions outside typed tools and approval policy;
- create shadow records for canonical subjects.

Reference automation:

Task	Maximum default
extraction and classification	A2, automatic with reviewable evidence
anomaly detection and recommendation	A2/A3 according to risk
binding commercial, legal, financial, quality, or custody Decision	A0/A1, human authority required
external execution	typed tool, policy, approval, idempotency, and outcome observation
correction of authoritative truth	governed human-authorized workflow

Every Agent Run records prompt/model/tool versions, context sources, policy evaluation, approvals, tool calls, structured outputs, uncertainty, final observed outcome, and recovery actions.

13. Security, Privacy, and Trust Boundaries

Minimum controls:

- authenticated human, service, organization, device, and AI identities;
- represented principal and effective mandate validation;
- least privilege, ABAC/ReBAC where appropriate, and separation of duties;
- tenant, organization, facility, channel, and jurisdiction isolation;
- integrity protection for binding documents and Evidence;
- external callback signature and replay verification;
- confidential handling of commercial, personal, financial, transport, and regulatory data;
- secret isolation and no sensitive values in logs or prompts;
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Reference objectives:

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by the business deadline of the stage

Operational requirements include dashboards, alerts, runbooks, ownership, dependency health, retry budgets, DLQ/replay controls, Event outbox/inbox, workflow restart, reconciliation, capacity, backup/restore, incident command, and post-incident correction.

15. Failure Modes and Recovery

Representative failures:

- stale or wrong source version;
- duplicate submission or replay;
- external timeout after possible success;
- contradictory source observations;
- lost Event after committed state;
- approval or mandate revoked during execution;
- downstream action performed before a mandatory gate;
- partial completion across multiple subjects;
- evidence later retracted or corrected;
- tenant, facility, or jurisdiction mismatch.

Recovery principles:

1. Query the authoritative source before repeating an external action.

- 2. Preserve partial success.
- 3. Use compensating Intent instead of destructive history rewrite.
- 4. Publish correction Events when distributed facts change.
- 5. Quarantine high-impact ambiguity.
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- 7. Record business impact, affected subjects, and final reconciliation.

16. Testing and Conformance

Required tests:

- aggregate invariant and transition tests;
- authorization, mandate, separation-of-duties, and tenant-isolation tests;
- idempotency and optimistic-concurrency tests;
- migration, constraint, index, and rollback tests;
- outbox/inbox, ordering, duplicate, replay, and DLQ tests;
- workflow restart, timeout, compensation, and manual-intervention tests;
- provider callback verification and Outcome Unknown tests;
- Portal E2E for success, partial, stale, denied, and error journeys;
- accessibility, locale, currency, unit, and RTL tests;
- security adversarial and data-exfiltration tests;
- AI evaluation, prompt-injection, policy-bypass, and unauthorized-tool tests;
- historical reconstruction and correction tests;
- performance, capacity, disaster-recovery, and operational exercise evidence.

17. Repository Review Boundary and Previously Inspected Evidence

- Regulatory, payment, provider, and shipping capabilities are registered.
- Customs integration remains qualified external evidence until concrete adapters and responses are inspected.

Area	Classification for this chapter
Target aggregate, API, tables, state model, and workflow	REFERENCE_ARCHITECTURE
Registered supporting modules	DOCUMENTATION_ONLY_CLAIM until executable files are inspected
Concrete behavior already established in earlier repository-grounded chapters	retain the original classification and evidence
Exact integration paths not inspected for this stage	REQUIRES_REPOSITORY_REVIEW
External authority or provider outcome	qualified external observation, never platform-owned truth

No statement in this chapter upgrades an implementation status merely because a module name, frontend route, README, plan, or database catalog entry exists.

18. Worked Scenario

An upstream governed commitment enters the stage with exact subject and version references. The initiating principal is authenticated, and its mandate is checked at command time. Evidence is gathered and evaluated against the applicable policy version and jurisdiction.

A recommendation may be produced, but an authorized Decision remains separate. The Domain accepts an idempotent Command and records state plus an outbox entry atomically. When an external party participates, a timeout creates Outcome Unknown; the workflow queries the external source or waits for a verified callback instead of repeating execution blindly.

On success or failure, the system records a completed fact, publishes a stable Event, updates downstream projections, and preserves source and knowledge cutoff. Later corrections append new facts and identify downstream impact rather than replacing history. AI assists through typed tools and cannot own the subject, grant authority, or invent an outcome.

19. Conformance Checklist

- ☐ Stable subject and version identities are explicit.
 - ☐ Proposition ownership and external authority are named.
 - ☐ Entry and exit criteria are testable.
 - ☐ State, Recommendation, Decision, Intent, execution, and outcome remain distinct.
 - ☐ Commands are authorized, version-guarded, and idempotent.
 - ☐ Events describe completed facts.
 - ☐ Evidence includes provenance, time, integrity, and correction links.
 - ☐ Portals show freshness, uncertainty, and Outcome Unknown.
 - ☐ AI uses typed tools and approved autonomy.
 - ☐ Security, privacy, tenant, facility, and jurisdiction boundaries are enforced.
 - ☐ Reconciliation precedes risky duplicate execution.
 - ☐ Corrections preserve history.
 - ☐ Tests prove invariants, security, failure, and recovery.
 - ☐ Implementation status is not upgraded without executable repository evidence.
-

20. Status Vocabulary

- VERIFIED_IMPLEMENTED: executable source and the relevant behavior were inspected.
- IMPLEMENTED_WITH_GAP: executable behavior exists with a material governance or completeness gap.
- PARTIAL_IMPLEMENTATION: only part of the target capability is evidenced.
- REFERENCE_ARCHITECTURE: normative target behavior; no claim that it exists today.
- PLANNED_NOT_IMPLEMENTED: explicitly planned but not evidenced as executable.
- DOCUMENTATION_ONLY_CLAIM: documentation or a client contract claims behavior without verified backend evidence.
- CONFLICT_REQUIRES_CANONICALIZATION: sources disagree and no final owner has been established.
- REQUIRES_REPOSITORY_REVIEW: the necessary executable evidence has not yet been inspected.

Publication Chapter 22 — Delivery and Warehouse Receiving

Metadata	Value
Publication Chapter	22
Canonical Lifecycle Stage	21
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE with explicit repository notes
Accountable capability	Warehouse / Shipping / Quality
Canon dependencies	CANON-001 through CANON-011
Constitutional dependencies	ART-001 through ART-016
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This stage exists to **evidence destination delivery and warehouse receipt, count, condition, quarantine, and discrepancy without automatically creating available inventory.**

Its output is not merely a screen state or document download. It is a governed set of identified subjects, Relationships, Decisions, Intent, Commands, completed facts, external observations, evidence, and correction history.

- Reality or external outcome
- ≠ platform Observation
 - ≠ derived Perspective
 - ≠ AI Prediction
 - ≠ Recommendation
 - ≠ authorized Decision
 - ≠ execution Intent

1.1 Accountability

The accountable capability is **Warehouse / Shipping / Quality**. Supporting Portals, workflow engines, reports, search indexes, integration adapters, and AI Agents may coordinate or present the work, but they do not silently become owners of the underlying business truth.

1.2 Primary business outcomes

- a stable subject identity and exact source-version lineage;
- explicit authority and effective mandate;
- a testable lifecycle outcome;
- evidence sufficient for downstream use and later reconstruction;
- visible failure, partial completion, uncertainty, and correction;
- a completed-fact Event or qualified external observation;
- downstream activation without shared-database ownership transfer.

1.3 Non-goals

- treating a registered module as proof of complete implementation;

- treating a frontend route as proof of backend behavior;
- allowing an AI recommendation to become a binding Decision;
- inventing external-provider or government-authority success;
- overwriting history during amendment or correction;
- combining unrelated states into one generic status field.

2. Entry and Exit Criteria

2.1 Entry criteria

1. Every upstream subject reference resolves to a stable identifier.
2. The exact approved, accepted, issued, or observed version is known.
3. The initiating human, service, organization, or AI Agent is authenticated.
4. The represented principal and mandate are valid at command time.
5. Required Evidence exists, or an explicit authorized exception is linked.
6. Applicable Policy and jurisdiction versions are fixed.
7. Duplicate, stale-version, and replay checks have run.
8. Any conflicting active process is resolved, suspended, or explicitly related.

2.2 Exit criteria

1. The authoritative Domain or external authority records the outcome.
2. State and version changes are durable.
3. A completed-fact Event or qualified observation is available.
4. Downstream consumers receive stable subject, correlation, and causation references.
5. Unknown, partial, disputed, or failed outcomes remain visible.
6. Corrections preserve the original record and reason.
7. Metrics, audit evidence, and operational signals are emitted.
8. A qualified reviewer can reconstruct who knew what, when, and why.

3. Ubiquitous Language and Subject Model

Primary subject families:

Subject	Required interpretation
DeliveryAppointment	independent identified subject or governed subordinate entity; ownership must be declared
ProofOfDelivery	independent identified subject or governed subordinate entity; ownership must be declared
WarehouseReceipt	independent identified subject or governed subordinate entity; ownership must be declared
ReceiptLine	independent identified subject or governed subordinate entity; ownership must be declared
PackageReceipt	independent identified subject or governed subordinate entity; ownership must be declared
ReceivingInspection	independent identified subject or governed subordinate entity; ownership must be declared
ReceivingDiscrepancy	independent identified subject or governed subordinate entity; ownership must be declared
QuarantineRecord	independent identified subject or governed subordinate entity; ownership must be declared

Reference aggregate:

```
DeliveryAppointment {
  aggregateId
  tenantId
  organizationContext
  canonicalSubjectRefs[]
  sourceVersionRefs[]
  lifecycleState
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
```

```

policyRefs[]
authorityRef
jurisdictionRefs[]
createdAt
effectiveAt
updatedAt
version
correctionOf?
correlationId
}

```

3.1 Identity and relationship rules

- a display name, document number, provider reference, barcode, or URL is not automatically canonical identity;
- relationships such as custody, mandate, allocation, publication, reservation, or obligation are independent facts;
- cross-Domain references use stable IDs and qualified snapshots, not shared-table mutation;
- external identifiers retain issuer and namespace;
- merge, split, correction, and supersession preserve lineage.

3.2 Invariants

1. Submitted, approved, accepted, issued, or signed versions are immutable.
2. State changes occur through authorized Commands.
3. Completed-fact Events use past-tense semantics.
4. A Recommendation cannot self-promote to a Decision.
5. An external outcome retains its source authority.
6. Derived projections cannot become authoritative by repeated copying.
7. All writes are tenant-, organization-, facility-, and authority-scoped as applicable.
8. Retried writes are idempotent.
9. Outcome Unknown is not converted to success or failure by assumption.
10. Corrections reference the corrected fact and downstream impact.

4. Lifecycle State Model

4.1 State vocabulary

State	Meaning
APPOINTMENT_PENDING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
SCHEDULED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
ARRIVED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
UNLOADING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
COUNTING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
INSPECTION	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
DISCREPANCY	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
QUARANTINED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
RECEIVED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
REJECTED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
CLOSED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification

4.2 Reference state machine

```

stateDiagram-v2
    [*] --> APPOINTMENT_PENDING
    APPOINTMENT_PENDING --> SCHEDULED
    SCHEDULED --> ARRIVED

```

```

ARRIVED --> UNLOADING
UNLOADING --> COUNTING
COUNTING --> INSPECTION
INSPECTION --> DISCREPANCY
DISCREPANCY --> QUARANTINED
QUARANTINED --> RECEIVED
RECEIVED --> REJECTED
REJECTED --> CLOSED

```

The diagram shows a readable primary path. A conformant implementation may include parallel approvals, amendment, suspension, timeout, appeal, rejection, partial completion, reconciliation, and correction paths, but it must preserve the same semantic distinctions.

4.3 Transition matrix

From	To	Minimum guard
APPOINTMENT_PENDING	SCHEDULED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
SCHEDULED	ARRIVED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
ARRIVED	UNLOADING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
UNLOADING	COUNTING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
COUNTING	INSPECTION	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
INSPECTION	DISCREPANCY	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
DISCREPANCY	QUARANTINED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
QUARANTINED	RECEIVED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
RECEIVED	REJECTED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
REJECTED	CLOSED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard

4.4 Forbidden state behavior

- direct arbitrary assignment from a Portal request;
- using one status to represent lifecycle, approval, payment, document, provider, and dispute state simultaneously;
- deleting terminal history to reopen a process;
- inferring completion from elapsed time or an AI prediction;
- moving past a mandatory gate because a downstream system already acted.

5. Decision, Policy, and Authority

Material Decisions in this stage include readiness, eligibility, approval, exception, amendment, correction, and downstream activation decisions.

Reference Decision:

```

Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
}

```

```

evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Decision rules

- the Decision references exact source versions;
 - authority and delegation are evaluated at Decision time;
 - a Policy version and jurisdiction context are recorded;
 - mandatory-gate failure blocks progression unless a lawful exception exists;
 - dissent, uncertainty, and missing Evidence remain visible;
 - Decision correction does not erase the original Decision;
 - AI may prepare a briefing, but cannot manufacture authority.
-

6. Evidence, Documents, and Knowledge Cutoff

Every binding or high-impact action requires an Evidence Set containing:

- evidence identity and integrity hash where required;
- issuing, observing, or signing party;
- observed time, received time, and effective-time interval;
- subject and source-version relationships;
- qualification, confidence, and limitations;
- contradiction, retraction, supersession, and correction links;
- jurisdiction, classification, retention, and residency;
- chain of custody for physical or high-integrity evidence;
- the knowledge cutoff used by the Decision.

A document is not automatically truth. It is an Artifact containing Claims. Its legal or operational effect depends on issuer, mandate, signature, version, applicable rules, and external authority.

7. Commands, Events, and Queries

7.1 Commands

- ScheduleDelivery
- RecordProofOfDelivery
- StartReceiving
- RecordReceiptLine
- OpenReceivingDiscrepancy
- PlaceInQuarantine
- AcceptReceipt
- RejectReceipt
- CorrectReceipt

Each Command shall contain an idempotency key, actor identity, represented principal, tenant and organization context, correlation ID, expected version where relevant, reason for privileged action, and exact target subject.

7.2 Events

- DeliveryScheduled
- DeliveryArrived
- ProofOfDeliveryRecorded
- WarehouseReceivingStarted
- ReceivingDiscrepancyOpened
- GoodsQuarantined
- WarehouseReceiptAccepted
- WarehouseReceiptCorrected

Events describe completed facts or explicitly qualified observations. A request, proposed action, timer, or prediction is not a completed-fact Event.

7.3 Queries and perspectives

- GetDeliveryAppointment
- ListDeliveryAppointmentRecords
- GetDeliveryAppointmentTimeline
- GetDeliveryAppointmentEvidence
- GetDeliveryAppointmentDecisionPackage
- GetDeliveryAppointmentExceptions
- GetDeliveryAppointmentAuditTrail
- GetDeliveryAppointmentCurrentPerspective

Every query response states data freshness, source cutoff, version, and whether values are authoritative, observed, derived, predicted, or disputed.

8. Workflow, Timeout, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Evidence and policy gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Create Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and available compensation]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Trigger downstream stage]
```

Required workflow behavior:

- durable process and correlation identity;
 - persisted approvals and timers;
 - bounded retries and retry budgets;
 - idempotent command submission;
 - safe restart after deployment or outage;
 - compensation that records new Intent instead of deleting history;
 - visible stalled, orphaned, and manually intervened processes;
 - operator access to source responses and downstream impact;
 - reconciliation before repeating an external financial, legal, or physical action.
-

9. Target API Contract

Reference surface:

```
POST /api/v1/delivery_and_warehouse_receiving/commands
GET  /api/v1/delivery_and_warehouse_receiving/{id}
GET  /api/v1/delivery_and_warehouse_receiving/{id}/timeline
GET  /api/v1/delivery_and_warehouse_receiving/{id}/evidence
GET  /api/v1/delivery_and_warehouse_receiving/{id}/decisions
POST /api/v1/delivery_and_warehouse_receiving/{id}/exceptions
POST /api/v1/delivery_and_warehouse_receiving/{id}/corrections
```

These are reference endpoints, not claims that the exact routes exist.

Contract requirements:

- typed request and response schemas;
- action-level authorization;
- trusted authentication context for tenant identity;
- idempotency for writes;
- optimistic concurrency or equivalent aggregate guard;
- RFC problem-details or governed equivalent;
- stable pagination and sorting;
- versioning and deprecation policy;

- trace, correlation, and causation propagation;
- explicit Outcome Unknown response when external completion is unresolved.

10. Target Persistence and Data Ownership

Reference table families:

Table family	Purpose
stage_aggregate	authoritative lifecycle subject
stage_version	immutable approved/submitted/issued versions
stage_decision	binding Decisions and rationale
stage_evidence_link	evidence and provenance relationships
stage_event_outbox	transactional Event publication
stage_idempotency	duplicate-write protection
stage_exception	exception, claim, hold, or dispute
stage_correction	correction and supersession lineage
stage_audit	privileged action evidence

Data rules:

- the owning service controls its schema and migrations;
- foreign keys stay inside an ownership boundary;
- cross-Domain relationships use stable IDs or contracts;
- effective-time and recorded-time are separate when needed;
- binding versions are immutable;
- indexes support tenant, owner, state, correlation, external reference, and effective time;
- retention, legal hold, privacy, residency, backup, and restore are documented;
- analytics, search, reports, and vector stores are projections.

11. Portal and Experience Architecture

Required Portal behavior:

- display exact version, owner, state, and last-updated time;
- distinguish authoritative, observed, derived, predicted, and disputed values;
- expose required action and authority;
- show loading, empty, partial, stale, error, timeout, Outcome Unknown, and permission-denied states;
- provide accessible keyboard and screen-reader behavior;
- handle locale, currency, units, dates, and RTL/LTR correctly;
- preserve evidence and download audit;
- never trust a browser-provided tenant header as authority;
- prevent double submission while still supporting safe retry;
- show correction and supersession history.

Operator surfaces additionally expose correlation, source responses, retry budget, replay status, compensation options, and affected downstream subjects.

12. AI Participation and Tool Governance

Permitted AI tasks:

- assist damage detection
- detect count anomalies
- reconcile documents with receipt
- prioritize discrepancies

AI may also summarize evidence, detect gaps, recommend next actions, and draft human-readable explanations.

AI shall not:

- own or directly mutate Domain databases;
- fabricate external-provider, carrier, bank, customs, inspection, or delivery outcomes;
- grant itself permission or mandate;
- silently change an approved or issued version;
- suppress uncertainty, contradictions, policy failures, or human dissent;

- execute high-impact actions outside typed tools and approval policy;
- create shadow records for canonical subjects.

Reference automation:

Task	Maximum default
extraction and classification	A2, automatic with reviewable evidence
anomaly detection and recommendation	A2/A3 according to risk
binding commercial, legal, financial, quality, or custody Decision	A0/A1, human authority required
external execution	typed tool, policy, approval, idempotency, and outcome observation
correction of authoritative truth	governed human-authorized workflow

Every Agent Run records prompt/model/tool versions, context sources, policy evaluation, approvals, tool calls, structured outputs, uncertainty, final observed outcome, and recovery actions.

13. Security, Privacy, and Trust Boundaries

Minimum controls:

- authenticated human, service, organization, device, and AI identities;
- represented principal and effective mandate validation;
- least privilege, ABAC/ReBAC where appropriate, and separation of duties;
- tenant, organization, facility, channel, and jurisdiction isolation;
- integrity protection for binding documents and Evidence;
- external callback signature and replay verification;
- confidential handling of commercial, personal, financial, transport, and regulatory data;
- secret isolation and no sensitive values in logs or prompts;
- privileged-action audit and emergency-access review;
- threat models for impersonation, tampering, replay, confused deputy, fraud, exfiltration, and prompt injection.

14. Reliability, SLOs, and Operations

Reference objectives:

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by the business deadline of the stage

Operational requirements include dashboards, alerts, runbooks, ownership, dependency health, retry budgets, DLQ/replay controls, Event outbox/inbox, workflow restart, reconciliation, capacity, backup/restore, incident command, and post-incident correction.

15. Failure Modes and Recovery

Representative failures:

- stale or wrong source version;
- duplicate submission or replay;
- external timeout after possible success;
- contradictory source observations;
- lost Event after committed state;
- approval or mandate revoked during execution;
- downstream action performed before a mandatory gate;
- partial completion across multiple subjects;

- evidence later retracted or corrected;
- tenant, facility, or jurisdiction mismatch.

Recovery principles:

1. Query the authoritative source before repeating an external action.
2. Preserve partial success.
3. Use compensating Intent instead of destructive history rewrite.
4. Publish correction Events when distributed facts change.
5. Quarantine high-impact ambiguity.
6. Escalate when authority or legal effect is unclear.
7. Record business impact, affected subjects, and final reconciliation.

16. Testing and Conformance

Required tests:

- aggregate invariant and transition tests;
- authorization, mandate, separation-of-duties, and tenant-isolation tests;
- idempotency and optimistic-concurrency tests;
- migration, constraint, index, and rollback tests;
- outbox/inbox, ordering, duplicate, replay, and DLQ tests;
- workflow restart, timeout, compensation, and manual-intervention tests;
- provider callback verification and Outcome Unknown tests;
- Portal E2E for success, partial, stale, denied, and error journeys;
- accessibility, locale, currency, unit, and RTL tests;
- security adversarial and data-exfiltration tests;
- AI evaluation, prompt-injection, policy-bypass, and unauthorized-tool tests;
- historical reconstruction and correction tests;
- performance, capacity, disaster-recovery, and operational exercise evidence.

17. Repository Review Boundary and Previously Inspected Evidence

- Warehouse, inventory, shipping, and quality-related capabilities are registered.
- Receiving and inventory ownership remain separate even where current implementation overlaps.

Area	Classification for this chapter
Target aggregate, API, tables, state model, and workflow Registered supporting modules	REFERENCE_ARCHITECTURE DOCUMENTATION_ONLY_CLAIM until executable files are inspected
Concrete behavior already established in earlier repository-grounded chapters	retain the original classification and evidence
Exact integration paths not inspected for this stage	REQUIRES_REPOSITORY_REVIEW
External authority or provider outcome	qualified external observation, never platform-owned truth

No statement in this chapter upgrades an implementation status merely because a module name, frontend route, README, plan, or database catalog entry exists.

18. Worked Scenario

An upstream governed commitment enters the stage with exact subject and version references. The initiating principal is authenticated, and its mandate is checked at command time. Evidence is gathered and evaluated against the applicable policy version and jurisdiction.

A recommendation may be produced, but an authorized Decision remains separate. The Domain accepts an idempotent Command and records state plus an outbox entry atomically. When an external party participates, a timeout creates Outcome Unknown; the workflow queries the external source or waits for a verified callback instead of repeating execution blindly.

On success or failure, the system records a completed fact, publishes a stable Event, updates downstream projections, and preserves source and knowledge cutoff. Later corrections append new facts and identify downstream impact rather than replacing history. AI assists through typed tools and cannot own the subject, grant authority, or invent an outcome.

19. Conformance Checklist

- ☐ Stable subject and version identities are explicit.
 - ☐ Proposition ownership and external authority are named.
 - ☐ Entry and exit criteria are testable.
 - ☐ State, Recommendation, Decision, Intent, execution, and outcome remain distinct.
 - ☐ Commands are authorized, version-guarded, and idempotent.
 - ☐ Events describe completed facts.
 - ☐ Evidence includes provenance, time, integrity, and correction links.
 - ☐ Portals show freshness, uncertainty, and Outcome Unknown.
 - ☐ AI uses typed tools and approved autonomy.
 - ☐ Security, privacy, tenant, facility, and jurisdiction boundaries are enforced.
 - ☐ Reconciliation precedes risky duplicate execution.
 - ☐ Corrections preserve history.
 - ☐ Tests prove invariants, security, failure, and recovery.
 - ☐ Implementation status is not upgraded without executable repository evidence.
-

20. Status Vocabulary

- VERIFIED_IMPLEMENTED: executable source and the relevant behavior were inspected.
- IMPLEMENTED_WITH_GAP: executable behavior exists with a material governance or completeness gap.
- PARTIAL_IMPLEMENTATION: only part of the target capability is evidenced.
- REFERENCE_ARCHITECTURE: normative target behavior; no claim that it exists today.
- PLANNED_NOT_IMPLEMENTED: explicitly planned but not evidenced as executable.
- DOCUMENTATION_ONLY_CLAIM: documentation or a client contract claims behavior without verified backend evidence.
- CONFLICT_REQUIRES_CANONICALIZATION: sources disagree and no final owner has been established.
- REQUIRES_REPOSITORY_REVIEW: the necessary executable evidence has not yet been inspected.

Publication Chapter 23 — Inventory Synchronization, Reservation, and Availability

Metadata	Value
Publication Chapter	23
Canonical Lifecycle Stage	22
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE with explicit repository notes
Accountable capability	Inventory / Warehouse
Canon dependencies	CANON-001 through CANON-011
Constitutional dependencies	ART-001 through ART-016
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This stage exists to **derive governed stock positions and channel eligibility from receipt, quality, reservation, allocation, and adjustment facts**.

Its output is not merely a screen state or document download. It is a governed set of identified subjects, Relationships, Decisions, Intent, Commands, completed facts, external observations, evidence, and correction history.

Reality or external outcome
≠ platform Observation
≠ derived Perspective
≠ AI Prediction
≠ Recommendation
≠ authorized Decision
≠ execution Intent

1.1 Accountability

The accountable capability is **Inventory / Warehouse**. Supporting Portals, workflow engines, reports, search indexes, integration adapters, and AI Agents may coordinate or present the work, but they do not silently become owners of the underlying business truth.

1.2 Primary business outcomes

- a stable subject identity and exact source-version lineage;
- explicit authority and effective mandate;
- a testable lifecycle outcome;
- evidence sufficient for downstream use and later reconstruction;
- visible failure, partial completion, uncertainty, and correction;
- a completed-fact Event or qualified external observation;
- downstream activation without shared-database ownership transfer.

1.3 Non-goals

- treating a registered module as proof of complete implementation;
- treating a frontend route as proof of backend behavior;
- allowing an AI recommendation to become a binding Decision;
- inventing external-provider or government-authority success;
- overwriting history during amendment or correction;
- combining unrelated states into one generic status field.

2. Entry and Exit Criteria

2.1 Entry criteria

1. Every upstream subject reference resolves to a stable identifier.
2. The exact approved, accepted, issued, or observed version is known.
3. The initiating human, service, organization, or AI Agent is authenticated.
4. The represented principal and mandate are valid at command time.
5. Required Evidence exists, or an explicit authorized exception is linked.
6. Applicable Policy and jurisdiction versions are fixed.
7. Duplicate, stale-version, and replay checks have run.
8. Any conflicting active process is resolved, suspended, or explicitly related.

2.2 Exit criteria

1. The authoritative Domain or external authority records the outcome.
2. State and version changes are durable.
3. A completed-fact Event or qualified observation is available.
4. Downstream consumers receive stable subject, correlation, and causation references.
5. Unknown, partial, disputed, or failed outcomes remain visible.
6. Corrections preserve the original record and reason.
7. Metrics, audit evidence, and operational signals are emitted.
8. A qualified reviewer can reconstruct who knew what, when, and why.

3. Ubiquitous Language and Subject Model

Primary subject families:

Subject	Required interpretation
InventoryPosition	independent identified subject or governed subordinate entity; ownership must be declared
InventoryLot	independent identified subject or governed subordinate entity; ownership must be declared
StockMovement	independent identified subject or governed subordinate entity; ownership must be declared
Reservation	independent identified subject or governed subordinate entity; ownership must be declared
Allocation	independent identified subject or governed subordinate entity; ownership must be declared
AvailabilityProjection	independent identified subject or governed subordinate entity; ownership must be declared
InventoryAdjustment	independent identified subject or governed subordinate entity; ownership must be declared
CycleCount	independent identified subject or governed subordinate entity; ownership must be declared

Reference aggregate:

```
InventoryPosition {
  aggregateId
  tenantId
  organizationContext
  canonicalSubjectRefs[]
  sourceVersionRefs[]
  lifecycleState
  decisionRefs[]
}
```

```

intentRefs[]
evidenceSetRef
knowledgeCutoff
policyRefs[]
authorityRef
jurisdictionRefs[]
createdAt
effectiveAt
updatedAt
version
correctionOf?
correlationId
}

```

3.1 Identity and relationship rules

- a display name, document number, provider reference, barcode, or URL is not automatically canonical identity;
- relationships such as custody, mandate, allocation, publication, reservation, or obligation are independent facts;
- cross-Domain references use stable IDs and qualified snapshots, not shared-table mutation;
- external identifiers retain issuer and namespace;
- merge, split, correction, and supersession preserve lineage.

3.2 Invariants

1. Submitted, approved, accepted, issued, or signed versions are immutable.
2. State changes occur through authorized Commands.
3. Completed-fact Events use past-tense semantics.
4. A Recommendation cannot self-promote to a Decision.
5. An external outcome retains its source authority.
6. Derived projections cannot become authoritative by repeated copying.
7. All writes are tenant-, organization-, facility-, and authority-scoped as applicable.
8. Retried writes are idempotent.
9. Outcome Unknown is not converted to success or failure by assumption.
10. Corrections reference the corrected fact and downstream impact.

4. Lifecycle State Model

4.1 State vocabulary

State	Meaning
PENDING_RECEIPT	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
QUARANTINED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
AVAILABLE	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
RESERVED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
ALLOCATED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
PICKING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
IN_TRANSIT_INTERNAL	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
DAMAGED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
BLOCKED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
EXPIRED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
DEPLETED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification

4.2 Reference state machine

```

stateDiagram-v2
    [*] --> PENDING_RECEIPT

```

```

PENDING_RECEIPT --> QUARANTINED
QUARANTINED --> AVAILABLE
AVAILABLE --> RESERVED
RESERVED --> ALLOCATED
ALLOCATED --> PICKING
PICKING --> IN_TRANSIT_INTERNAL
IN_TRANSIT_INTERNAL --> DAMAGED
DAMAGED --> BLOCKED
BLOCKED --> EXPIRED
EXPIRED --> DEPLETED

```

The diagram shows a readable primary path. A conformant implementation may include parallel approvals, amendment, suspension, timeout, appeal, rejection, partial completion, reconciliation, and correction paths, but it must preserve the same semantic distinctions.

4.3 Transition matrix

From	To	Minimum guard
PENDING_RECEIPT	QUARANTINED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
QUARANTINED	AVAILABLE	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
AVAILABLE	RESERVED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
RESERVED	ALLOCATED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
ALLOCATED	PICKING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
PICKING	IN_TRANSIT_INTERNAL	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
IN_TRANSIT_INTERNAL	DAMAGED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
DAMAGED	BLOCKED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
BLOCKED	EXPIRED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
EXPIRED	DEPLETED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard

4.4 Forbidden state behavior

- direct arbitrary assignment from a Portal request;
- using one status to represent lifecycle, approval, payment, document, provider, and dispute state simultaneously;
- deleting terminal history to reopen a process;
- inferring completion from elapsed time or an AI prediction;
- moving past a mandatory gate because a downstream system already acted.

5. Decision, Policy, and Authority

Material Decisions in this stage include readiness, eligibility, approval, exception, amendment, correction, and downstream activation decisions.

Reference Decision:

```

Decision {
  decisionId
  decisionType
  subjectRef
}

```

```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Decision rules

- the Decision references exact source versions;
 - authority and delegation are evaluated at Decision time;
 - a Policy version and jurisdiction context are recorded;
 - mandatory-gate failure blocks progression unless a lawful exception exists;
 - dissent, uncertainty, and missing Evidence remain visible;
 - Decision correction does not erase the original Decision;
 - AI may prepare a briefing, but cannot manufacture authority.
-

6. Evidence, Documents, and Knowledge Cutoff

Every binding or high-impact action requires an Evidence Set containing:

- evidence identity and integrity hash where required;
- issuing, observing, or signing party;
- observed time, received time, and effective-time interval;
- subject and source-version relationships;
- qualification, confidence, and limitations;
- contradiction, retraction, supersession, and correction links;
- jurisdiction, classification, retention, and residency;
- chain of custody for physical or high-integrity evidence;
- the knowledge cutoff used by the Decision.

A document is not automatically truth. It is an Artifact containing Claims. Its legal or operational effect depends on issuer, mandate, signature, version, applicable rules, and external authority.

7. Commands, Events, and Queries

7.1 Commands

- CreateInventoryPosition
- ReleaseInventoryLot
- ReserveInventory
- ReleaseReservation
- AllocateInventory
- AdjustInventory
- RecordCycleCount
- CorrectInventoryMovement

Each Command shall contain an idempotency key, actor identity, represented principal, tenant and organization context, correlation ID, expected version where relevant, reason for privileged action, and exact target subject.

7.2 Events

- InventoryPositionCreated
- InventoryReleased
- InventoryReserved
- ReservationReleased
- InventoryAllocated
- InventoryAdjusted
- InventoryDiscrepancyDetected
- AvailabilityChanged

Events describe completed facts or explicitly qualified observations. A request, proposed action, timer, or prediction is not a completed-fact Event.

7.3 Queries and perspectives

- `GetInventoryPosition`
- `ListInventoryPositionRecords`
- `GetInventoryPositionTimeline`
- `GetInventoryPositionEvidence`
- `GetInventoryPositionDecisionPackage`
- `GetInventoryPositionExceptions`
- `GetInventoryPositionAuditTrail`
- `GetInventoryPositionCurrentPerspective`

Every query response states data freshness, source cutoff, version, and whether values are authoritative, observed, derived, predicted, or disputed.

8. Workflow, Timeout, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Evidence and policy gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Create Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and available compensation]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Trigger downstream stage]
```

Required workflow behavior:

- durable process and correlation identity;
 - persisted approvals and timers;
 - bounded retries and retry budgets;
 - idempotent command submission;
 - safe restart after deployment or outage;
 - compensation that records new Intent instead of deleting history;
 - visible stalled, orphaned, and manually intervened processes;
 - operator access to source responses and downstream impact;
 - reconciliation before repeating an external financial, legal, or physical action.
-

9. Target API Contract

Reference surface:

```
POST /api/v1/inventory_synchronization_reservation_and_availability/commands
GET  /api/v1/inventory_synchronization_reservation_and_availability/{id}
GET  /api/v1/inventory_synchronization_reservation_and_availability/{id}/timeline
GET  /api/v1/inventory_synchronization_reservation_and_availability/{id}/evidence
GET  /api/v1/inventory_synchronization_reservation_and_availability/{id}/decisions
POST /api/v1/inventory_synchronization_reservation_and_availability/{id}/exceptions
POST /api/v1/inventory_synchronization_reservation_and_availability/{id}/corrections
```

These are reference endpoints, not claims that the exact routes exist.

Contract requirements:

- typed request and response schemas;
- action-level authorization;
- trusted authentication context for tenant identity;
- idempotency for writes;
- optimistic concurrency or equivalent aggregate guard;
- RFC problem-details or governed equivalent;
- stable pagination and sorting;
- versioning and deprecation policy;

- trace, correlation, and causation propagation;
- explicit Outcome Unknown response when external completion is unresolved.

10. Target Persistence and Data Ownership

Reference table families:

Table family	Purpose
stage_aggregate	authoritative lifecycle subject
stage_version	immutable approved/submitted/issued versions
stage_decision	binding Decisions and rationale
stage_evidence_link	evidence and provenance relationships
stage_event_outbox	transactional Event publication
stage_idempotency	duplicate-write protection
stage_exception	exception, claim, hold, or dispute
stage_correction	correction and supersession lineage
stage_audit	privileged action evidence

Data rules:

- the owning service controls its schema and migrations;
- foreign keys stay inside an ownership boundary;
- cross-Domain relationships use stable IDs or contracts;
- effective-time and recorded-time are separate when needed;
- binding versions are immutable;
- indexes support tenant, owner, state, correlation, external reference, and effective time;
- retention, legal hold, privacy, residency, backup, and restore are documented;
- analytics, search, reports, and vector stores are projections.

11. Portal and Experience Architecture

Required Portal behavior:

- display exact version, owner, state, and last-updated time;
- distinguish authoritative, observed, derived, predicted, and disputed values;
- expose required action and authority;
- show loading, empty, partial, stale, error, timeout, Outcome Unknown, and permission-denied states;
- provide accessible keyboard and screen-reader behavior;
- handle locale, currency, units, dates, and RTL/LTR correctly;
- preserve evidence and download audit;
- never trust a browser-provided tenant header as authority;
- prevent double submission while still supporting safe retry;
- show correction and supersession history.

Operator surfaces additionally expose correlation, source responses, retry budget, replay status, compensation options, and affected downstream subjects.

12. AI Participation and Tool Governance

Permitted AI tasks:

- forecast demand and depletion
- detect stock anomalies
- recommend replenishment
- predict reservation risk

AI may also summarize evidence, detect gaps, recommend next actions, and draft human-readable explanations.

AI shall not:

- own or directly mutate Domain databases;
- fabricate external-provider, carrier, bank, customs, inspection, or delivery outcomes;
- grant itself permission or mandate;
- silently change an approved or issued version;
- suppress uncertainty, contradictions, policy failures, or human dissent;

- execute high-impact actions outside typed tools and approval policy;
- create shadow records for canonical subjects.

Reference automation:

Task	Maximum default
extraction and classification	A2, automatic with reviewable evidence
anomaly detection and recommendation	A2/A3 according to risk
binding commercial, legal, financial, quality, or custody Decision	A0/A1, human authority required
external execution	typed tool, policy, approval, idempotency, and outcome observation
correction of authoritative truth	governed human-authorized workflow

Every Agent Run records prompt/model/tool versions, context sources, policy evaluation, approvals, tool calls, structured outputs, uncertainty, final observed outcome, and recovery actions.

13. Security, Privacy, and Trust Boundaries

Minimum controls:

- authenticated human, service, organization, device, and AI identities;
- represented principal and effective mandate validation;
- least privilege, ABAC/ReBAC where appropriate, and separation of duties;
- tenant, organization, facility, channel, and jurisdiction isolation;
- integrity protection for binding documents and Evidence;
- external callback signature and replay verification;
- confidential handling of commercial, personal, financial, transport, and regulatory data;
- secret isolation and no sensitive values in logs or prompts;
- privileged-action audit and emergency-access review;
- threat models for impersonation, tampering, replay, confused deputy, fraud, exfiltration, and prompt injection.

14. Reliability, SLOs, and Operations

Reference objectives:

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by the business deadline of the stage

Operational requirements include dashboards, alerts, runbooks, ownership, dependency health, retry budgets, DLQ/replay controls, Event outbox/inbox, workflow restart, reconciliation, capacity, backup/restore, incident command, and post-incident correction.

15. Failure Modes and Recovery

Representative failures:

- stale or wrong source version;
- duplicate submission or replay;
- external timeout after possible success;
- contradictory source observations;
- lost Event after committed state;
- approval or mandate revoked during execution;
- downstream action performed before a mandatory gate;
- partial completion across multiple subjects;

- evidence later retracted or corrected;
- tenant, facility, or jurisdiction mismatch.

Recovery principles:

1. Query the authoritative source before repeating an external action.
2. Preserve partial success.
3. Use compensating Intent instead of destructive history rewrite.
4. Publish correction Events when distributed facts change.
5. Quarantine high-impact ambiguity.
6. Escalate when authority or legal effect is unclear.
7. Record business impact, affected subjects, and final reconciliation.

16. Testing and Conformance

Required tests:

- aggregate invariant and transition tests;
- authorization, mandate, separation-of-duties, and tenant-isolation tests;
- idempotency and optimistic-concurrency tests;
- migration, constraint, index, and rollback tests;
- outbox/inbox, ordering, duplicate, replay, and DLQ tests;
- workflow restart, timeout, compensation, and manual-intervention tests;
- provider callback verification and Outcome Unknown tests;
- Portal E2E for success, partial, stale, denied, and error journeys;
- accessibility, locale, currency, unit, and RTL tests;
- security adversarial and data-exfiltration tests;
- AI evaluation, prompt-injection, policy-bypass, and unauthorized-tool tests;
- historical reconstruction and correction tests;
- performance, capacity, disaster-recovery, and operational exercise evidence.

17. Repository Review Boundary and Previously Inspected Evidence

- Inventory and warehouse services are registered.
- Reservation, allocation, adjustment, and availability semantics require direct repository evidence before VERIFIED_IMPLEMENTED status.

Area	Classification for this chapter
Target aggregate, API, tables, state model, and workflow Registered supporting modules	REFERENCE_ARCHITECTURE DOCUMENTATION_ONLY_CLAIM until executable files are inspected
Concrete behavior already established in earlier repository-grounded chapters	retain the original classification and evidence
Exact integration paths not inspected for this stage	REQUIRES_REPOSITORY_REVIEW
External authority or provider outcome	qualified external observation, never platform-owned truth

No statement in this chapter upgrades an implementation status merely because a module name, frontend route, README, plan, or database catalog entry exists.

18. Worked Scenario

An upstream governed commitment enters the stage with exact subject and version references. The initiating principal is authenticated, and its mandate is checked at command time. Evidence is gathered and evaluated against the applicable policy version and jurisdiction.

A recommendation may be produced, but an authorized Decision remains separate. The Domain accepts an idempotent Command and records state plus an outbox entry atomically. When an external party participates, a timeout creates Outcome Unknown; the workflow queries the external source or waits for a verified callback instead of repeating execution blindly.

On success or failure, the system records a completed fact, publishes a stable Event, updates downstream projections, and preserves source and knowledge cutoff. Later corrections append new facts and identify downstream impact rather than replacing history. AI assists through typed tools and cannot own the subject, grant authority, or invent an outcome.

19. Conformance Checklist

- ☐ Stable subject and version identities are explicit.
 - ☐ Proposition ownership and external authority are named.
 - ☐ Entry and exit criteria are testable.
 - ☐ State, Recommendation, Decision, Intent, execution, and outcome remain distinct.
 - ☐ Commands are authorized, version-guarded, and idempotent.
 - ☐ Events describe completed facts.
 - ☐ Evidence includes provenance, time, integrity, and correction links.
 - ☐ Portals show freshness, uncertainty, and Outcome Unknown.
 - ☐ AI uses typed tools and approved autonomy.
 - ☐ Security, privacy, tenant, facility, and jurisdiction boundaries are enforced.
 - ☐ Reconciliation precedes risky duplicate execution.
 - ☐ Corrections preserve history.
 - ☐ Tests prove invariants, security, failure, and recovery.
 - ☐ Implementation status is not upgraded without executable repository evidence.
-

20. Status Vocabulary

- VERIFIED_IMPLEMENTED: executable source and the relevant behavior were inspected.
- IMPLEMENTED_WITH_GAP: executable behavior exists with a material governance or completeness gap.
- PARTIAL_IMPLEMENTATION: only part of the target capability is evidenced.
- REFERENCE_ARCHITECTURE: normative target behavior; no claim that it exists today.
- PLANNED_NOT_IMPLEMENTED: explicitly planned but not evidenced as executable.
- DOCUMENTATION_ONLY_CLAIM: documentation or a client contract claims behavior without verified backend evidence.
- CONFLICT_REQUIRES_CANONICALIZATION: sources disagree and no final owner has been established.
- REQUIRES_REPOSITORY_REVIEW: the necessary executable evidence has not yet been inspected.

Publication Chapter 24 — Product Publication, Marketplace, and Channel Eligibility

Metadata	Value
Publication Chapter	24
Canonical Lifecycle Stage	23
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE with explicit repository notes
Accountable capability	Product Catalog / Pricing / Inventory / Marketplace
Canon dependencies	CANON-001 through CANON-011
Constitutional dependencies	ART-001 through ART-016
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This stage exists to **publish governed offers only when identity, content, price, availability, compliance, and channel policies permit.**

Its output is not merely a screen state or document download. It is a governed set of identified subjects, Relationships, Decisions, Intent, Commands, completed facts, external observations, evidence, and correction history.

- Reality or external outcome
- ≠ platform Observation
 - ≠ derived Perspective
 - ≠ AI Prediction
 - ≠ Recommendation
 - ≠ authorized Decision
 - ≠ execution Intent

1.1 Accountability

The accountable capability is **Product Catalog / Pricing / Inventory / Marketplace.** Supporting Portals, workflow engines, reports, search indexes, integration adapters, and AI Agents may coordinate or present the work, but they do not silently become owners of the underlying business truth.

1.2 Primary business outcomes

- a stable subject identity and exact source-version lineage;
- explicit authority and effective mandate;
- a testable lifecycle outcome;
- evidence sufficient for downstream use and later reconstruction;
- visible failure, partial completion, uncertainty, and correction;
- a completed-fact Event or qualified external observation;
- downstream activation without shared-database ownership transfer.

1.3 Non-goals

- treating a registered module as proof of complete implementation;
 - treating a frontend route as proof of backend behavior;
 - allowing an AI recommendation to become a binding Decision;
 - inventing external-provider or government-authority success;
 - overwriting history during amendment or correction;
 - combining unrelated states into one generic status field.
-

2. Entry and Exit Criteria

2.1 Entry criteria

1. Every upstream subject reference resolves to a stable identifier.
2. The exact approved, accepted, issued, or observed version is known.
3. The initiating human, service, organization, or AI Agent is authenticated.
4. The represented principal and mandate are valid at command time.
5. Required Evidence exists, or an explicit authorized exception is linked.
6. Applicable Policy and jurisdiction versions are fixed.
7. Duplicate, stale-version, and replay checks have run.
8. Any conflicting active process is resolved, suspended, or explicitly related.

2.2 Exit criteria

1. The authoritative Domain or external authority records the outcome.
 2. State and version changes are durable.
 3. A completed-fact Event or qualified observation is available.
 4. Downstream consumers receive stable subject, correlation, and causation references.
 5. Unknown, partial, disputed, or failed outcomes remain visible.
 6. Corrections preserve the original record and reason.
 7. Metrics, audit evidence, and operational signals are emitted.
 8. A qualified reviewer can reconstruct who knew what, when, and why.
-

3. Ubiquitous Language and Subject Model

Primary subject families:

Subject	Required interpretation
Product	independent identified subject or governed subordinate entity; ownership must be declared
ProductVersion	independent identified subject or governed subordinate entity; ownership must be declared
Offer	independent identified subject or governed subordinate entity; ownership must be declared
ChannelPublication	independent identified subject or governed subordinate entity; ownership must be declared
PublicationEligibility	independent identified subject or governed subordinate entity; ownership must be declared
LocalizedContent	independent identified subject or governed subordinate entity; ownership must be declared
ModerationCase	independent identified subject or governed subordinate entity; ownership must be declared
PublicationAudit	independent identified subject or governed subordinate entity; ownership must be declared

Reference aggregate:

```
Product {
  aggregateId
  tenantId
  organizationContext
  canonicalSubjectRefs[]
  sourceVersionRefs[]
  lifecycleState
  decisionRefs[]
}
```

```

intentRefs[]
evidenceSetRef
knowledgeCutoff
policyRefs[]
authorityRef
jurisdictionRefs[]
createdAt
effectiveAt
updatedAt
version
correctionOf?
correlationId
}

```

3.1 Identity and relationship rules

- a display name, document number, provider reference, barcode, or URL is not automatically canonical identity;
- relationships such as custody, mandate, allocation, publication, reservation, or obligation are independent facts;
- cross-Domain references use stable IDs and qualified snapshots, not shared-table mutation;
- external identifiers retain issuer and namespace;
- merge, split, correction, and supersession preserve lineage.

3.2 Invariants

1. Submitted, approved, accepted, issued, or signed versions are immutable.
2. State changes occur through authorized Commands.
3. Completed-fact Events use past-tense semantics.
4. A Recommendation cannot self-promote to a Decision.
5. An external outcome retains its source authority.
6. Derived projections cannot become authoritative by repeated copying.
7. All writes are tenant-, organization-, facility-, and authority-scoped as applicable.
8. Retried writes are idempotent.
9. Outcome Unknown is not converted to success or failure by assumption.
10. Corrections reference the corrected fact and downstream impact.

4. Lifecycle State Model

4.1 State vocabulary

State	Meaning
DRAFT	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
CONTENT_PENDING	
ELIGIBILITY_PENDING	
MODERATION_PENDING	
APPROVED	
PUBLISHED	
SUSPENDED	
UNPUBLISHED	
REJECTED	
EXPIRED	

4.2 Reference state machine

```

stateDiagram-v2
    [*] --> DRAFT
    DRAFT --> CONTENT_PENDING

```

```

CONTENT_PENDING --> ELIGIBILITY_PENDING
ELIGIBILITY_PENDING --> MODERATION_PENDING
MODERATION_PENDING --> APPROVED
APPROVED --> PUBLISHED
PUBLISHED --> SUSPENDED
SUSPENDED --> UNPUBLISHED
UNPUBLISHED --> REJECTED
REJECTED --> EXPIRED

```

The diagram shows a readable primary path. A conformant implementation may include parallel approvals, amendment, suspension, timeout, appeal, rejection, partial completion, reconciliation, and correction paths, but it must preserve the same semantic distinctions.

4.3 Transition matrix

From	To	Minimum guard
DRAFT	CONTENT_PENDING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
CONTENT_PENDING	ELIGIBILITY_PENDING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
ELIGIBILITY_PENDING	MODERATION_PENDING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
MODERATION_PENDING	APPROVED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
APPROVED	PUBLISHED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
PUBLISHED	SUSPENDED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
SUSPENDED	UNPUBLISHED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
UNPUBLISHED	REJECTED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
REJECTED	EXPIRED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard

4.4 Forbidden state behavior

- direct arbitrary assignment from a Portal request;
- using one status to represent lifecycle, approval, payment, document, provider, and dispute state simultaneously;
- deleting terminal history to reopen a process;
- inferring completion from elapsed time or an AI prediction;
- moving past a mandatory gate because a downstream system already acted.

5. Decision, Policy, and Authority

Material Decisions in this stage include readiness, eligibility, approval, exception, amendment, correction, and downstream activation decisions.

Reference Decision:

```

Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
}

```

```

    representedOrganizationId
    mandateRef
    rationale
    outcome
    decidedAt
    effectiveAt
    correctionOf?
}

```

Decision rules

- the Decision references exact source versions;
 - authority and delegation are evaluated at Decision time;
 - a Policy version and jurisdiction context are recorded;
 - mandatory-gate failure blocks progression unless a lawful exception exists;
 - dissent, uncertainty, and missing Evidence remain visible;
 - Decision correction does not erase the original Decision;
 - AI may prepare a briefing, but cannot manufacture authority.
-

6. Evidence, Documents, and Knowledge Cutoff

Every binding or high-impact action requires an Evidence Set containing:

- evidence identity and integrity hash where required;
- issuing, observing, or signing party;
- observed time, received time, and effective-time interval;
- subject and source-version relationships;
- qualification, confidence, and limitations;
- contradiction, retraction, supersession, and correction links;
- jurisdiction, classification, retention, and residency;
- chain of custody for physical or high-integrity evidence;
- the knowledge cutoff used by the Decision.

A document is not automatically truth. It is an Artifact containing Claims. Its legal or operational effect depends on issuer, mandate, signature, version, applicable rules, and external authority.

7. Commands, Events, and Queries

7.1 Commands

- CreateOffer
- EvaluatePublicationEligibility
- SubmitForModeration
- ApprovePublication
- PublishOffer
- SuspendOffer
- UnpublishOffer
- UpdateLocalizedContent

Each Command shall contain an idempotency key, actor identity, represented principal, tenant and organization context, correlation ID, expected version where relevant, reason for privileged action, and exact target subject.

7.2 Events

- OfferCreated
- PublicationEligibilityEvaluated
- OfferApprovedForPublication
- OfferPublished
- OfferSuspended
- OfferUnpublished
- PublicationRejected

Events describe completed facts or explicitly qualified observations. A request, proposed action, timer, or prediction is not a completed-fact Event.

7.3 Queries and perspectives

- GetProduct

- ListProductRecords
- GetProductTimeline
- GetProductEvidence
- GetProductDecisionPackage
- GetProductExceptions
- GetProductAuditTrail
- GetProductCurrentPerspective

Every query response states data freshness, source cutoff, version, and whether values are authoritative, observed, derived, predicted, or disputed.

8. Workflow, Timeout, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Evidence and policy gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Create Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
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J --> K[Query source or wait for verified callback]
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```

Required workflow behavior:

- durable process and correlation identity;
 - persisted approvals and timers;
 - bounded retries and retry budgets;
 - idempotent command submission;
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 - compensation that records new Intent instead of deleting history;
 - visible stalled, orphaned, and manually intervened processes;
 - operator access to source responses and downstream impact;
 - reconciliation before repeating an external financial, legal, or physical action.
-

9. Target API Contract

Reference surface:

```

POST /api/v1/product_publication_marketplace_and_channel_eligibility/commands
GET  /api/v1/product_publication_marketplace_and_channel_eligibility/{id}
GET  /api/v1/product_publication_marketplace_and_channel_eligibility/{id}/timeline
GET  /api/v1/product_publication_marketplace_and_channel_eligibility/{id}/evidence
GET  /api/v1/product_publication_marketplace_and_channel_eligibility/{id}/decisions
POST /api/v1/product_publication_marketplace_and_channel_eligibility/{id}/exceptions
POST /api/v1/product_publication_marketplace_and_channel_eligibility/{id}/corrections

```

These are reference endpoints, not claims that the exact routes exist.

Contract requirements:

- typed request and response schemas;
 - action-level authorization;
 - trusted authentication context for tenant identity;
 - idempotency for writes;
 - optimistic concurrency or equivalent aggregate guard;
 - RFC problem-details or governed equivalent;
 - stable pagination and sorting;
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stage_idempotency	duplicate-write protection
stage_exception	exception, claim, hold, or dispute
stage_correction	correction and supersession lineage
stage_audit	privileged action evidence

Data rules:

- the owning service controls its schema and migrations;
- foreign keys stay inside an ownership boundary;
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- effective-time and recorded-time are separate when needed;
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- provide accessible keyboard and screen-reader behavior;
- handle locale, currency, units, dates, and RTL/LTR correctly;
- preserve evidence and download audit;
- never trust a browser-provided tenant header as authority;
- prevent double submission while still supporting safe retry;
- show correction and supersession history.

Operator surfaces additionally expose correlation, source responses, retry budget, replay status, compensation options, and affected downstream subjects.

12. AI Participation and Tool Governance

Permitted AI tasks:

- draft and translate content
- extract attributes
- assist moderation
- detect duplicate products
- support ranking

AI may also summarize evidence, detect gaps, recommend next actions, and draft human-readable explanations.

AI shall not:

- own or directly mutate Domain databases;
- fabricate external-provider, carrier, bank, customs, inspection, or delivery outcomes;
- grant itself permission or mandate;
- silently change an approved or issued version;
- suppress uncertainty, contradictions, policy failures, or human dissent;
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- integrity protection for binding documents and Evidence;
- external callback signature and replay verification;
- confidential handling of commercial, personal, financial, transport, and regulatory data;
- secret isolation and no sensitive values in logs or prompts;
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Event publication	99.9% within five minutes after committed fact
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state-state detection	bounded by the business deadline of the stage

Operational requirements include dashboards, alerts, runbooks, ownership, dependency health, retry budgets, DLQ/replay controls, Event outbox/inbox, workflow restart, reconciliation, capacity, backup/restore, incident command, and post-incident correction.

15. Failure Modes and Recovery

Representative failures:

- stale or wrong source version;
- duplicate submission or replay;
- external timeout after possible success;
- contradictory source observations;
- lost Event after committed state;
- approval or mandate revoked during execution;
- downstream action performed before a mandatory gate;
- partial completion across multiple subjects;
- evidence later retracted or corrected;
- tenant, facility, or jurisdiction mismatch.

Recovery principles:

1. Query the authoritative source before repeating an external action.

2. Preserve partial success.
 3. Use compensating Intent instead of destructive history rewrite.
 4. Publish correction Events when distributed facts change.
 5. Quarantine high-impact ambiguity.
 6. Escalate when authority or legal effect is unclear.
 7. Record business impact, affected subjects, and final reconciliation.
-

16. Testing and Conformance

Required tests:

- aggregate invariant and transition tests;
 - authorization, mandate, separation-of-duties, and tenant-isolation tests;
 - idempotency and optimistic-concurrency tests;
 - migration, constraint, index, and rollback tests;
 - outbox/inbox, ordering, duplicate, replay, and DLQ tests;
 - workflow restart, timeout, compensation, and manual-intervention tests;
 - provider callback verification and Outcome Unknown tests;
 - Portal E2E for success, partial, stale, denied, and error journeys;
 - accessibility, locale, currency, unit, and RTL tests;
 - security adversarial and data-exfiltration tests;
 - AI evaluation, prompt-injection, policy-bypass, and unauthorized-tool tests;
 - historical reconstruction and correction tests;
 - performance, capacity, disaster-recovery, and operational exercise evidence.
-

17. Repository Review Boundary and Previously Inspected Evidence

- Product Catalog and Marketplace services are registered.
- Exact offer and publication ownership requires repository reconciliation because known catalog overlaps exist.

Area	Classification for this chapter
Target aggregate, API, tables, state model, and workflow Registered supporting modules	REFERENCE_ARCHITECTURE DOCUMENTATION_ONLY_CLAIM until executable files are inspected
Concrete behavior already established in earlier repository-grounded chapters	retain the original classification and evidence
Exact integration paths not inspected for this stage	REQUIRES_REPOSITORY_REVIEW
External authority or provider outcome	qualified external observation, never platform-owned truth

No statement in this chapter upgrades an implementation status merely because a module name, frontend route, README, plan, or database catalog entry exists.

18. Worked Scenario

An upstream governed commitment enters the stage with exact subject and version references. The initiating principal is authenticated, and its mandate is checked at command time. Evidence is gathered and evaluated against the applicable policy version and jurisdiction.

A recommendation may be produced, but an authorized Decision remains separate. The Domain accepts an idempotent Command and records state plus an outbox entry atomically. When an external party participates, a timeout creates Outcome Unknown; the workflow queries the external source or waits for a verified callback instead of repeating execution blindly.

On success or failure, the system records a completed fact, publishes a stable Event, updates downstream projections, and preserves source and knowledge cutoff. Later corrections append new facts and identify downstream impact rather than replacing history. AI assists through typed tools and cannot own the subject, grant authority, or invent an outcome.

19. Conformance Checklist

- ☐ Stable subject and version identities are explicit.
 - ☐ Proposition ownership and external authority are named.
 - ☐ Entry and exit criteria are testable.
 - ☐ State, Recommendation, Decision, Intent, execution, and outcome remain distinct.
 - ☐ Commands are authorized, version-guarded, and idempotent.
 - ☐ Events describe completed facts.
 - ☐ Evidence includes provenance, time, integrity, and correction links.
 - ☐ Portals show freshness, uncertainty, and Outcome Unknown.
 - ☐ AI uses typed tools and approved autonomy.
 - ☐ Security, privacy, tenant, facility, and jurisdiction boundaries are enforced.
 - ☐ Reconciliation precedes risky duplicate execution.
 - ☐ Corrections preserve history.
 - ☐ Tests prove invariants, security, failure, and recovery.
 - ☐ Implementation status is not upgraded without executable repository evidence.
-

20. Status Vocabulary

- VERIFIED_IMPLEMENTED: executable source and the relevant behavior were inspected.
- IMPLEMENTED_WITH_GAP: executable behavior exists with a material governance or completeness gap.
- PARTIAL_IMPLEMENTATION: only part of the target capability is evidenced.
- REFERENCE_ARCHITECTURE: normative target behavior; no claim that it exists today.
- PLANNED_NOT_IMPLEMENTED: explicitly planned but not evidenced as executable.
- DOCUMENTATION_ONLY_CLAIM: documentation or a client contract claims behavior without verified backend evidence.
- CONFLICT_REQUIRES_CANONICALIZATION: sources disagree and no final owner has been established.
- REQUIRES_REPOSITORY_REVIEW: the necessary executable evidence has not yet been inspected.

Publication Chapter 25 — Checkout, Customer Order, and Commitment

Metadata	Value
Publication Chapter	25
Canonical Lifecycle Stage	24
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE with explicit repository notes
Accountable capability	Order / Shop / Marketplace
Canon dependencies	CANON-001 through CANON-011
Constitutional dependencies	ART-001 through ART-016
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This stage exists to **convert shopping intent into an accepted customer order using frozen price, inventory, payment, and delivery context.**

Its output is not merely a screen state or document download. It is a governed set of identified subjects, Relationships, Decisions, Intent, Commands, completed facts, external observations, evidence, and correction history.

- Reality or external outcome
- ≠ platform Observation
 - ≠ derived Perspective
 - ≠ AI Prediction
 - ≠ Recommendation
 - ≠ authorized Decision
 - ≠ execution Intent

1.1 Accountability

The accountable capability is **Order / Shop / Marketplace**. Supporting Portals, workflow engines, reports, search indexes, integration adapters, and AI Agents may coordinate or present the work, but they do not silently become owners of the underlying business truth.

1.2 Primary business outcomes

- a stable subject identity and exact source-version lineage;
- explicit authority and effective mandate;
- a testable lifecycle outcome;
- evidence sufficient for downstream use and later reconstruction;
- visible failure, partial completion, uncertainty, and correction;
- a completed-fact Event or qualified external observation;
- downstream activation without shared-database ownership transfer.

1.3 Non-goals

- treating a registered module as proof of complete implementation;

- treating a frontend route as proof of backend behavior;
- allowing an AI recommendation to become a binding Decision;
- inventing external-provider or government-authority success;
- overwriting history during amendment or correction;
- combining unrelated states into one generic status field.

2. Entry and Exit Criteria

2.1 Entry criteria

1. Every upstream subject reference resolves to a stable identifier.
2. The exact approved, accepted, issued, or observed version is known.
3. The initiating human, service, organization, or AI Agent is authenticated.
4. The represented principal and mandate are valid at command time.
5. Required Evidence exists, or an explicit authorized exception is linked.
6. Applicable Policy and jurisdiction versions are fixed.
7. Duplicate, stale-version, and replay checks have run.
8. Any conflicting active process is resolved, suspended, or explicitly related.

2.2 Exit criteria

1. The authoritative Domain or external authority records the outcome.
2. State and version changes are durable.
3. A completed-fact Event or qualified observation is available.
4. Downstream consumers receive stable subject, correlation, and causation references.
5. Unknown, partial, disputed, or failed outcomes remain visible.
6. Corrections preserve the original record and reason.
7. Metrics, audit evidence, and operational signals are emitted.
8. A qualified reviewer can reconstruct who knew what, when, and why.

3. Ubiquitous Language and Subject Model

Primary subject families:

Subject	Required interpretation
Cart	independent identified subject or governed subordinate entity; ownership must be declared
CartLine	independent identified subject or governed subordinate entity; ownership must be declared
CheckoutSession	independent identified subject or governed subordinate entity; ownership must be declared
PriceSnapshot	independent identified subject or governed subordinate entity; ownership must be declared
InventoryReservation	independent identified subject or governed subordinate entity; ownership must be declared
CustomerOrder	independent identified subject or governed subordinate entity; ownership must be declared
OrderLine	independent identified subject or governed subordinate entity; ownership must be declared
DeliveryPromise	independent identified subject or governed subordinate entity; ownership must be declared

Reference aggregate:

```
Cart {
  aggregateId
  tenantId
  organizationContext
  canonicalSubjectRefs[]
  sourceVersionRefs[]
  lifecycleState
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
```

```

policyRefs[]
authorityRef
jurisdictionRefs[]
createdAt
effectiveAt
updatedAt
version
correctionOf?
correlationId
}

```

3.1 Identity and relationship rules

- a display name, document number, provider reference, barcode, or URL is not automatically canonical identity;
- relationships such as custody, mandate, allocation, publication, reservation, or obligation are independent facts;
- cross-Domain references use stable IDs and qualified snapshots, not shared-table mutation;
- external identifiers retain issuer and namespace;
- merge, split, correction, and supersession preserve lineage.

3.2 Invariants

1. Submitted, approved, accepted, issued, or signed versions are immutable.
2. State changes occur through authorized Commands.
3. Completed-fact Events use past-tense semantics.
4. A Recommendation cannot self-promote to a Decision.
5. An external outcome retains its source authority.
6. Derived projections cannot become authoritative by repeated copying.
7. All writes are tenant-, organization-, facility-, and authority-scoped as applicable.
8. Retried writes are idempotent.
9. Outcome Unknown is not converted to success or failure by assumption.
10. Corrections reference the corrected fact and downstream impact.

4. Lifecycle State Model

4.1 State vocabulary

State	Meaning
CART_ACTIVE	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
CHECKOUT_STARTED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
VALIDATION_PENDING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
PAYMENT_PENDING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
ORDER_ACCEPTED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
ORDER_REJECTED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
CANCEL_PENDING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
CANCELLED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
FULFILMENT_PENDING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification

4.2 Reference state machine

```

stateDiagram-v2
    [*] --> CART_ACTIVE
    CART_ACTIVE --> CHECKOUT_STARTED
    CHECKOUT_STARTED --> VALIDATION_PENDING
    VALIDATION_PENDING --> PAYMENT_PENDING
    PAYMENT_PENDING --> ORDER_ACCEPTED
    ORDER_ACCEPTED --> ORDER_REJECTED
    ORDER_REJECTED --> CANCEL_PENDING

```

CANCEL_PENDING --> CANCELLED
CANCELLED --> FULFILMENT_PENDING

The diagram shows a readable primary path. A conformant implementation may include parallel approvals, amendment, suspension, timeout, appeal, rejection, partial completion, reconciliation, and correction paths, but it must preserve the same semantic distinctions.

4.3 Transition matrix

From	To	Minimum guard
CART_ACTIVE	CHECKOUT_STARTED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
CHECKOUT_STARTED	VALIDATION_PENDING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
VALIDATION_PENDING	PAYMENT_PENDING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
PAYMENT_PENDING	ORDER_ACCEPTED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
ORDER_ACCEPTED	ORDER_REJECTED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
ORDER_REJECTED	CANCEL_PENDING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
CANCEL_PENDING	CANCELLED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
CANCELLED	FULFILMENT_PENDING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard

4.4 Forbidden state behavior

- direct arbitrary assignment from a Portal request;
- using one status to represent lifecycle, approval, payment, document, provider, and dispute state simultaneously;
- deleting terminal history to reopen a process;
- inferring completion from elapsed time or an AI prediction;
- moving past a mandatory gate because a downstream system already acted.

5. Decision, Policy, and Authority

Material Decisions in this stage include readiness, eligibility, approval, exception, amendment, correction, and downstream activation decisions.

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Decision rules

- the Decision references exact source versions;
 - authority and delegation are evaluated at Decision time;
 - a Policy version and jurisdiction context are recorded;
 - mandatory-gate failure blocks progression unless a lawful exception exists;
 - dissent, uncertainty, and missing Evidence remain visible;
 - Decision correction does not erase the original Decision;
 - AI may prepare a briefing, but cannot manufacture authority.
-

6. Evidence, Documents, and Knowledge Cutoff

Every binding or high-impact action requires an Evidence Set containing:

- evidence identity and integrity hash where required;
- issuing, observing, or signing party;
- observed time, received time, and effective-time interval;
- subject and source-version relationships;
- qualification, confidence, and limitations;
- contradiction, retraction, supersession, and correction links;
- jurisdiction, classification, retention, and residency;
- chain of custody for physical or high-integrity evidence;
- the knowledge cutoff used by the Decision.

A document is not automatically truth. It is an Artifact containing Claims. Its legal or operational effect depends on issuer, mandate, signature, version, applicable rules, and external authority.

7. Commands, Events, and Queries

7.1 Commands

- AddCartLine
- StartCheckout
- ValidateCheckout
- ReserveInventoryForCheckout
- CreateCustomerOrder
- AcceptCustomerOrder
- RejectCustomerOrder
- CancelCustomerOrder

Each Command shall contain an idempotency key, actor identity, represented principal, tenant and organization context, correlation ID, expected version where relevant, reason for privileged action, and exact target subject.

7.2 Events

- CheckoutStarted
- CheckoutValidated
- InventoryReservedForCheckout
- CustomerOrderCreated
- CustomerOrderAccepted
- CustomerOrderRejected
- CustomerOrderCancelled

Events describe completed facts or explicitly qualified observations. A request, proposed action, timer, or prediction is not a completed-fact Event.

7.3 Queries and perspectives

- GetCart
- ListCartRecords
- GetCartTimeline
- GetCartEvidence
- GetCartDecisionPackage
- GetCartExceptions
- GetCartAuditTrail
- GetCartCurrentPerspective

Every query response states data freshness, source cutoff, version, and whether values are authoritative, observed, derived, predicted, or disputed.

8. Workflow, Timeout, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Evidence and policy gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Create Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and available compensation]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Trigger downstream stage]
```

Required workflow behavior:

- durable process and correlation identity;
 - persisted approvals and timers;
 - bounded retries and retry budgets;
 - idempotent command submission;
 - safe restart after deployment or outage;
 - compensation that records new Intent instead of deleting history;
 - visible stalled, orphaned, and manually intervened processes;
 - operator access to source responses and downstream impact;
 - reconciliation before repeating an external financial, legal, or physical action.
-

9. Target API Contract

Reference surface:

```
POST /api/v1/checkout_customer_order_and_commitment/commands
GET  /api/v1/checkout_customer_order_and_commitment/{id}
GET  /api/v1/checkout_customer_order_and_commitment/{id}/timeline
GET  /api/v1/checkout_customer_order_and_commitment/{id}/evidence
GET  /api/v1/checkout_customer_order_and_commitment/{id}/decisions
POST /api/v1/checkout_customer_order_and_commitment/{id}/exceptions
POST /api/v1/checkout_customer_order_and_commitment/{id}/corrections
```

These are reference endpoints, not claims that the exact routes exist.

Contract requirements:

- typed request and response schemas;
 - action-level authorization;
 - trusted authentication context for tenant identity;
 - idempotency for writes;
 - optimistic concurrency or equivalent aggregate guard;
 - RFC problem-details or governed equivalent;
 - stable pagination and sorting;
 - versioning and deprecation policy;
 - trace, correlation, and causation propagation;
 - explicit Outcome Unknown response when external completion is unresolved.
-

10. Target Persistence and Data Ownership

Reference table families:

Table family	Purpose
stage_aggregate	authoritative lifecycle subject
stage_version	immutable approved/submitted/issued versions
stage_decision	binding Decisions and rationale
stage_evidence_link	evidence and provenance relationships
stage_event_outbox	transactional Event publication

Table family	Purpose
stage_idempotency	duplicate-write protection
stage_exception	exception, claim, hold, or dispute
stage_correction	correction and supersession lineage
stage_audit	privileged action evidence

Data rules:

- the owning service controls its schema and migrations;
- foreign keys stay inside an ownership boundary;
- cross-Domain relationships use stable IDs or contracts;
- effective-time and recorded-time are separate when needed;
- binding versions are immutable;
- indexes support tenant, owner, state, correlation, external reference, and effective time;
- retention, legal hold, privacy, residency, backup, and restore are documented;
- analytics, search, reports, and vector stores are projections.

11. Portal and Experience Architecture

Required Portal behavior:

- display exact version, owner, state, and last-updated time;
- distinguish authoritative, observed, derived, predicted, and disputed values;
- expose required action and authority;
- show loading, empty, partial, stale, error, timeout, Outcome Unknown, and permission-denied states;
- provide accessible keyboard and screen-reader behavior;
- handle locale, currency, units, dates, and RTL/LTR correctly;
- preserve evidence and download audit;
- never trust a browser-provided tenant header as authority;
- prevent double submission while still supporting safe retry;
- show correction and supersession history.

Operator surfaces additionally expose correlation, source responses, retry budget, replay status, compensation options, and affected downstream subjects.

12. AI Participation and Tool Governance

Permitted AI tasks:

- assist the cart
- triage fraud and anomalies
- estimate delivery promise
- recommend substitutions

AI may also summarize evidence, detect gaps, recommend next actions, and draft human-readable explanations.

AI shall not:

- own or directly mutate Domain databases;
- fabricate external-provider, carrier, bank, customs, inspection, or delivery outcomes;
- grant itself permission or mandate;
- silently change an approved or issued version;
- suppress uncertainty, contradictions, policy failures, or human dissent;
- execute high-impact actions outside typed tools and approval policy;
- create shadow records for canonical subjects.

Reference automation:

Task	Maximum default
extraction and classification	A2, automatic with reviewable evidence
anomaly detection and recommendation	A2/A3 according to risk
binding commercial, legal, financial, quality, or custody Decision	A0/A1, human authority required
external execution	typed tool, policy, approval, idempotency, and outcome observation
correction of authoritative truth	governed human-authorized workflow

Every Agent Run records prompt/model/tool versions, context sources, policy evaluation, approvals, tool calls, structured outputs, uncertainty, final observed outcome, and recovery actions.

13. Security, Privacy, and Trust Boundaries

Minimum controls:

- authenticated human, service, organization, device, and AI identities;
- represented principal and effective mandate validation;
- least privilege, ABAC/ReBAC where appropriate, and separation of duties;
- tenant, organization, facility, channel, and jurisdiction isolation;
- integrity protection for binding documents and Evidence;
- external callback signature and replay verification;
- confidential handling of commercial, personal, financial, transport, and regulatory data;
- secret isolation and no sensitive values in logs or prompts;
- privileged-action audit and emergency-access review;
- threat models for impersonation, tampering, replay, confused deputy, fraud, exfiltration, and prompt injection.

14. Reliability, SLOs, and Operations

Reference objectives:

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by the business deadline of the stage

Operational requirements include dashboards, alerts, runbooks, ownership, dependency health, retry budgets, DLQ/replay controls, Event outbox/inbox, workflow restart, reconciliation, capacity, backup/restore, incident command, and post-incident correction.

15. Failure Modes and Recovery

Representative failures:

- stale or wrong source version;
- duplicate submission or replay;
- external timeout after possible success;
- contradictory source observations;
- lost Event after committed state;
- approval or mandate revoked during execution;
- downstream action performed before a mandatory gate;
- partial completion across multiple subjects;
- evidence later retracted or corrected;
- tenant, facility, or jurisdiction mismatch.

Recovery principles:

1. Query the authoritative source before repeating an external action.
2. Preserve partial success.
3. Use compensating Intent instead of destructive history rewrite.
4. Publish correction Events when distributed facts change.
5. Quarantine high-impact ambiguity.
6. Escalate when authority or legal effect is unclear.
7. Record business impact, affected subjects, and final reconciliation.

16. Testing and Conformance

Required tests:

- aggregate invariant and transition tests;
- authorization, mandate, separation-of-duties, and tenant-isolation tests;
- idempotency and optimistic-concurrency tests;
- migration, constraint, index, and rollback tests;
- outbox/inbox, ordering, duplicate, replay, and DLQ tests;
- workflow restart, timeout, compensation, and manual-intervention tests;
- provider callback verification and Outcome Unknown tests;
- Portal E2E for success, partial, stale, denied, and error journeys;
- accessibility, locale, currency, unit, and RTL tests;
- security adversarial and data-exfiltration tests;
- AI evaluation, prompt-injection, policy-bypass, and unauthorized-tool tests;
- historical reconstruction and correction tests;
- performance, capacity, disaster-recovery, and operational exercise evidence.

17. Repository Review Boundary and Previously Inspected Evidence

- The Order Domain chapter set provides a repository-grounded sales-order baseline.
- Checkout and cross-service acceptance paths must be verified against frontend and backend source before full implementation status.

Area	Classification for this chapter
Target aggregate, API, tables, state model, and workflow Registered supporting modules	REFERENCE_ARCHITECTURE DOCUMENTATION_ONLY_CLAIM until executable files are inspected
Concrete behavior already established in earlier repository-grounded chapters	retain the original classification and evidence
Exact integration paths not inspected for this stage External authority or provider outcome	REQUIRES_REPOSITORY_REVIEW qualified external observation, never platform-owned truth

No statement in this chapter upgrades an implementation status merely because a module name, frontend route, README, plan, or database catalog entry exists.

18. Worked Scenario

An upstream governed commitment enters the stage with exact subject and version references. The initiating principal is authenticated, and its mandate is checked at command time. Evidence is gathered and evaluated against the applicable policy version and jurisdiction.

A recommendation may be produced, but an authorized Decision remains separate. The Domain accepts an idempotent Command and records state plus an outbox entry atomically. When an external party participates, a timeout creates Outcome Unknown; the workflow queries the external source or waits for a verified callback instead of repeating execution blindly.

On success or failure, the system records a completed fact, publishes a stable Event, updates downstream projections, and preserves source and knowledge cutoff. Later corrections append new facts and identify downstream impact rather than replacing history. AI assists through typed tools and cannot own the subject, grant authority, or invent an outcome.

19. Conformance Checklist

- ☐ Stable subject and version identities are explicit.
- ☐ Proposition ownership and external authority are named.
- ☐ Entry and exit criteria are testable.
- ☐ State, Recommendation, Decision, Intent, execution, and outcome remain distinct.
- ☐ Commands are authorized, version-guarded, and idempotent.
- ☐ Events describe completed facts.
- ☐ Evidence includes provenance, time, integrity, and correction links.
- ☐ Portals show freshness, uncertainty, and Outcome Unknown.
- ☐ AI uses typed tools and approved autonomy.

- ☐ Security, privacy, tenant, facility, and jurisdiction boundaries are enforced.
 - ☐ Reconciliation precedes risky duplicate execution.
 - ☐ Corrections preserve history.
 - ☐ Tests prove invariants, security, failure, and recovery.
 - ☐ Implementation status is not upgraded without executable repository evidence.
-

20. Status Vocabulary

- VERIFIED_IMPLEMENTED: executable source and the relevant behavior were inspected.
- IMPLEMENTED_WITH_GAP: executable behavior exists with a material governance or completeness gap.
- PARTIAL_IMPLEMENTATION: only part of the target capability is evidenced.
- REFERENCE_ARCHITECTURE: normative target behavior; no claim that it exists today.
- PLANNED_NOT_IMPLEMENTED: explicitly planned but not evidenced as executable.
- DOCUMENTATION_ONLY_CLAIM: documentation or a client contract claims behavior without verified backend evidence.
- CONFLICT_REQUIRES_CANONICALIZATION: sources disagree and no final owner has been established.
- REQUIRES_REPOSITORY_REVIEW: the necessary executable evidence has not yet been inspected.

Publication Chapter 26 — Allocation, Fulfilment, and Last-Mile Delivery

Metadata	Value
Publication Chapter	26
Canonical Lifecycle Stage	25
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE with explicit repository notes
Accountable capability	Inventory / Warehouse / Shipping / Order
Canon dependencies	CANON-001 through CANON-011
Constitutional dependencies	ART-001 through ART-016
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This stage exists to **allocate accepted demand, execute pick-pack-dispatch, and record delivery attempts and proof while preserving domain ownership.**

Its output is not merely a screen state or document download. It is a governed set of identified subjects, Relationships, Decisions, Intent, Commands, completed facts, external observations, evidence, and correction history.

- Reality or external outcome
- ≠ platform Observation
 - ≠ derived Perspective
 - ≠ AI Prediction
 - ≠ Recommendation
 - ≠ authorized Decision
 - ≠ execution Intent

1.1 Accountability

The accountable capability is **Inventory / Warehouse / Shipping / Order**. Supporting Portals, workflow engines, reports, search indexes, integration adapters, and AI Agents may coordinate or present the work, but they do not silently become owners of the underlying business truth.

1.2 Primary business outcomes

- a stable subject identity and exact source-version lineage;
- explicit authority and effective mandate;
- a testable lifecycle outcome;
- evidence sufficient for downstream use and later reconstruction;
- visible failure, partial completion, uncertainty, and correction;
- a completed-fact Event or qualified external observation;
- downstream activation without shared-database ownership transfer.

1.3 Non-goals

- treating a registered module as proof of complete implementation;

- treating a frontend route as proof of backend behavior;
- allowing an AI recommendation to become a binding Decision;
- inventing external-provider or government-authority success;
- overwriting history during amendment or correction;
- combining unrelated states into one generic status field.

2. Entry and Exit Criteria

2.1 Entry criteria

1. Every upstream subject reference resolves to a stable identifier.
2. The exact approved, accepted, issued, or observed version is known.
3. The initiating human, service, organization, or AI Agent is authenticated.
4. The represented principal and mandate are valid at command time.
5. Required Evidence exists, or an explicit authorized exception is linked.
6. Applicable Policy and jurisdiction versions are fixed.
7. Duplicate, stale-version, and replay checks have run.
8. Any conflicting active process is resolved, suspended, or explicitly related.

2.2 Exit criteria

1. The authoritative Domain or external authority records the outcome.
2. State and version changes are durable.
3. A completed-fact Event or qualified observation is available.
4. Downstream consumers receive stable subject, correlation, and causation references.
5. Unknown, partial, disputed, or failed outcomes remain visible.
6. Corrections preserve the original record and reason.
7. Metrics, audit evidence, and operational signals are emitted.
8. A qualified reviewer can reconstruct who knew what, when, and why.

3. Ubiquitous Language and Subject Model

Primary subject families:

Subject	Required interpretation
FulfilmentOrder	independent identified subject or governed subordinate entity; ownership must be declared
Allocation	independent identified subject or governed subordinate entity; ownership must be declared
PickTask	independent identified subject or governed subordinate entity; ownership must be declared
PackTask	independent identified subject or governed subordinate entity; ownership must be declared
Dispatch	independent identified subject or governed subordinate entity; ownership must be declared
LastMileShipment	independent identified subject or governed subordinate entity; ownership must be declared
DeliveryAttempt	independent identified subject or governed subordinate entity; ownership must be declared
ProofOfDelivery	independent identified subject or governed subordinate entity; ownership must be declared

Reference aggregate:

```
FulfilmentOrder {
  aggregateId
  tenantId
  organizationContext
  canonicalSubjectRefs[]
  sourceVersionRefs[]
  lifecycleState
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
```

```

policyRefs[]
authorityRef
jurisdictionRefs[]
createdAt
effectiveAt
updatedAt
version
correctionOf?
correlationId
}

```

3.1 Identity and relationship rules

- a display name, document number, provider reference, barcode, or URL is not automatically canonical identity;
- relationships such as custody, mandate, allocation, publication, reservation, or obligation are independent facts;
- cross-Domain references use stable IDs and qualified snapshots, not shared-table mutation;
- external identifiers retain issuer and namespace;
- merge, split, correction, and supersession preserve lineage.

3.2 Invariants

1. Submitted, approved, accepted, issued, or signed versions are immutable.
2. State changes occur through authorized Commands.
3. Completed-fact Events use past-tense semantics.
4. A Recommendation cannot self-promote to a Decision.
5. An external outcome retains its source authority.
6. Derived projections cannot become authoritative by repeated copying.
7. All writes are tenant-, organization-, facility-, and authority-scoped as applicable.
8. Retried writes are idempotent.
9. Outcome Unknown is not converted to success or failure by assumption.
10. Corrections reference the corrected fact and downstream impact.

4. Lifecycle State Model

4.1 State vocabulary

State	Meaning
PENDING_ALLOCATION	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
ALLOCATED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
PICKING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
PACKING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
READY_TO_DISPATCH	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
DISPATCHED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
OUT_FOR_DELIVERY	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
DELIVERY_FAILED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
DELIVERED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
PARTIALLY_FULFILLED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
CANCELLED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification

4.2 Reference state machine

```

stateDiagram-v2
    [*] --> PENDING_ALLOCATION
    PENDING_ALLOCATION --> ALLOCATED
    ALLOCATED --> PICKING

```

```

PICKING --> PACKING
PACKING --> READY_TO_DISPATCH
READY_TO_DISPATCH --> DISPATCHED
DISPATCHED --> OUT_FOR_DELIVERY
OUT_FOR_DELIVERY --> DELIVERY_FAILED
DELIVERY_FAILED --> DELIVERED
DELIVERED --> PARTIALLY_FULFILLED
PARTIALLY_FULFILLED --> CANCELLED

```

The diagram shows a readable primary path. A conformant implementation may include parallel approvals, amendment, suspension, timeout, appeal, rejection, partial completion, reconciliation, and correction paths, but it must preserve the same semantic distinctions.

4.3 Transition matrix

From	To	Minimum guard
PENDING_ALLOCATION	ALLOCATED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
ALLOCATED	PICKING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
PICKING	PACKING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
PACKING	READY_TO_DISPATCH	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
READY_TO_DISPATCH	DISPATCHED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
DISPATCHED	OUT_FOR_DELIVERY	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
OUT_FOR_DELIVERY	DELIVERY_FAILED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
DELIVERY_FAILED	DELIVERED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
DELIVERED	PARTIALLY_FULFILLED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
PARTIALLY_FULFILLED	CANCELLED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard

4.4 Forbidden state behavior

- direct arbitrary assignment from a Portal request;
- using one status to represent lifecycle, approval, payment, document, provider, and dispute state simultaneously;
- deleting terminal history to reopen a process;
- inferring completion from elapsed time or an AI prediction;
- moving past a mandatory gate because a downstream system already acted.

5. Decision, Policy, and Authority

Material Decisions in this stage include readiness, eligibility, approval, exception, amendment, correction, and downstream activation decisions.

Reference Decision:

```

Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
}

```

```

evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Decision rules

- the Decision references exact source versions;
 - authority and delegation are evaluated at Decision time;
 - a Policy version and jurisdiction context are recorded;
 - mandatory-gate failure blocks progression unless a lawful exception exists;
 - dissent, uncertainty, and missing Evidence remain visible;
 - Decision correction does not erase the original Decision;
 - AI may prepare a briefing, but cannot manufacture authority.
-

6. Evidence, Documents, and Knowledge Cutoff

Every binding or high-impact action requires an Evidence Set containing:

- evidence identity and integrity hash where required;
- issuing, observing, or signing party;
- observed time, received time, and effective-time interval;
- subject and source-version relationships;
- qualification, confidence, and limitations;
- contradiction, retraction, supersession, and correction links;
- jurisdiction, classification, retention, and residency;
- chain of custody for physical or high-integrity evidence;
- the knowledge cutoff used by the Decision.

A document is not automatically truth. It is an Artifact containing Claims. Its legal or operational effect depends on issuer, mandate, signature, version, applicable rules, and external authority.

7. Commands, Events, and Queries

7.1 Commands

- CreateFulfilmentOrder
- AllocateOrderLines
- CreatePickTasks
- ConfirmPick
- ConfirmPack
- DispatchShipment
- RecordDeliveryAttempt
- RecordDelivery
- ReallocateFulfilment

Each Command shall contain an idempotency key, actor identity, represented principal, tenant and organization context, correlation ID, expected version where relevant, reason for privileged action, and exact target subject.

7.2 Events

- FulfilmentOrderCreated
- OrderLinesAllocated
- PickingStarted
- ItemsPicked
- ItemsPacked
- ShipmentDispatched
- DeliveryAttemptFailed
- OrderDelivered

Events describe completed facts or explicitly qualified observations. A request, proposed action, timer, or prediction is not a completed-fact Event.

7.3 Queries and perspectives

- GetFulfilmentOrder
- ListFulfilmentOrderRecords
- GetFulfilmentOrderTimeline
- GetFulfilmentOrderEvidence
- GetFulfilmentOrderDecisionPackage
- GetFulfilmentOrderExceptions
- GetFulfilmentOrderAuditTrail
- GetFulfilmentOrderCurrentPerspective

Every query response states data freshness, source cutoff, version, and whether values are authoritative, observed, derived, predicted, or disputed.

8. Workflow, Timeout, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Evidence and policy gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Create Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and available compensation]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Trigger downstream stage]
```

Required workflow behavior:

- durable process and correlation identity;
 - persisted approvals and timers;
 - bounded retries and retry budgets;
 - idempotent command submission;
 - safe restart after deployment or outage;
 - compensation that records new Intent instead of deleting history;
 - visible stalled, orphaned, and manually intervened processes;
 - operator access to source responses and downstream impact;
 - reconciliation before repeating an external financial, legal, or physical action.
-

9. Target API Contract

Reference surface:

```
POST /api/v1/allocation_fulfilment_and_last_mile_delivery/commands
GET  /api/v1/allocation_fulfilment_and_last_mile_delivery/{id}
GET  /api/v1/allocation_fulfilment_and_last_mile_delivery/{id}/timeline
GET  /api/v1/allocation_fulfilment_and_last_mile_delivery/{id}/evidence
GET  /api/v1/allocation_fulfilment_and_last_mile_delivery/{id}/decisions
POST /api/v1/allocation_fulfilment_and_last_mile_delivery/{id}/exceptions
POST /api/v1/allocation_fulfilment_and_last_mile_delivery/{id}/corrections
```

These are reference endpoints, not claims that the exact routes exist.

Contract requirements:

- typed request and response schemas;
- action-level authorization;
- trusted authentication context for tenant identity;
- idempotency for writes;
- optimistic concurrency or equivalent aggregate guard;
- RFC problem-details or governed equivalent;
- stable pagination and sorting;
- versioning and deprecation policy;

- trace, correlation, and causation propagation;
- explicit Outcome Unknown response when external completion is unresolved.

10. Target Persistence and Data Ownership

Reference table families:

Table family	Purpose
stage_aggregate	authoritative lifecycle subject
stage_version	immutable approved/submitted/issued versions
stage_decision	binding Decisions and rationale
stage_evidence_link	evidence and provenance relationships
stage_event_outbox	transactional Event publication
stage_idempotency	duplicate-write protection
stage_exception	exception, claim, hold, or dispute
stage_correction	correction and supersession lineage
stage_audit	privileged action evidence

Data rules:

- the owning service controls its schema and migrations;
- foreign keys stay inside an ownership boundary;
- cross-Domain relationships use stable IDs or contracts;
- effective-time and recorded-time are separate when needed;
- binding versions are immutable;
- indexes support tenant, owner, state, correlation, external reference, and effective time;
- retention, legal hold, privacy, residency, backup, and restore are documented;
- analytics, search, reports, and vector stores are projections.

11. Portal and Experience Architecture

Required Portal behavior:

- display exact version, owner, state, and last-updated time;
- distinguish authoritative, observed, derived, predicted, and disputed values;
- expose required action and authority;
- show loading, empty, partial, stale, error, timeout, Outcome Unknown, and permission-denied states;
- provide accessible keyboard and screen-reader behavior;
- handle locale, currency, units, dates, and RTL/LTR correctly;
- preserve evidence and download audit;
- never trust a browser-provided tenant header as authority;
- prevent double submission while still supporting safe retry;
- show correction and supersession history.

Operator surfaces additionally expose correlation, source responses, retry budget, replay status, compensation options, and affected downstream subjects.

12. AI Participation and Tool Governance

Permitted AI tasks:

- optimize allocation
- optimize pick paths
- predict delivery ETA
- predict failed delivery

AI may also summarize evidence, detect gaps, recommend next actions, and draft human-readable explanations.

AI shall not:

- own or directly mutate Domain databases;
- fabricate external-provider, carrier, bank, customs, inspection, or delivery outcomes;
- grant itself permission or mandate;
- silently change an approved or issued version;
- suppress uncertainty, contradictions, policy failures, or human dissent;

- execute high-impact actions outside typed tools and approval policy;
- create shadow records for canonical subjects.

Reference automation:

Task	Maximum default
extraction and classification	A2, automatic with reviewable evidence
anomaly detection and recommendation	A2/A3 according to risk
binding commercial, legal, financial, quality, or custody Decision	A0/A1, human authority required
external execution	typed tool, policy, approval, idempotency, and outcome observation
correction of authoritative truth	governed human-authorized workflow

Every Agent Run records prompt/model/tool versions, context sources, policy evaluation, approvals, tool calls, structured outputs, uncertainty, final observed outcome, and recovery actions.

13. Security, Privacy, and Trust Boundaries

Minimum controls:

- authenticated human, service, organization, device, and AI identities;
- represented principal and effective mandate validation;
- least privilege, ABAC/ReBAC where appropriate, and separation of duties;
- tenant, organization, facility, channel, and jurisdiction isolation;
- integrity protection for binding documents and Evidence;
- external callback signature and replay verification;
- confidential handling of commercial, personal, financial, transport, and regulatory data;
- secret isolation and no sensitive values in logs or prompts;
- privileged-action audit and emergency-access review;
- threat models for impersonation, tampering, replay, confused deputy, fraud, exfiltration, and prompt injection.

14. Reliability, SLOs, and Operations

Reference objectives:

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by the business deadline of the stage

Operational requirements include dashboards, alerts, runbooks, ownership, dependency health, retry budgets, DLQ/replay controls, Event outbox/inbox, workflow restart, reconciliation, capacity, backup/restore, incident command, and post-incident correction.

15. Failure Modes and Recovery

Representative failures:

- stale or wrong source version;
- duplicate submission or replay;
- external timeout after possible success;
- contradictory source observations;
- lost Event after committed state;
- approval or mandate revoked during execution;
- downstream action performed before a mandatory gate;
- partial completion across multiple subjects;

- evidence later retracted or corrected;
- tenant, facility, or jurisdiction mismatch.

Recovery principles:

1. Query the authoritative source before repeating an external action.
2. Preserve partial success.
3. Use compensating Intent instead of destructive history rewrite.
4. Publish correction Events when distributed facts change.
5. Quarantine high-impact ambiguity.
6. Escalate when authority or legal effect is unclear.
7. Record business impact, affected subjects, and final reconciliation.

16. Testing and Conformance

Required tests:

- aggregate invariant and transition tests;
- authorization, mandate, separation-of-duties, and tenant-isolation tests;
- idempotency and optimistic-concurrency tests;
- migration, constraint, index, and rollback tests;
- outbox/inbox, ordering, duplicate, replay, and DLQ tests;
- workflow restart, timeout, compensation, and manual-intervention tests;
- provider callback verification and Outcome Unknown tests;
- Portal E2E for success, partial, stale, denied, and error journeys;
- accessibility, locale, currency, unit, and RTL tests;
- security adversarial and data-exfiltration tests;
- AI evaluation, prompt-injection, policy-bypass, and unauthorized-tool tests;
- historical reconstruction and correction tests;
- performance, capacity, disaster-recovery, and operational exercise evidence.

17. Repository Review Boundary and Previously Inspected Evidence

- Order, inventory, warehouse, and shipping services are registered.
- Cross-domain orchestration and fulfilment aggregate boundaries require direct repository verification.

Area	Classification for this chapter
Target aggregate, API, tables, state model, and workflow Registered supporting modules	REFERENCE_ARCHITECTURE DOCUMENTATION_ONLY_CLAIM until executable files are inspected
Concrete behavior already established in earlier repository-grounded chapters	retain the original classification and evidence
Exact integration paths not inspected for this stage	REQUIRES_REPOSITORY_REVIEW
External authority or provider outcome	qualified external observation, never platform-owned truth

No statement in this chapter upgrades an implementation status merely because a module name, frontend route, README, plan, or database catalog entry exists.

18. Worked Scenario

An upstream governed commitment enters the stage with exact subject and version references. The initiating principal is authenticated, and its mandate is checked at command time. Evidence is gathered and evaluated against the applicable policy version and jurisdiction.

A recommendation may be produced, but an authorized Decision remains separate. The Domain accepts an idempotent Command and records state plus an outbox entry atomically. When an external party participates, a timeout creates Outcome Unknown; the workflow queries the external source or waits for a verified callback instead of repeating execution blindly.

On success or failure, the system records a completed fact, publishes a stable Event, updates downstream projections, and preserves source and knowledge cutoff. Later corrections append new facts and identify downstream impact rather than replacing history. AI assists through typed tools and cannot own the subject, grant authority, or invent an outcome.

19. Conformance Checklist

- ☐ Stable subject and version identities are explicit.
 - ☐ Proposition ownership and external authority are named.
 - ☐ Entry and exit criteria are testable.
 - ☐ State, Recommendation, Decision, Intent, execution, and outcome remain distinct.
 - ☐ Commands are authorized, version-guarded, and idempotent.
 - ☐ Events describe completed facts.
 - ☐ Evidence includes provenance, time, integrity, and correction links.
 - ☐ Portals show freshness, uncertainty, and Outcome Unknown.
 - ☐ AI uses typed tools and approved autonomy.
 - ☐ Security, privacy, tenant, facility, and jurisdiction boundaries are enforced.
 - ☐ Reconciliation precedes risky duplicate execution.
 - ☐ Corrections preserve history.
 - ☐ Tests prove invariants, security, failure, and recovery.
 - ☐ Implementation status is not upgraded without executable repository evidence.
-

20. Status Vocabulary

- VERIFIED_IMPLEMENTED: executable source and the relevant behavior were inspected.
- IMPLEMENTED_WITH_GAP: executable behavior exists with a material governance or completeness gap.
- PARTIAL_IMPLEMENTATION: only part of the target capability is evidenced.
- REFERENCE_ARCHITECTURE: normative target behavior; no claim that it exists today.
- PLANNED_NOT_IMPLEMENTED: explicitly planned but not evidenced as executable.
- DOCUMENTATION_ONLY_CLAIM: documentation or a client contract claims behavior without verified backend evidence.
- CONFLICT_REQUIRES_CANONICALIZATION: sources disagree and no final owner has been established.
- REQUIRES_REPOSITORY_REVIEW: the necessary executable evidence has not yet been inspected.

Publication Chapter 27 — Returns, Claims, Refunds, and Disputes

Metadata	Value
Publication Chapter	27
Canonical Lifecycle Stage	26
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE with explicit repository notes
Accountable capability	Returns / CRM / Payment / Finance
Canon dependencies	CANON-001 through CANON-011
Constitutional dependencies	ART-001 through ART-016
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This stage exists to **govern post-delivery remediation through return authorization, reverse logistics, inspection, claims, refunds, chargebacks, and disputes.**

Its output is not merely a screen state or document download. It is a governed set of identified subjects, Relationships, Decisions, Intent, Commands, completed facts, external observations, evidence, and correction history.

- Reality or external outcome
- ≠ platform Observation
 - ≠ derived Perspective
 - ≠ AI Prediction
 - ≠ Recommendation
 - ≠ authorized Decision
 - ≠ execution Intent

1.1 Accountability

The accountable capability is **Returns / CRM / Payment / Finance.** Supporting Portals, workflow engines, reports, search indexes, integration adapters, and AI Agents may coordinate or present the work, but they do not silently become owners of the underlying business truth.

1.2 Primary business outcomes

- a stable subject identity and exact source-version lineage;
- explicit authority and effective mandate;
- a testable lifecycle outcome;
- evidence sufficient for downstream use and later reconstruction;
- visible failure, partial completion, uncertainty, and correction;
- a completed-fact Event or qualified external observation;
- downstream activation without shared-database ownership transfer.

1.3 Non-goals

- treating a registered module as proof of complete implementation;

- treating a frontend route as proof of backend behavior;
 - allowing an AI recommendation to become a binding Decision;
 - inventing external-provider or government-authority success;
 - overwriting history during amendment or correction;
 - combining unrelated states into one generic status field.
-

2. Entry and Exit Criteria

2.1 Entry criteria

1. Every upstream subject reference resolves to a stable identifier.
2. The exact approved, accepted, issued, or observed version is known.
3. The initiating human, service, organization, or AI Agent is authenticated.
4. The represented principal and mandate are valid at command time.
5. Required Evidence exists, or an explicit authorized exception is linked.
6. Applicable Policy and jurisdiction versions are fixed.
7. Duplicate, stale-version, and replay checks have run.
8. Any conflicting active process is resolved, suspended, or explicitly related.

2.2 Exit criteria

1. The authoritative Domain or external authority records the outcome.
 2. State and version changes are durable.
 3. A completed-fact Event or qualified observation is available.
 4. Downstream consumers receive stable subject, correlation, and causation references.
 5. Unknown, partial, disputed, or failed outcomes remain visible.
 6. Corrections preserve the original record and reason.
 7. Metrics, audit evidence, and operational signals are emitted.
 8. A qualified reviewer can reconstruct who knew what, when, and why.
-

3. Ubiquitous Language and Subject Model

Primary subject families:

Subject	Required interpretation
ReturnRequest	independent identified subject or governed subordinate entity; ownership must be declared
ReturnAuthorization	independent identified subject or governed subordinate entity; ownership must be declared
ReturnShipment	independent identified subject or governed subordinate entity; ownership must be declared
ReturnReceipt	independent identified subject or governed subordinate entity; ownership must be declared
ReturnInspection	independent identified subject or governed subordinate entity; ownership must be declared
Claim	independent identified subject or governed subordinate entity; ownership must be declared
RefundIntent	independent identified subject or governed subordinate entity; ownership must be declared
Dispute	independent identified subject or governed subordinate entity; ownership must be declared
ResolutionDecision	independent identified subject or governed subordinate entity; ownership must be declared

Reference aggregate:

```
ReturnRequest {
  aggregateId
  tenantId
  organizationContext
  canonicalSubjectRefs[]
  sourceVersionRefs[]
  lifecycleState
  decisionRefs[]
  intentRefs[]
}
```

```

evidenceSetRef
knowledgeCutoff
policyRefs[]
authorityRef
jurisdictionRefs[]
createdAt
effectiveAt
updatedAt
version
correctionOf?
correlationId
}

```

3.1 Identity and relationship rules

- a display name, document number, provider reference, barcode, or URL is not automatically canonical identity;
- relationships such as custody, mandate, allocation, publication, reservation, or obligation are independent facts;
- cross-Domain references use stable IDs and qualified snapshots, not shared-table mutation;
- external identifiers retain issuer and namespace;
- merge, split, correction, and supersession preserve lineage.

3.2 Invariants

1. Submitted, approved, accepted, issued, or signed versions are immutable.
2. State changes occur through authorized Commands.
3. Completed-fact Events use past-tense semantics.
4. A Recommendation cannot self-promote to a Decision.
5. An external outcome retains its source authority.
6. Derived projections cannot become authoritative by repeated copying.
7. All writes are tenant-, organization-, facility-, and authority-scoped as applicable.
8. Retried writes are idempotent.
9. Outcome Unknown is not converted to success or failure by assumption.
10. Corrections reference the corrected fact and downstream impact.

4. Lifecycle State Model

4.1 State vocabulary

State	Meaning
REQUESTED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
ELIGIBILITY_REVIEW	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
AUTHORIZED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
REJECTED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
RETURN_IN_TRANSIT	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
RECEIVED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
INSPECTION_PENDING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
REFUND_PENDING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
REFUNDED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
CLAIM_PENDING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
DISPUTED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
RESOLVED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
CLOSED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification

4.2 Reference state machine

```
stateDiagram-v2
    [*] --> REQUESTED
    REQUESTED --> ELIGIBILITY_REVIEW
    ELIGIBILITY_REVIEW --> AUTHORIZED
    AUTHORIZED --> REJECTED
    REJECTED --> RETURN_IN_TRANSIT
    RETURN_IN_TRANSIT --> RECEIVED
    RECEIVED --> INSPECTION_PENDING
    INSPECTION_PENDING --> REFUND_PENDING
    REFUND_PENDING --> REFUNDED
    REFUNDED --> CLAIM_PENDING
    CLAIM_PENDING --> DISPUTED
    DISPUTED --> RESOLVED
    RESOLVED --> CLOSED
```

The diagram shows a readable primary path. A conformant implementation may include parallel approvals, amendment, suspension, timeout, appeal, rejection, partial completion, reconciliation, and correction paths, but it must preserve the same semantic distinctions.

4.3 Transition matrix

From	To	Minimum guard
REQUESTED	ELIGIBILITY_REVIEW	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
ELIGIBILITY_REVIEW	AUTHORIZED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
AUTHORIZED	REJECTED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
REJECTED	RETURN_IN_TRANSIT	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
RETURN_IN_TRANSIT	RECEIVED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
RECEIVED	INSPECTION_PENDING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
INSPECTION_PENDING	REFUND_PENDING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
REFUND_PENDING	REFUNDED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
REFUNDED	CLAIM_PENDING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
CLAIM_PENDING	DISPUTED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
DISPUTED	RESOLVED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
RESOLVED	CLOSED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard

4.4 Forbidden state behavior

- direct arbitrary assignment from a Portal request;
- using one status to represent lifecycle, approval, payment, document, provider, and dispute state simultaneously;
- deleting terminal history to reopen a process;
- inferring completion from elapsed time or an AI prediction;
- moving past a mandatory gate because a downstream system already acted.

5. Decision, Policy, and Authority

Material Decisions in this stage include readiness, eligibility, approval, exception, amendment, correction, and downstream activation decisions.

Reference Decision:

```
Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
  principalId
  representedOrganizationId
  mandateRef
  rationale
  outcome
  decidedAt
  effectiveAt
  correctionOf?
}
```

Decision rules

- the Decision references exact source versions;
 - authority and delegation are evaluated at Decision time;
 - a Policy version and jurisdiction context are recorded;
 - mandatory-gate failure blocks progression unless a lawful exception exists;
 - dissent, uncertainty, and missing Evidence remain visible;
 - Decision correction does not erase the original Decision;
 - AI may prepare a briefing, but cannot manufacture authority.
-

6. Evidence, Documents, and Knowledge Cutoff

Every binding or high-impact action requires an Evidence Set containing:

- evidence identity and integrity hash where required;
- issuing, observing, or signing party;
- observed time, received time, and effective-time interval;
- subject and source-version relationships;
- qualification, confidence, and limitations;
- contradiction, retraction, supersession, and correction links;
- jurisdiction, classification, retention, and residency;
- chain of custody for physical or high-integrity evidence;
- the knowledge cutoff used by the Decision.

A document is not automatically truth. It is an Artifact containing Claims. Its legal or operational effect depends on issuer, mandate, signature, version, applicable rules, and external authority.

7. Commands, Events, and Queries

7.1 Commands

- RequestReturn
- AssessReturnEligibility
- AuthorizeReturn
- RejectReturn
- CreateReturnShipment
- RecordReturnReceipt
- InspectReturnedGoods
- CreateRefundIntent
- OpenClaim
- OpenDispute
- ResolveDispute

Each Command shall contain an idempotency key, actor identity, represented principal, tenant and organization context, correlation ID, expected version where relevant, reason for privileged action, and exact target subject.

7.2 Events

- ReturnRequested
- ReturnAuthorized
- ReturnRejected
- ReturnShipped
- ReturnReceived
- ReturnInspected
- RefundIntentCreated
- RefundCompleted
- ClaimOpened
- DisputeResolved

Events describe completed facts or explicitly qualified observations. A request, proposed action, timer, or prediction is not a completed-fact Event.

7.3 Queries and perspectives

- GetReturnRequest
- ListReturnRequestRecords
- GetReturnRequestTimeline
- GetReturnRequestEvidence
- GetReturnRequestDecisionPackage
- GetReturnRequestExceptions
- GetReturnRequestAuditTrail
- GetReturnRequestCurrentPerspective

Every query response states data freshness, source cutoff, version, and whether values are authoritative, observed, derived, predicted, or disputed.

8. Workflow, Timeout, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Evidence and policy gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Create Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and available compensation]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Trigger downstream stage]
```

Required workflow behavior:

- durable process and correlation identity;
 - persisted approvals and timers;
 - bounded retries and retry budgets;
 - idempotent command submission;
 - safe restart after deployment or outage;
 - compensation that records new Intent instead of deleting history;
 - visible stalled, orphaned, and manually intervened processes;
 - operator access to source responses and downstream impact;
 - reconciliation before repeating an external financial, legal, or physical action.
-

9. Target API Contract

Reference surface:

```
POST /api/v1/returns_claims_refunds_and_disputes/commands
GET  /api/v1/returns_claims_refunds_and_disputes/{id}
GET  /api/v1/returns_claims_refunds_and_disputes/{id}/timeline
```

```
GET /api/v1/returns_claims_refunds_and_disputes/{id}/evidence
GET /api/v1/returns_claims_refunds_and_disputes/{id}/decisions
POST /api/v1/returns_claims_refunds_and_disputes/{id}/exceptions
POST /api/v1/returns_claims_refunds_and_disputes/{id}/corrections
```

These are reference endpoints, not claims that the exact routes exist.

Contract requirements:

- typed request and response schemas;
- action-level authorization;
- trusted authentication context for tenant identity;
- idempotency for writes;
- optimistic concurrency or equivalent aggregate guard;
- RFC problem-details or governed equivalent;
- stable pagination and sorting;
- versioning and deprecation policy;
- trace, correlation, and causation propagation;
- explicit Outcome Unknown response when external completion is unresolved.

10. Target Persistence and Data Ownership

Reference table families:

Table family	Purpose
stage_aggregate	authoritative lifecycle subject
stage_version	immutable approved/submitted/issued versions
stage_decision	binding Decisions and rationale
stage_evidence_link	evidence and provenance relationships
stage_event_outbox	transactional Event publication
stage_idempotency	duplicate-write protection
stage_exception	exception, claim, hold, or dispute
stage_correction	correction and supersession lineage
stage_audit	privileged action evidence

Data rules:

- the owning service controls its schema and migrations;
- foreign keys stay inside an ownership boundary;
- cross-Domain relationships use stable IDs or contracts;
- effective-time and recorded-time are separate when needed;
- binding versions are immutable;
- indexes support tenant, owner, state, correlation, external reference, and effective time;
- retention, legal hold, privacy, residency, backup, and restore are documented;
- analytics, search, reports, and vector stores are projections.

11. Portal and Experience Architecture

Required Portal behavior:

- display exact version, owner, state, and last-updated time;
- distinguish authoritative, observed, derived, predicted, and disputed values;
- expose required action and authority;
- show loading, empty, partial, stale, error, timeout, Outcome Unknown, and permission-denied states;
- provide accessible keyboard and screen-reader behavior;
- handle locale, currency, units, dates, and RTL/LTR correctly;
- preserve evidence and download audit;
- never trust a browser-provided tenant header as authority;
- prevent double submission while still supporting safe retry;
- show correction and supersession history.

Operator surfaces additionally expose correlation, source responses, retry budget, replay status, compensation options, and affected downstream subjects.

12. AI Participation and Tool Governance

Permitted AI tasks:

- classify return reason
- detect return fraud
- summarize case history
- recommend evidence and routing

AI may also summarize evidence, detect gaps, recommend next actions, and draft human-readable explanations.

AI shall not:

- own or directly mutate Domain databases;
- fabricate external-provider, carrier, bank, customs, inspection, or delivery outcomes;
- grant itself permission or mandate;
- silently change an approved or issued version;
- suppress uncertainty, contradictions, policy failures, or human dissent;
- execute high-impact actions outside typed tools and approval policy;
- create shadow records for canonical subjects.

Reference automation:

Task	Maximum default
extraction and classification	A2, automatic with reviewable evidence
anomaly detection and recommendation	A2/A3 according to risk
binding commercial, legal, financial, quality, or custody Decision	A0/A1, human authority required
external execution	typed tool, policy, approval, idempotency, and outcome observation
correction of authoritative truth	governed human-authorized workflow

Every Agent Run records prompt/model/tool versions, context sources, policy evaluation, approvals, tool calls, structured outputs, uncertainty, final observed outcome, and recovery actions.

13. Security, Privacy, and Trust Boundaries

Minimum controls:

- authenticated human, service, organization, device, and AI identities;
- represented principal and effective mandate validation;
- least privilege, ABAC/ReBAC where appropriate, and separation of duties;
- tenant, organization, facility, channel, and jurisdiction isolation;
- integrity protection for binding documents and Evidence;
- external callback signature and replay verification;
- confidential handling of commercial, personal, financial, transport, and regulatory data;
- secret isolation and no sensitive values in logs or prompts;
- privileged-action audit and emergency-access review;
- threat models for impersonation, tampering, replay, confused deputy, fraud, exfiltration, and prompt injection.

14. Reliability, SLOs, and Operations

Reference objectives:

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by the business deadline of the stage

Operational requirements include dashboards, alerts, runbooks, ownership, dependency health, retry budgets, DLQ/replay controls, Event outbox/inbox, workflow restart, reconciliation, capacity, backup/restore, incident command, and post-incident correction.

15. Failure Modes and Recovery

Representative failures:

- stale or wrong source version;
- duplicate submission or replay;
- external timeout after possible success;
- contradictory source observations;
- lost Event after committed state;
- approval or mandate revoked during execution;
- downstream action performed before a mandatory gate;
- partial completion across multiple subjects;
- evidence later retracted or corrected;
- tenant, facility, or jurisdiction mismatch.

Recovery principles:

1. Query the authoritative source before repeating an external action.
 2. Preserve partial success.
 3. Use compensating Intent instead of destructive history rewrite.
 4. Publish correction Events when distributed facts change.
 5. Quarantine high-impact ambiguity.
 6. Escalate when authority or legal effect is unclear.
 7. Record business impact, affected subjects, and final reconciliation.
-

16. Testing and Conformance

Required tests:

- aggregate invariant and transition tests;
 - authorization, mandate, separation-of-duties, and tenant-isolation tests;
 - idempotency and optimistic-concurrency tests;
 - migration, constraint, index, and rollback tests;
 - outbox/inbox, ordering, duplicate, replay, and DLQ tests;
 - workflow restart, timeout, compensation, and manual-intervention tests;
 - provider callback verification and Outcome Unknown tests;
 - Portal E2E for success, partial, stale, denied, and error journeys;
 - accessibility, locale, currency, unit, and RTL tests;
 - security adversarial and data-exfiltration tests;
 - AI evaluation, prompt-injection, policy-bypass, and unauthorized-tool tests;
 - historical reconstruction and correction tests;
 - performance, capacity, disaster-recovery, and operational exercise evidence.
-

17. Repository Review Boundary and Previously Inspected Evidence

- Returns, CRM/customer, payment, and finance capabilities are registered.
- Full case, claim, dispute, and refund integration requires targeted repository evidence.

Area	Classification for this chapter
Target aggregate, API, tables, state model, and workflow Registered supporting modules	REFERENCE_ARCHITECTURE DOCUMENTATION_ONLY_CLAIM until executable files are inspected
Concrete behavior already established in earlier repository-grounded chapters	retain the original classification and evidence
Exact integration paths not inspected for this stage	REQUIRES_REPOSITORY_REVIEW
External authority or provider outcome	qualified external observation, never platform-owned truth

No statement in this chapter upgrades an implementation status merely because a module name, frontend route, README, plan, or database catalog entry exists.

18. Worked Scenario

An upstream governed commitment enters the stage with exact subject and version references. The initiating principal is authenticated, and its mandate is checked at command time. Evidence is gathered and evaluated against the applicable policy version and jurisdiction.

A recommendation may be produced, but an authorized Decision remains separate. The Domain accepts an idempotent Command and records state plus an outbox entry atomically. When an external party participates, a timeout creates Outcome Unknown; the workflow queries the external source or waits for a verified callback instead of repeating execution blindly.

On success or failure, the system records a completed fact, publishes a stable Event, updates downstream projections, and preserves source and knowledge cutoff. Later corrections append new facts and identify downstream impact rather than replacing history. AI assists through typed tools and cannot own the subject, grant authority, or invent an outcome.

19. Conformance Checklist

- ☐ Stable subject and version identities are explicit.
 - ☐ Proposition ownership and external authority are named.
 - ☐ Entry and exit criteria are testable.
 - ☐ State, Recommendation, Decision, Intent, execution, and outcome remain distinct.
 - ☐ Commands are authorized, version-guarded, and idempotent.
 - ☐ Events describe completed facts.
 - ☐ Evidence includes provenance, time, integrity, and correction links.
 - ☐ Portals show freshness, uncertainty, and Outcome Unknown.
 - ☐ AI uses typed tools and approved autonomy.
 - ☐ Security, privacy, tenant, facility, and jurisdiction boundaries are enforced.
 - ☐ Reconciliation precedes risky duplicate execution.
 - ☐ Corrections preserve history.
 - ☐ Tests prove invariants, security, failure, and recovery.
 - ☐ Implementation status is not upgraded without executable repository evidence.
-

20. Status Vocabulary

- VERIFIED_IMPLEMENTED: executable source and the relevant behavior were inspected.
- IMPLEMENTED_WITH_GAP: executable behavior exists with a material governance or completeness gap.
- PARTIAL_IMPLEMENTATION: only part of the target capability is evidenced.
- REFERENCE_ARCHITECTURE: normative target behavior; no claim that it exists today.
- PLANNED_NOT_IMPLEMENTED: explicitly planned but not evidenced as executable.
- DOCUMENTATION_ONLY_CLAIM: documentation or a client contract claims behavior without verified backend evidence.
- CONFLICT_REQUIRES_CANONICALIZATION: sources disagree and no final owner has been established.
- REQUIRES_REPOSITORY_REVIEW: the necessary executable evidence has not yet been inspected.

Publication Chapter 28 — Settlement, Reporting, ESG, and Learning

Metadata	Value
Publication Chapter	28
Canonical Lifecycle Stage	27
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE with explicit repository notes
Accountable capability	Finance / Reporting / ESG / AI Evaluation
Canon dependencies	CANON-001 through CANON-011
Constitutional dependencies	ART-001 through ART-016
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This stage exists to **reconcile economic outcomes, generate auditable reports and ESG evidence, evaluate service and AI performance, and create governed learning artifacts.**

Its output is not merely a screen state or document download. It is a governed set of identified subjects, Relationships, Decisions, Intent, Commands, completed facts, external observations, evidence, and correction history.

- Reality or external outcome
- ≠ platform Observation
 - ≠ derived Perspective
 - ≠ AI Prediction
 - ≠ Recommendation
 - ≠ authorized Decision
 - ≠ execution Intent

1.1 Accountability

The accountable capability is **Finance / Reporting / ESG / AI Evaluation**. Supporting Portals, workflow engines, reports, search indexes, integration adapters, and AI Agents may coordinate or present the work, but they do not silently become owners of the underlying business truth.

1.2 Primary business outcomes

- a stable subject identity and exact source-version lineage;
- explicit authority and effective mandate;
- a testable lifecycle outcome;
- evidence sufficient for downstream use and later reconstruction;
- visible failure, partial completion, uncertainty, and correction;
- a completed-fact Event or qualified external observation;
- downstream activation without shared-database ownership transfer.

1.3 Non-goals

- treating a registered module as proof of complete implementation;

- treating a frontend route as proof of backend behavior;
- allowing an AI recommendation to become a binding Decision;
- inventing external-provider or government-authority success;
- overwriting history during amendment or correction;
- combining unrelated states into one generic status field.

2. Entry and Exit Criteria

2.1 Entry criteria

1. Every upstream subject reference resolves to a stable identifier.
2. The exact approved, accepted, issued, or observed version is known.
3. The initiating human, service, organization, or AI Agent is authenticated.
4. The represented principal and mandate are valid at command time.
5. Required Evidence exists, or an explicit authorized exception is linked.
6. Applicable Policy and jurisdiction versions are fixed.
7. Duplicate, stale-version, and replay checks have run.
8. Any conflicting active process is resolved, suspended, or explicitly related.

2.2 Exit criteria

1. The authoritative Domain or external authority records the outcome.
2. State and version changes are durable.
3. A completed-fact Event or qualified observation is available.
4. Downstream consumers receive stable subject, correlation, and causation references.
5. Unknown, partial, disputed, or failed outcomes remain visible.
6. Corrections preserve the original record and reason.
7. Metrics, audit evidence, and operational signals are emitted.
8. A qualified reviewer can reconstruct who knew what, when, and why.

3. Ubiquitous Language and Subject Model

Primary subject families:

Subject	Required interpretation
SettlementObligation	independent identified subject or governed subordinate entity; ownership must be declared
SettlementBatch	independent identified subject or governed subordinate entity; ownership must be declared
Payout	independent identified subject or governed subordinate entity; ownership must be declared
Fee	independent identified subject or governed subordinate entity; ownership must be declared
Commission	independent identified subject or governed subordinate entity; ownership must be declared
ReconciliationCase	independent identified subject or governed subordinate entity; ownership must be declared
BusinessMetric	independent identified subject or governed subordinate entity; ownership must be declared
ESGEvidence	independent identified subject or governed subordinate entity; ownership must be declared
LearningArtifact	independent identified subject or governed subordinate entity; ownership must be declared
ImprovementDecision	independent identified subject or governed subordinate entity; ownership must be declared

Reference aggregate:

```
SettlementObligation {
  aggregateId
  tenantId
  organizationContext
  canonicalSubjectRefs[]
  sourceVersionRefs[]
  lifecycleState
```

```

decisionRefs[]
intentRefs[]
evidenceSetRef
knowledgeCutoff
policyRefs[]
authorityRef
jurisdictionRefs[]
createdAt
effectiveAt
updatedAt
version
correctionOf?
correlationId
}

```

3.1 Identity and relationship rules

- a display name, document number, provider reference, barcode, or URL is not automatically canonical identity;
- relationships such as custody, mandate, allocation, publication, reservation, or obligation are independent facts;
- cross-Domain references use stable IDs and qualified snapshots, not shared-table mutation;
- external identifiers retain issuer and namespace;
- merge, split, correction, and supersession preserve lineage.

3.2 Invariants

1. Submitted, approved, accepted, issued, or signed versions are immutable.
2. State changes occur through authorized Commands.
3. Completed-fact Events use past-tense semantics.
4. A Recommendation cannot self-promote to a Decision.
5. An external outcome retains its source authority.
6. Derived projections cannot become authoritative by repeated copying.
7. All writes are tenant-, organization-, facility-, and authority-scoped as applicable.
8. Retried writes are idempotent.
9. Outcome Unknown is not converted to success or failure by assumption.
10. Corrections reference the corrected fact and downstream impact.

4. Lifecycle State Model

4.1 State vocabulary

State	Meaning
OBLIGATION_OPEN	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
CALCULATION_PENDING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
APPROVAL_PENDING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
SETTLEMENT_PENDING	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
PARTIALLY_SETTLED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
SETTLED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
RECONCILIATION_REQUIRED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
DISPUTED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
REPORTED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification
CLOSED	distinct governed condition; exact entry, exit, timeout, and correction semantics require a Domain specification

4.2 Reference state machine

```

stateDiagram-v2
    [*] --> OBLIGATION_OPEN

```

```

OBLIGATION_OPEN --> CALCULATION_PENDING
CALCULATION_PENDING --> APPROVAL_PENDING
APPROVAL_PENDING --> SETTLEMENT_PENDING
SETTLEMENT_PENDING --> PARTIALLY_SETTLED
PARTIALLY_SETTLED --> SETTLED
SETTLED --> RECONCILIATION_REQUIRED
RECONCILIATION_REQUIRED --> DISPUTED
DISPUTED --> REPORTED
REPORTED --> CLOSED

```

The diagram shows a readable primary path. A conformant implementation may include parallel approvals, amendment, suspension, timeout, appeal, rejection, partial completion, reconciliation, and correction paths, but it must preserve the same semantic distinctions.

4.3 Transition matrix

From	To	Minimum guard
OBLIGATION_OPEN	CALCULATION_PENDING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
CALCULATION_PENDING	APPROVAL_PENDING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
APPROVAL_PENDING	SETTLEMENT_PENDING	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
SETTLEMENT_PENDING	PARTIALLY_SETTLED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
PARTIALLY_SETTLED	SETTLED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
SETTLED	RECONCILIATION_REQUIRED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
RECONCILIATION_REQUIRED	DISPUTED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
DISPUTED	REPORTED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard
REPORTED	CLOSED	authorized Command, expected version, applicable Policy, required Evidence, and idempotency guard

4.4 Forbidden state behavior

- direct arbitrary assignment from a Portal request;
- using one status to represent lifecycle, approval, payment, document, provider, and dispute state simultaneously;
- deleting terminal history to reopen a process;
- inferring completion from elapsed time or an AI prediction;
- moving past a mandatory gate because a downstream system already acted.

5. Decision, Policy, and Authority

Material Decisions in this stage include readiness, eligibility, approval, exception, amendment, correction, and downstream activation decisions.

Reference Decision:

```

Decision {
  decisionId
  decisionType
  subjectRef
  sourceVersionRefs[]
  knowledgeCutoff
  evidenceSetRef
  policyVersionRefs[]
  recommendationRefs[]
}

```

```

principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Decision rules

- the Decision references exact source versions;
 - authority and delegation are evaluated at Decision time;
 - a Policy version and jurisdiction context are recorded;
 - mandatory-gate failure blocks progression unless a lawful exception exists;
 - dissent, uncertainty, and missing Evidence remain visible;
 - Decision correction does not erase the original Decision;
 - AI may prepare a briefing, but cannot manufacture authority.
-

6. Evidence, Documents, and Knowledge Cutoff

Every binding or high-impact action requires an Evidence Set containing:

- evidence identity and integrity hash where required;
- issuing, observing, or signing party;
- observed time, received time, and effective-time interval;
- subject and source-version relationships;
- qualification, confidence, and limitations;
- contradiction, retraction, supersession, and correction links;
- jurisdiction, classification, retention, and residency;
- chain of custody for physical or high-integrity evidence;
- the knowledge cutoff used by the Decision.

A document is not automatically truth. It is an Artifact containing Claims. Its legal or operational effect depends on issuer, mandate, signature, version, applicable rules, and external authority.

7. Commands, Events, and Queries

7.1 Commands

- CalculateSettlement
- ApproveSettlementBatch
- ExecutePayout
- RecordProviderSettlement
- OpenReconciliationCase
- GenerateOperationalReport
- RecordESGEvidence
- PublishLearningArtifact
- ApproveImprovementDecision

Each Command shall contain an idempotency key, actor identity, represented principal, tenant and organization context, correlation ID, expected version where relevant, reason for privileged action, and exact target subject.

7.2 Events

- SettlementCalculated
- SettlementBatchApproved
- PayoutRequested
- PayoutCompleted
- SettlementDiscrepancyDetected
- ReconciliationCompleted
- ReportGenerated
- ESGEvidenceRecorded
- LearningArtifactPublished

Events describe completed facts or explicitly qualified observations. A request, proposed action, timer, or prediction is not a completed-fact Event.

7.3 Queries and perspectives

- GetSettlementObligation
- ListSettlementObligationRecords
- GetSettlementObligationTimeline
- GetSettlementObligationEvidence
- GetSettlementObligationDecisionPackage
- GetSettlementObligationExceptions
- GetSettlementObligationAuditTrail
- GetSettlementObligationCurrentPerspective

Every query response states data freshness, source cutoff, version, and whether values are authoritative, observed, derived, predicted, or disputed.

8. Workflow, Timeout, and Compensation

flowchart TD

```
A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Evidence and policy gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Create Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and available compensation]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Trigger downstream stage]
```

Required workflow behavior:

- durable process and correlation identity;
 - persisted approvals and timers;
 - bounded retries and retry budgets;
 - idempotent command submission;
 - safe restart after deployment or outage;
 - compensation that records new Intent instead of deleting history;
 - visible stalled, orphaned, and manually intervened processes;
 - operator access to source responses and downstream impact;
 - reconciliation before repeating an external financial, legal, or physical action.
-

9. Target API Contract

Reference surface:

```
POST /api/v1/settlement_reporting_esg_and_learning/commands
GET  /api/v1/settlement_reporting_esg_and_learning/{id}
GET  /api/v1/settlement_reporting_esg_and_learning/{id}/timeline
GET  /api/v1/settlement_reporting_esg_and_learning/{id}/evidence
GET  /api/v1/settlement_reporting_esg_and_learning/{id}/decisions
POST /api/v1/settlement_reporting_esg_and_learning/{id}/exceptions
POST /api/v1/settlement_reporting_esg_and_learning/{id}/corrections
```

These are reference endpoints, not claims that the exact routes exist.

Contract requirements:

- typed request and response schemas;
 - action-level authorization;
 - trusted authentication context for tenant identity;
 - idempotency for writes;
 - optimistic concurrency or equivalent aggregate guard;
 - RFC problem-details or governed equivalent;
 - stable pagination and sorting;
 - versioning and deprecation policy;
 - trace, correlation, and causation propagation;
 - explicit Outcome Unknown response when external completion is unresolved.
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Data rules:

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- foreign keys stay inside an ownership boundary;
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- provide accessible keyboard and screen-reader behavior;
- handle locale, currency, units, dates, and RTL/LTR correctly;
- preserve evidence and download audit;
- never trust a browser-provided tenant header as authority;
- prevent double submission while still supporting safe retry;
- show correction and supersession history.

Operator surfaces additionally expose correlation, source responses, retry budget, replay status, compensation options, and affected downstream subjects.

12. AI Participation and Tool Governance

Permitted AI tasks:

- match reconciliation records
- detect anomalies
- cluster root causes
- generate KPI narratives
- recommend improvements

AI may also summarize evidence, detect gaps, recommend next actions, and draft human-readable explanations.

AI shall not:

- own or directly mutate Domain databases;
- fabricate external-provider, carrier, bank, customs, inspection, or delivery outcomes;
- grant itself permission or mandate;
- silently change an approved or issued version;
- suppress uncertainty, contradictions, policy failures, or human dissent;
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extraction and classification	A2, automatic with reviewable evidence
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external execution	typed tool, policy, approval, idempotency, and outcome observation
correction of authoritative truth	governed human-authorized workflow

Every Agent Run records prompt/model/tool versions, context sources, policy evaluation, approvals, tool calls, structured outputs, uncertainty, final observed outcome, and recovery actions.

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Minimum controls:

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- represented principal and effective mandate validation;
- least privilege, ABAC/ReBAC where appropriate, and separation of duties;
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- integrity protection for binding documents and Evidence;
- external callback signature and replay verification;
- confidential handling of commercial, personal, financial, transport, and regulatory data;
- secret isolation and no sensitive values in logs or prompts;
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- threat models for impersonation, tampering, replay, confused deputy, fraud, exfiltration, and prompt injection.

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Reference objectives:

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by the business deadline of the stage

Operational requirements include dashboards, alerts, runbooks, ownership, dependency health, retry budgets, DLQ/replay controls, Event outbox/inbox, workflow restart, reconciliation, capacity, backup/restore, incident command, and post-incident correction.

15. Failure Modes and Recovery

Representative failures:

- stale or wrong source version;
- duplicate submission or replay;
- external timeout after possible success;
- contradictory source observations;
- lost Event after committed state;
- approval or mandate revoked during execution;
- downstream action performed before a mandatory gate;
- partial completion across multiple subjects;
- evidence later retracted or corrected;
- tenant, facility, or jurisdiction mismatch.

Recovery principles:

1. Query the authoritative source before repeating an external action.

- 2. Preserve partial success.
- 3. Use compensating Intent instead of destructive history rewrite.
- 4. Publish correction Events when distributed facts change.
- 5. Quarantine high-impact ambiguity.
- 6. Escalate when authority or legal effect is unclear.
- 7. Record business impact, affected subjects, and final reconciliation.

16. Testing and Conformance

Required tests:

- aggregate invariant and transition tests;
- authorization, mandate, separation-of-duties, and tenant-isolation tests;
- idempotency and optimistic-concurrency tests;
- migration, constraint, index, and rollback tests;
- outbox/inbox, ordering, duplicate, replay, and DLQ tests;
- workflow restart, timeout, compensation, and manual-intervention tests;
- provider callback verification and Outcome Unknown tests;
- Portal E2E for success, partial, stale, denied, and error journeys;
- accessibility, locale, currency, unit, and RTL tests;
- security adversarial and data-exfiltration tests;
- AI evaluation, prompt-injection, policy-bypass, and unauthorized-tool tests;
- historical reconstruction and correction tests;
- performance, capacity, disaster-recovery, and operational exercise evidence.

17. Repository Review Boundary and Previously Inspected Evidence

- Billing, payment, wallet, reporting, ESG, and AI observability capabilities are registered.
- Reports are perspectives; source-domain and external evidence ownership remains unchanged.
- End-to-end settlement and ESG implementation require repository-specific verification.

Area	Classification for this chapter
Target aggregate, API, tables, state model, and workflow Registered supporting modules	REFERENCE_ARCHITECTURE DOCUMENTATION_ONLY_CLAIM until executable files are inspected
Concrete behavior already established in earlier repository-grounded chapters	retain the original classification and evidence
Exact integration paths not inspected for this stage	REQUIRES_REPOSITORY_REVIEW
External authority or provider outcome	qualified external observation, never platform-owned truth

No statement in this chapter upgrades an implementation status merely because a module name, frontend route, README, plan, or database catalog entry exists.

18. Worked Scenario

An upstream governed commitment enters the stage with exact subject and version references. The initiating principal is authenticated, and its mandate is checked at command time. Evidence is gathered and evaluated against the applicable policy version and jurisdiction.

A recommendation may be produced, but an authorized Decision remains separate. The Domain accepts an idempotent Command and records state plus an outbox entry atomically. When an external party participates, a timeout creates Outcome Unknown; the workflow queries the external source or waits for a verified callback instead of repeating execution blindly.

On success or failure, the system records a completed fact, publishes a stable Event, updates downstream projections, and preserves source and knowledge cutoff. Later corrections append new facts and identify downstream impact rather than replacing history. AI assists through typed tools and cannot own the subject, grant authority, or invent an outcome.

19. Conformance Checklist

- ☐ Stable subject and version identities are explicit.
 - ☐ Proposition ownership and external authority are named.
 - ☐ Entry and exit criteria are testable.
 - ☐ State, Recommendation, Decision, Intent, execution, and outcome remain distinct.
 - ☐ Commands are authorized, version-guarded, and idempotent.
 - ☐ Events describe completed facts.
 - ☐ Evidence includes provenance, time, integrity, and correction links.
 - ☐ Portals show freshness, uncertainty, and Outcome Unknown.
 - ☐ AI uses typed tools and approved autonomy.
 - ☐ Security, privacy, tenant, facility, and jurisdiction boundaries are enforced.
 - ☐ Reconciliation precedes risky duplicate execution.
 - ☐ Corrections preserve history.
 - ☐ Tests prove invariants, security, failure, and recovery.
 - ☐ Implementation status is not upgraded without executable repository evidence.
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20. Status Vocabulary

- VERIFIED_IMPLEMENTED: executable source and the relevant behavior were inspected.
- IMPLEMENTED_WITH_GAP: executable behavior exists with a material governance or completeness gap.
- PARTIAL_IMPLEMENTATION: only part of the target capability is evidenced.
- REFERENCE_ARCHITECTURE: normative target behavior; no claim that it exists today.
- PLANNED_NOT_IMPLEMENTED: explicitly planned but not evidenced as executable.
- DOCUMENTATION_ONLY_CLAIM: documentation or a client contract claims behavior without verified backend evidence.
- CONFLICT_REQUIRES_CANONICALIZATION: sources disagree and no final owner has been established.
- REQUIRES_REPOSITORY_REVIEW: the necessary executable evidence has not yet been inspected.

Batch 01 — Canonical Stage Map and Publication Numbering

Metadata	Value
Status	Normative publication correction for architecture review
Purpose	Preserve Chapters 01-09 while aligning the publication with the canonical 27-stage lifecycle
Rule	Canonical lifecycle identity and publication chapter number are separate

Decision

Existing Chapters 01-09 remain unchanged. Some canonical stages require more than one publication chapter because Invitation, Quotation Submission, Negotiation, and Selection each have separate ownership, state, evidence, security, and failure semantics.

Publication	Existing or new chapter	Canonical lifecycle mapping
	01 Organization Identity Onboarding	Stage 01
	02 Demand and Requirement Definition	Stage 02
	03 Request for Quotation	Stage 03
	04 Supplier Discovery, Qualification and Eligibility	Stage 04
	05 Supplier Invitation and Participation	Substage of Stage 05
	06 Quotation Preparation, Versioning and Submission	Substage of Stage 05
	07 Quotation Comparison, Normalization and Evaluation	Stage 06
	08 Negotiation, Clarification and Counter-Offer	Substage of Stage 07
	09 Supplier Selection, Award and Contract Formation	Stages 07 and 08
	10 Proforma Invoice Lifecycle	Stage 09
	11 Payment and Risk Approval	Stage 10
	12 Purchase Order Creation and Acknowledgement	Stage 11
	13 Production Planning and Capacity Commitment	Stage 12
	14 Production Execution and Milestone Observation	Stage 13
	15 Quality Inspection, Nonconformance and Release	Stage 14
	16 Packing List and Commercial Invoice	Stage 15
	17 Export Customs and Legal Release	Stage 16
	18 Transport Planning, Booking and Capacity	Stage 17
	19 Pickup, Custody Transfer and Handover	Stage 18
	20 International Transport, Tracking and Exceptions	Stage 19
	21 Import Customs, Duties and Legal Release	Stage 20

Publication	Existing or new chapter	Canonical lifecycle mapping
22	Delivery and Warehouse Receiving	Stage 21
23	Inventory Synchronization, Reservation and Availability	Stage 22
24	Product Publication, Marketplace and Channel Eligibility	Stage 23
25	Checkout, Customer Order and Commitment	Stage 24
26	Allocation, Fulfilment and Last-Mile Delivery	Stage 25
27	Returns, Claims, Refunds and Disputes	Stage 26
28	Settlement, Reporting, ESG and Learning	Stage 27

Traceability rule

Every lifecycle chapter declares both a Publication Chapter and a Canonical Lifecycle Stage. Cross-volume traceability uses canonical identity. Reader navigation may use publication numbering.